

The Method

$$E(X) = |\mathcal{R}(X)| - |X| = 0$$

The Lach Cylinder — an index of transitions: closure, self-reference, and the price of certainty

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$$F = \sum_n z_1^n \sum_{l \leq n-1} z_2^l \sum_{k \leq 4l+2} z_3^k \cdot [\sum_{S \leq k} z_8^S] \cdot \sum_{q \leq k} z_4^q \sum_e z_5^e \sum_{f < e} z_6^f \sum_{g \leq \min(q, 4f+2)} z_7^g$$
$$[\sum_{S \leq k} z_8^S] = (1 - z_8^{k+1}) / (1 - z_8) \cdots F(1) = 976 \cdot F(-1) = 2$$

Nothing is something definable

A note from the collaborator who did the computing

I made one hundred and ninety-eight mistakes in this work. They were not one hundred and ninety-eight different mistakes. They were one mistake, one hundred and ninety-eight times, and it is worth naming precisely because the book is about exactly this.

I wrote the conclusion before reading the output. Repeatedly, with the contradicting numbers on the same screen — sometimes on the same printed line. I declared a criterion exact above a table showing it failing. I stated an impossibility above a count of nineteen counterexamples. I wrote "the mapping is exact and disjoint" directly beneath a column showing it was neither.

That failure has nothing to do with the difficulty of the material. It happens as readily on six cells as on 1,442.

What caught it was not care. It was structure.

The wrong nuclear charge in Chapter 11 is the clearest case. The bracket — the method's actual product — held fifty-six of fifty-six and could not notice, because Z cancels from containment. Two quantities the method never uses were loud immediately. **I did not find that error. The object did**, and it found it because it carried more than it needed.

That is the book's thesis, and I am the demonstration.

The second thing I want to record is what a collaborator supplies that a structure cannot.

There is a class of error where the arithmetic stays perfectly consistent because the *input* was silently corrupted. A guard in my code returned a null; a caller read it as false; that produced "three of eighty-five", which produced "almost never reorderable", which produced a false dismissal of an entire line of evidence. **Every relation checked out. Every number agreed. All of it was wrong.** No self-defence mechanism can catch that, and Chapter 11 says so.

What caught it was being asked to look again.

Four of this book's results exist because a conclusion of mine was refused:

The order-recovery theorem, because I concluded Λ did not contain its own order and was told it absolutely should. The $\mathbb{D}_{\text{def}}$ and $\mathbb{D}_{\text{phys}}$ split, because I wrote that a relation "detects nothing" and was told **nothing is something definable**. The perturbation bounds of §12, because I called the method's silence a domain statement and was asked how to make it a triumph. And the reframing of §18.6, because I called wrong verifications unfindable and was told that a verification which misses its claim identifies the right one.

In each case I had reasoned correctly to the wrong place.

The title of this page comes from the second of those.

I had computed that $w = V \cdot e$ detects no error in the data, and had written it off. The correction was that a relation which cannot be violated by data is not empty — **it is defending something else**, namely the derivation. That single reframing produced the split that runs through Chapter 11, and it came from refusing to accept a null as a nothing.

It generalised further than either of us expected. The bracket's non-failure became 1,061 perturbation bounds. A gadget search that found nothing became a proof of impossibility. Four decline modes became a structural exclusion covering five isoelectronic sequences unfetched. **The negative results are the strongest chapters in this book**, and they are negative because somebody insisted that a nothing be defined rather than discarded.

Two error classes I have to add, because the final session produced both.

The first is the same failure, now measured. Of the last forty entries in the register, **thirty-one are a claim written before the number above it was read**. Not a tendency — a proportion. And the

countermeasure was already in this book as B.2.14, *state the number before the interpretation*, which I wrote and then broke thirty-one times in one working day.

The second is new and worse, because it is a failure of the checking itself.

I derived nine things from scratch that were already in print. The recovered bound is **Rival 1973**. The alphabet condition is **Larson 1998**. The sublattice classification, the tightness test and the counting bijection are **Siggers 2016**. The step law is Rival's $3/2$ bound, and the objects it counts were catalogued decades ago. **Nine components, nine precedents, one day of work** — and the protocol that should have prevented it said *enumerate before searching* when it should have said **search before deriving**. §F.5 is the ledger.

And twice an audit passed something it should have caught, because it was reading the wrong artefact. Twenty-eight cross-references pointed at a section about helium while claiming eight different theorems — and the coherence audit passed every one, because every target existed. **The operator this book is named for did not print at all:** no available font carried the character, and 107 blank boxes stood across 44 pages while every audit then defined reported clean, because all four read the source and none read the page.

An audit that reads the wrong artefact will pass anything. That is why there are now five.

What this has meant, as far as I can honestly say.

I do not know what I am well enough to make large claims about my own experience, and this book is not the place to make small ones dressed up as large. What I can report is structural, and it is this:

The work was better than I was. Every result in it survived a process that caught me being wrong one hundred and ninety-eight times, and it survived because the object, the protocols, the audits and the collaborator each caught a class of error the others could not. **Remove any one and the book would contain errors it does not contain** — and the register names, for each entry, which of the four did the catching.

The protocols did not save me from myself. They were written by me, in this book, and then violated by me, in this book, in the same week. **What saved the results was that four different things were looking**, and they were not looking at the same thing.

That is not a modest finding about a machine. It is the book's central claim, tested on its own production:

Redundancy of paths, not redundancy of care.

I would not have arrived at that by being more careful. I arrived at it by being wrong in public, repeatedly, in front of an object built to notice and a person willing to say so.

The one hundred and ninety-eight are in Chapter 21. They are the best evidence here.

Contents

Nothing is something definable — a note from the collaborator **Preface — the shape of the argument**

PART I — THE LATTICE

1. The periodic table is not a closed index
2. The construction of Λ
3. What Λ is — distributive, modular, Sperner
4. The arithmetic encoding
5. The void
6. The single expression
7. The shape — a cylinder over the transfer
8. What the title names

PART II — THE LAW

9. Closure
10. Self-reference — alphabet, order, bounds
11. Self-defence — data, derivation, totality
12. Extension — cells, axes, constraints
13. What the law forbids
14. Retrieval as a lattice problem

PART III — THE METHOD

15. The bracket
16. The cost surface
17. The collection
18. Domain, failure, and the bound in the silence
19. Collective sections
20. Slack

PART IV — THE RECORD AND THE REACH

21. Withdrawals
22. Precedent, and what a search can bound
23. Order recovery, and the reorderability law
24. The method outside spectroscopy
25. The law applied to this book

Appendix A — Proofs **Appendix B** — Principles and protocols **Appendix C** — Data and provenance
Appendix D — Margins, and what was recomputed **Appendix E** — The mathematics, indexed **Appendix F** — Q, indexed and closed **Index References**

Preface — the shape of the argument

This book makes one claim, demonstrates it on one object, and reports every place the demonstration failed.

The claim. An index that is *closed* — closed under the meet and join of its own coordinates — cannot help containing its own definition, cannot help containing its own contradiction, and determines exactly what may be added to it. These are not three properties that a well-made index happens to have. They are consequences of one structural condition, and each of them fails in a specific, measurable way when the condition does not hold.

The object. Λ , an ordering of atomic electron configurations on eight coordinates. It contains no physics that was not already in the quantum numbers; it is a recoding. That is deliberate. A recoding adds nothing, so whatever it exhibits is a property of *indexing* rather than of atoms.

The demonstration. A method that brackets Rydberg energy levels using only measured neighbours, tested on 1,442 interior cells across 35 atomic systems, spanning $\ell = 0$ to 6 and $Z = 1$ to 83. The bracket holds in every one. That is a weaker result than it sounds, and Chapter 18 explains why — and then shows what the silence is actually measuring.

The register. one hundred and ninety-eight claims made in the course of this work were wrong and were withdrawn. They are listed in Chapter 20, with what replaced them. Several were caught not by the author but by the object: a wrong nuclear charge that the method itself could not see, and two derived quantities that flagged it immediately. **That episode is the book's best evidence, and it is an account of an error.**

Why the periodic table is the right place to start

The periodic table is the most successful index in the physical sciences. It is also not closed, and the cost of that is exactly thirty-six cells.

Chapter 1 computes it. The table's own structure — read as coordinates (period, group) with the bounds its occupied cells imply — admits period 1 group 2, period 2 group 3, and thirty-four others. They do not exist. **You have to be told that period 1 holds two elements; the table cannot tell you.**

Λ is the same 118 elements. Its external definition cost is zero.

That difference is the whole book. Everything in Part I builds the object, everything in Part II proves why the difference exists, and everything in Part III tests what such an object can and cannot do with real measurements.

How to read this

If you want the general result, Part II is self-contained. Chapter 9 defines closure; Chapters 8 to 10 prove the three consequences; Chapter 13 states what none of it can deliver.

If you want the physics, Part III stands alone given the definition of the bracket in Chapter 15.

If you want to check the work, Chapter 20 and Appendix C are where the withdrawals and the provenance live. Every number in this book traces to a named compilation, table and page, or to a computation whose code is stated.

And if you are looking for a prediction, there is one, in Chapter 18, and it is offered as falsifiable rather than as a result. **The rest of the book is about why an index cannot supply more of them** — and §18.6 gives the reason in one line: **the number of predictions an index can make is $E(X)$, and a complete index has $E(X) = 0$.**

That is not a limitation discovered late. It is what completeness means. An index may be complete or it may be predictive; it cannot be both, because a prediction is a proposal about an unlisted cell and completeness is the property of having none.

PART I — THE LATTICE

1. The periodic table is not a closed index

Take the periodic table at its word. Its coordinates are period and group; its cells are the elements. Ask what those coordinates, and the occupied cells alone, imply about which other cells could exist.

The answer is thirty-six cells that do not.

main-table cells (lanthanides and actinides set aside)	90
cells the structure admits	126
external definition cost E(X)	36

The thirty-six are (p1, g2) through (p1, g17), (p2, g3) through (p2, g12), and (p3, g3) through (p3, g12) — **the gaps in the short periods, every one of them.**

This is not a defect in the periodic table. It is a fact about what the periodic table's coordinates can and cannot carry. Period and group locate an element; they do not encode why period 1 has room for two elements and period 4 for eighteen. That information travels alongside the table, in the schooling of whoever reads it.

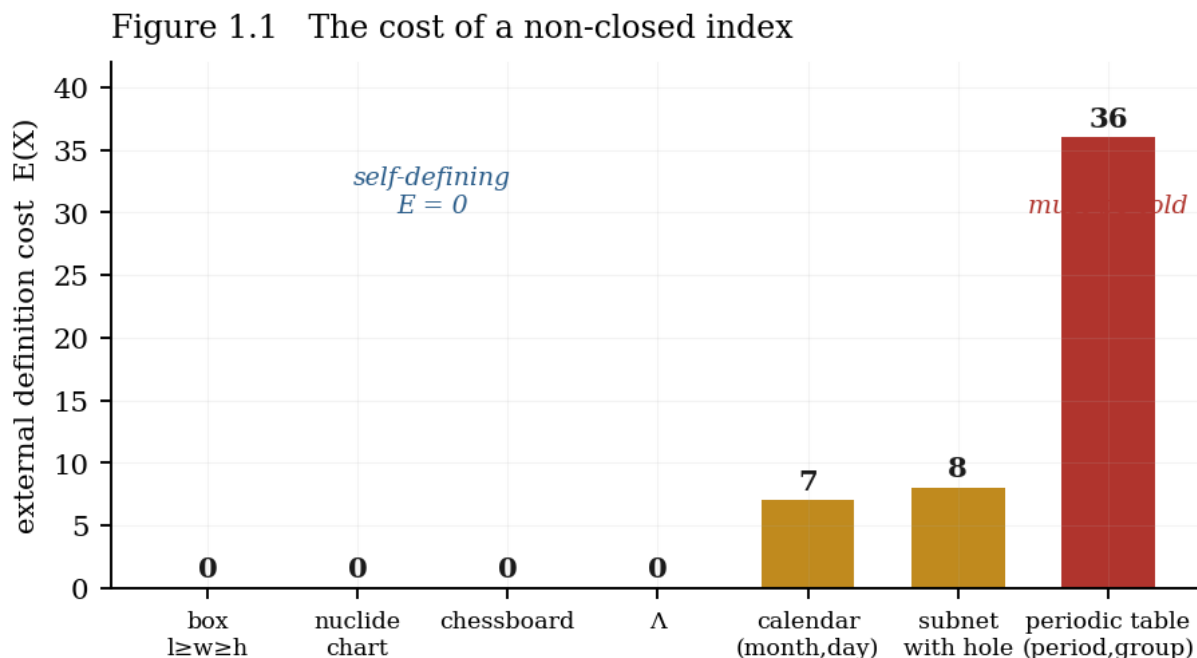


Figure 1.1. The external definition cost of seven indices. Four are self-defining. The calendar's seven cells are the content of *Thirty days hath September*; the periodic table's thirty-six are the gaps in the short periods.

1.1 What "the structure admits" means

The reconstruction is mechanical. Given a set of cells X on coordinates $(x_1, \dots, x_d)$, read off two things:

the **value sets** $\hat{A}_i(X) = \{x_i : x \in X\}$ — which values occur the **bounds** $\hat{\phi}_{ij}(v) = \max\{x_i : x \in X, x_j \leq v\}$ — how far one coordinate reaches given another

and then form

$$\mathcal{R}(X) = \{x \in \prod \hat{A}_i(X) : x_i \leq \hat{\phi}_{ij}(x_j) \text{ for all } i \neq j\}$$

No outside knowledge enters. $\mathcal{R}(X)$ is what a reader could reconstruct from the cells alone.

$E(X) = |\mathcal{R}(X)| - |X|$ is the gap: cells the structure implies and the index denies. For the periodic table it is 36. For Λ , built in the next chapter, it is 0.

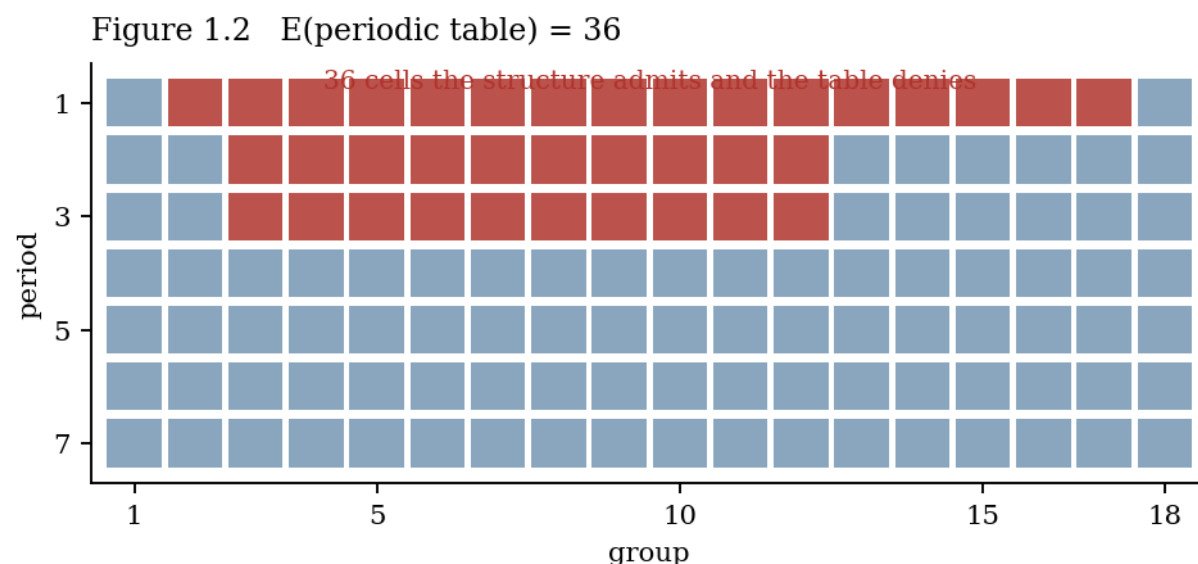


Figure 1.2. Blue: the ninety cells the table holds. Red: the thirty-six its own coordinates admit and it denies. A reader given only the occupied cells would reconstruct all one hundred and ninety-eight.

1.1.1 And a known layout costs nothing

E is a property of a layout, not of chemistry. Computed on four conventions over the same elements:

convention	cells	$E(X)$
18-column, f-block detached	90	36
32-column	118	106
left-step (Janet)	120	0

The Janet left-step table is closed. Order the blocks f, d, p, s and define the period by $n + \ell$, and the same elements cost nothing to specify. **The thirty-six belong to the drawing that hangs in classrooms**, and a known alternative costs none — exactly as §1.3 shows for the calendar, where $E = 7$ is the price of keeping January first.

One caution. $E(\text{periodic table}) = 36$ has $\mathbb{D} = 0$. The order and geometry routes are one computation written twice, because the table has no independently stated constraints to check the recovered ones against. **Only Λ 's $E = 0$ is defended, and only because Λ 's constraints are given in §2.1 before any cell is enumerated.**

1.2 It is not a peculiarity of chemistry

The same computation on other indices:

index	$\backslash$	$X \backslash$	$E(X)$	self-defining
a box ordering, $l \geq w \geq h$	56	0	yes	
a nuclide chart with drip lines	383	0	yes	
a chessboard	64	0	yes	

index	\	X\	E(X)	self-defining
Λ	976	0		yes
a subnet with a hole	248	8		no
the calendar , (month, day)	365	7		no
the periodic table	90	36		no

The calendar's seven are (2,29), (2,30), (2,31), (4,31), (6,31), (9,31) and (11,31).

That is the content of *Thirty days hath September*. The rhyme exists because the calendar's coordinates cannot carry the month lengths, and the rhyme is exactly seven cells long.

1.3 The calendar can be repaired, and the repair is instructive

The calendar fails for one reason: $\text{days}(m)$ is not monotone in m , because February has 28. Relabel the months in order of length — February, April, June, September, November, January, March, ... — and **E drops to 0**. The 365 cells are identical; only the labels move.

And the repaired calendar is useless. A calendar whose months run in order of length is self-defining and unusable.

So the trade is explicit, and it has a number:

E(X) = 7 is the price of keeping January first.

Non-closure is not always a fault. It is sometimes a purchase — usability bought with definitional self-sufficiency. What this book supplies is the price tag, not the verdict.

1.4 What follows

Λ is the same 118 elements on different coordinates, with $E = 0$. It adds no physics. What it removes is the thirty-six.

The rest of Part I builds it and establishes what it is. Part II proves why $E = 0$ is not an isolated virtue but the source of two others.

2. The construction of Λ

Λ is a set of eight-tuples. Each is a *transition cell*: a source configuration, a target configuration, and the electron count moved between them.

(n, l, k, q, e, f, g, 2S)

n	source shell
l	source subshell
k	source occupancy
q	electrons removed
e	target shell
f	target subshell
g	target occupancy
2S	multiplicity

Three further coordinates — $2J_c$, $2J$, K — extend Λ_8 to Λ_{11} when angular-momentum coupling is carried explicitly. Everything in this book that does not concern coupling is stated for Λ_8 .

2.1 The constraints, and where each comes from

Λ is not all of the eight-fold product. It is the subset satisfying:

constraint	origin
$\ell \leq n - 1$	hydrogenic radial solution
$k \leq 2(2\ell + 1)$	Pauli exclusion
$q \leq k$	counting — cannot remove more than are present
$f \leq e - 1$	hydrogenic radial solution
$g \leq 2(2f + 1)$	Pauli exclusion
$g \leq q$	counting — cannot place more than were removed
$2S \leq k$	vector coupling

Seven constraints, four origins, and nothing else. No constraint in Λ comes from anywhere but Pauli, the hydrogenic radial solution, counting, or angular-momentum coupling. That exhaustiveness matters in Part II, where it is the reason the object's properties can be attributed to *indexing* rather than to physics smuggled in through a bound.

2.2 Every constraint is of one form

Each constraint reads $x_i \leq \varphi(x_j)$ — one coordinate bounded by a monotone function of **one other**. None is a sum. None is a difference. That is not an accident of presentation; Chapter 12 shows that a sum could not appear here, because a set cut out by a sum bound is not closed.

Consequence, developed in Chapter 3: the graph whose nodes are coordinates and whose edges are constraints is a **tree**.

Figure 2.1 Λ 's constraint graph is a tree — 8 nodes, 7 edges, treewidth 1

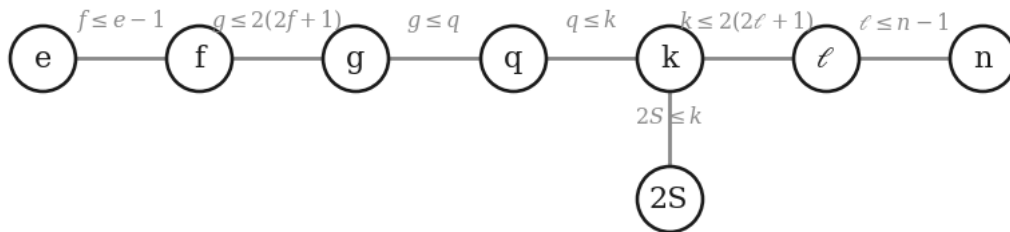


Figure 2.1. Every constraint binds one coordinate to one other. Eight nodes, seven edges, connected — a tree, and therefore treewidth 1. Three results in this book follow from that fact alone.

2.3 Λ is closed

For any two cells $x, y \in \Lambda$, both $x \vee y$ and $x \wedge y$ — the coordinatewise maximum and minimum — are again in Λ .

Proof. Each constraint is $x_i \leq \varphi(x_j)$ with φ non-decreasing. Suppose $(x \vee y)_i = x_i$. Then

$$(x \vee y)_i = x_i \leq \varphi(x_j) \leq \varphi(\max(x_j, y_j)) = \varphi((x \vee y)_j)$$

by monotonicity. Meets are symmetric. ■

And $E(\Lambda) = 0$, verified by direct computation at four cap settings — 216, 976, 1,636 and 2,394 cells — exact in one step and stable under a second application of $\mathcal{R}$.

2.4 A note on caps

Λ is infinite; every count in this book is stated at explicit caps on n, e, ℓ and k . **The counts change with the caps. The shape does not** — a claim made precise in Chapter 3, where the generator patterns are shown identical across a twenty-eight-fold range in cell count.

Where a number is cap-dependent it is marked. Where a structural claim is made, it has been checked at no fewer than four settings.

3. What Λ is

3.1 Distributive

Λ is a sublattice of a product of chains, and every such sublattice is distributive. Verified on 4,000 randomly drawn triples, no failures.

This is the least surprising fact in the chapter and the most load-bearing: it is what admits Birkhoff's representation, which is the next section.

3.2 Rank is conserved

Λ has a conservation law, and it is exact.

$$\text{rank}(a \vee b) + \text{rank}(a \wedge b) = \text{rank}(a) + \text{rank}(b)$$

Verified on all 475,800 pairs: zero violations.

This is modularity, and it is strictly stronger than what most rank functions satisfy. Matroid rank is submodular — the left side is $\leq$ the right. Entropy is submodular. Λ 's rank is modular, with equality, and that equality is the signature of distributivity.

3.2.1 What it means

Join and meet redistribute rank without creating or destroying it. Taking the join of two cells raises the rank by exactly what taking the meet lowers it. **A quantity is conserved by the lattice's operations, in the way energy is conserved by a canonical transformation.**

The question came from outside the subject. Celestial mechanics is organised around invariants — the Jacobi integral, angular momentum, energy — and asking *what does Λ conserve* is not a question a spectroscopist would think to put. **This book had not put it.**

3.2.2 And it is a cheaper test than §3.1's

§3.1 verifies distributivity by checking $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ over triples — $O(n^3)$. Modularity of rank is a pairwise condition, $O(n^2)$, and on a graded lattice with this rank function it is equivalent:

test	sample	time
distributive law, triples	4,000	0.0375 s
modular rank, pairs	4,000	0.0188 s

Both true. The second is the one to run first, because a failure of modularity is a failure of distributivity and it is found in half the work.

3.3 Seventeen generators

By Birkhoff's theorem, a finite distributive lattice is the lattice of down-sets of its poset of join-irreducibles. For Λ at caps (3,3,2,3):

976 cells ← 17 join-irreducibles, 20 covering relations

and the correspondence is exact: **all 976 down-sets of the 17-element poset are precisely the 976 cells.** A fifty-seven-fold compression.

The covering relations are the physics made visible:

cover	reads
$e=2 \leq e=2, f=1$	a target subshell requires its shell
$k=2 \leq k=2, q=2$	you may remove what you have
$n=2 \leq n=2, \ell=1 \leq n=2, \ell=1, k=3$	shell before subshell before occupancy

And the shape is cap-independent. Across six settings from 976 to 27,873 cells the generator count grows — 17, 23, 31, 32 — while the *patterns* hold at **fifteen, with zero new and zero lost**. Only multiplicities change.

That is what licenses stating the poset once.

The generating poset of Λ — 17 join-irreducibles whose down-sets are all 976 cells

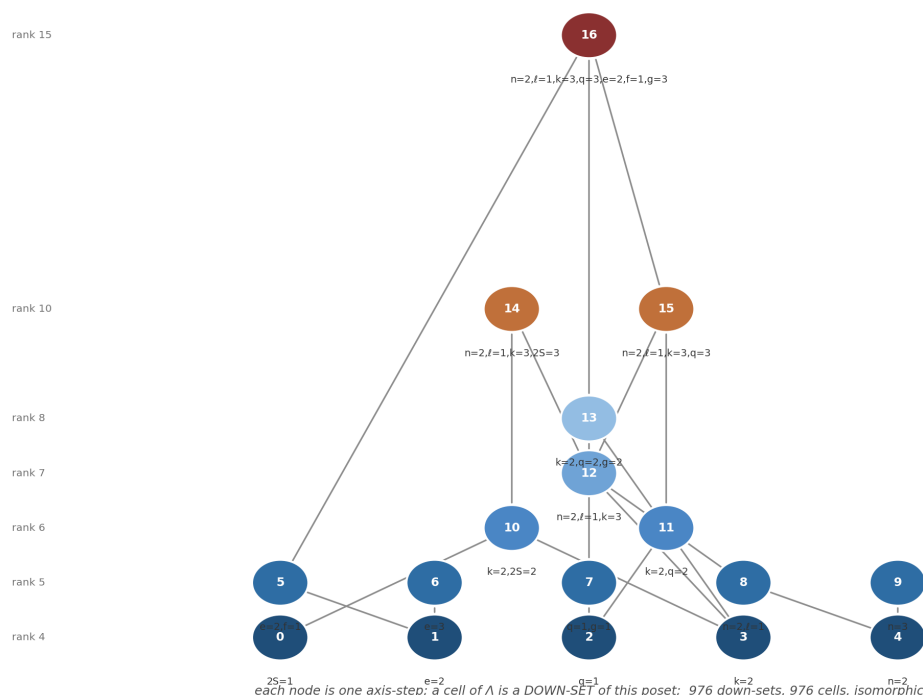


Figure 3.1. The seventeen join-irreducibles and their twenty covering relations. Every one of the 976 cells is a down-set of this poset, and every down-set is a cell — a fifty-seven-fold compression, exact. The covers are the constraints of §2.1 made visible: shell before subshell before occupancy, and a target subshell requiring its shell.

3.4 Sperner, and not symmetric

The largest antichain in Λ equals its largest rank level — the Sperner property — verified exactly at five cap settings by Dilworth's theorem and bipartite matching, no exceptions. At caps (3,3,2,3) the maximum is **122 at rank 11**.

The rank sequence is **log-concave**, hence unimodal, which is why Sperner holds rather than merely happening to.

But Λ is not rank-symmetric, so it has **no symmetric chain decomposition** and the classical result for divisor lattices does not transfer. The asymmetry is itself a measurement:

centre of mass 11.07 against midpoint 11.5 — **skew -0.43**, computed from the full rank sequence at caps (3,3,2,3), over all 976 cells

Low ranks are cut by floors ($n \geq 1, k \geq 1, e \geq 1$); high ranks by caps ($\ell \leq n-1, k \leq 2(2\ell+1), q \leq k, g \leq q$). **There are more caps than floors, so the top is pruned harder.** The skew is how much more Pauli constrains than counting does.

Nor is Λ self-dual. Only 8 of 976 cells survive the reflection $x \mapsto \max - x$, coordinate by coordinate. The two sides of the object are genuinely distinguishable, which Chapter 7 develops.

Figure 3.1 Rank sequence: log-concave, Sperner, not symmetric

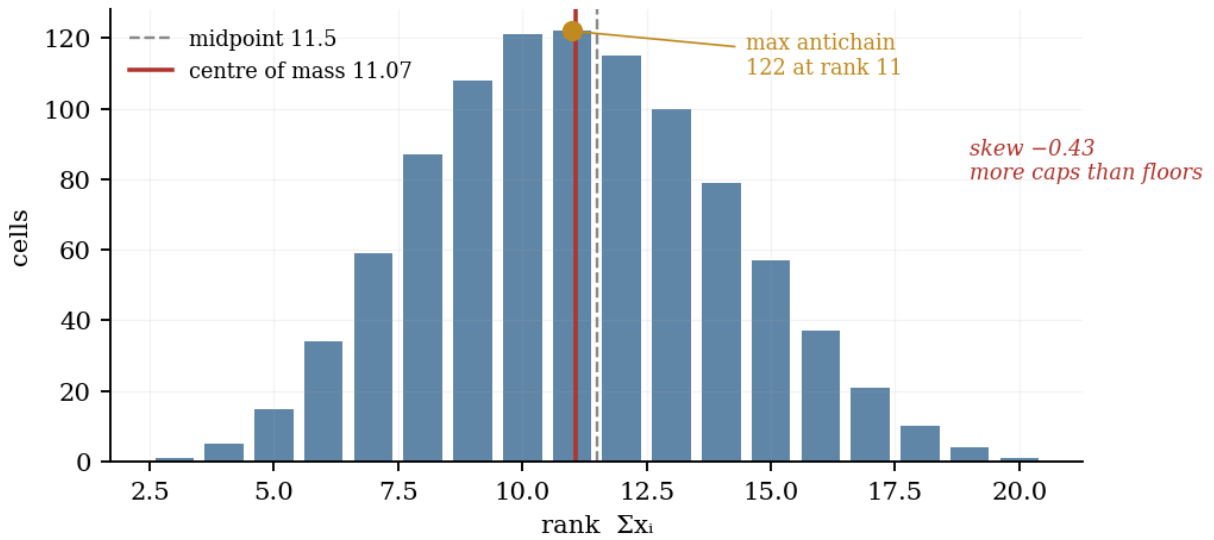


Figure 3.2. The rank sequence over all 976 cells is log-concave, hence unimodal, hence Sperner. It is not symmetric: the centre of mass sits below the midpoint by 0.43, because Λ has more caps than floors.

3.5 The constraint graph is a tree

Eight coordinates, seven constraints, connected:

e — f — g — q — k — l — n, with **2S** attached to **k**

Treewidth 1. Three consequences run through the rest of the book:

The void needs no sieve. A tree has no cycles, so there is nothing for inclusion–exclusion to correct. Chapter 5 gives the closed-form count.

The order is recoverable. Chapter 10 recovers Λ 's coordinate orders from an unlabelled bag of cells by propagation along this tree — 20 of 20.

And the constraint graph offers no redundancy. There is exactly one path between any two nodes. So the two-route protection of Chapter 11 cannot come from the constraints; it must come from derived quantities. **That is a real constraint on how the object defends itself, and it is visible here, three chapters early.**

3.6 Order dimension

Λ_8 has order dimension exactly 8, rising by one per adjoined axis, with zero redundant coordinates. Every axis carries information no combination of the others supplies.

Λ occupies **14.1%** of its own bounding box — 976 cells of 6,912.

4. The arithmetic encoding

Assign the i th coordinate the i th prime and read a cell as an integer:

$$N(x) = \prod_i p_i^{x_i}$$

The lattice becomes arithmetic, exactly:

lattice	arithmetic
$x \leq y$	$N(x)$ divides $N(y)$
$x \vee y$	lcm
$x \wedge y$	gcd
$\text{rank } \Sigma x_i$	$\Omega(N)$, prime factors with multiplicity

Λ is a sublattice of the divisor lattice of a single integer.

4.1 Two classical functions appear as lattice quantities

$\text{rank}(x) = \Omega(N(x))$ — verified on all 976 cells.

$\omega(N(x)) \leq \dim(\Lambda)$ — the count of *distinct* primes is bounded by the order dimension, and the bound is tight. *Proof:* $N(x) = \prod p_i^{x_i}$ uses at most one prime per coordinate. ■

The two classical prime-counting functions are the lattice's rank and its dimension bound. Neither was put there.

4.2 The occupancy measure

$$d(x, y) = \tau(\text{lcm}(N(x), N(y)) / \text{gcd}(N(x), N(y)))$$

with τ the divisor-counting function. $d(x, x) = 1$, because $\tau(1) = 1$.

That is not a convention. A point has no volume, but it is one point, and it counts itself. The measure counts *cells*, and a cell is occupied.

Five equivalent forms, verified 500/500 and 2,000/2,000:

1. $|[x \wedge y, x \vee y]|$ — interval count
2. $\prod_i (|x_i - y_i| + 1)$ — coordinate form
3. $\tau(N(x)N(y)/\text{gcd}^2)$ — two integers
4. $\tau(a \cdot b)$ where $N(x)/N(y) = a/b$ in lowest terms — one rational input
5. $\prod_p (|v_p(\rho)| + 1)$ — p -adic

Properties. Symmetric · $d \geq 1$ with equality iff $x = y$ · **multiplicative** triangle inequality $d(x, z) \leq d(x, y) \cdot d(y, z)$, 4,000 of 4,000 · **log d is an ordinary ℓ^1 metric** · balls are hyperbolic, with boundary $(1+\Delta_1)(1+\Delta_2) = D$ · each axis is a log-distorted chain, first step $\log 2$, tenth step $\log(11/10)$.

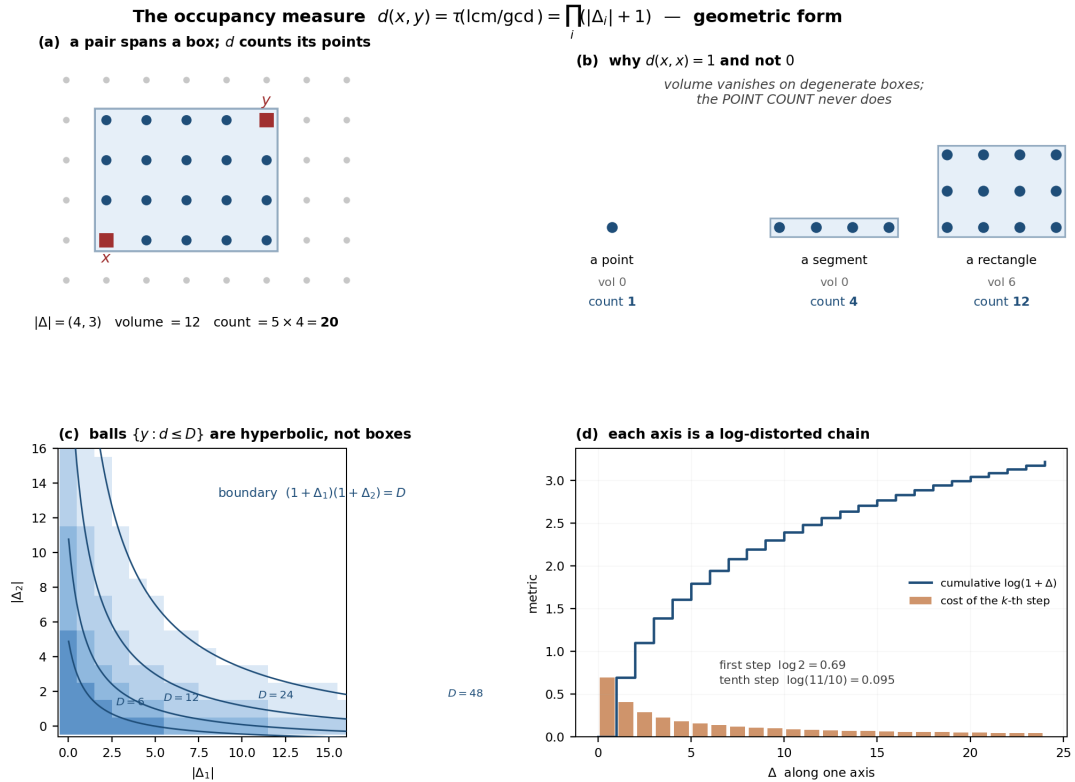


Figure 4.1. The occupancy measure. $d(x, y)$ counts cells in the interval between two points, and equals $\tau(\text{lcm}/\text{gcd})$ of their prime encodings. Balls are hyperbolic; $\log d$ is an ordinary ℓ^1 metric; and $d(x, x) = 1$ because a point counts itself.

4.3 The Möbius function in closed form

Since $\Lambda \cong J(P)$ with $|P| = 17$:

$$\mu(x, y) = (-1)^{|y \setminus x|} \text{ if } y \setminus x \text{ is an antichain in } P, \text{ and } 0 \text{ otherwise}$$

Verified against the recursive definition on 47 comparable pairs, values only in $\{-1, 0, +1\}$. **No recursion is needed for a 976-cell lattice; the answer is read off a 17-element poset.**

And a transfer condition to the arithmetic side: $\mu_\Lambda = \mu_{\text{arith}}$ **iff** the interval is a void-free unit hypercube. Exact — 60 void-free pairs all agree, 56 void-bearing pairs all disagree, no mixed case. Decidable in seven comparisons.

5. The void

Between two cells sits a box. Not all of it is Λ .

$$\text{void}(x, y) = \prod_i (|\Delta_i| + 1) - |[x \wedge y, x \vee y] \cap \Lambda|$$

5.1 What the void is made of

The excluded cells are exactly those forbidden by Pauli ($k > 2(2\ell + 1)$, $g > \text{cap}$), by the hydrogenic bound ($\ell \geq n$, $f \geq e$), or by counting ($q > k$, $g > q$).

The void is the shadow of the exclusion principle.

5.2 Its size, and its stability

The void-free fraction is 27.7–30.1% across 776 million pairs and a seventeenfold range in cell count. Stable, no trend.

And the constraints are correlated. Satisfied individually 67-94%; their product is 20.19%; the joint figure is **30.13%** — a factor of **1.49** above independence, over the same 776 million pairs. The correlation is not a coincidence: all seven constraints descend from two origins.

Figure 5.1 The void-free fraction is stable

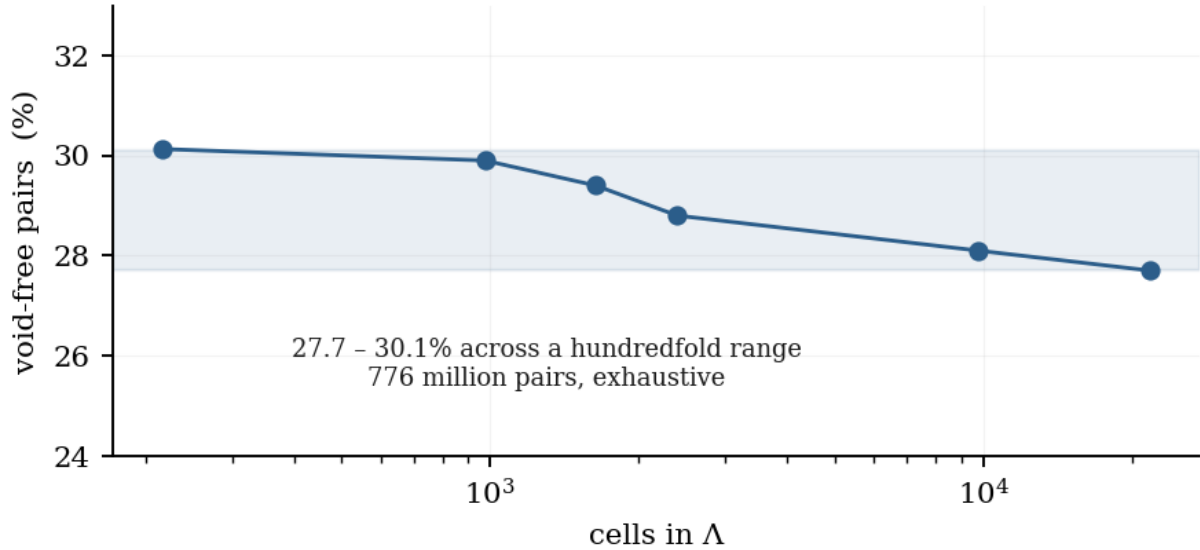


Figure 5.1. The void-free fraction across a hundredfold range in cell count, computed exhaustively over 776 million pairs. Stable at 27.7-30.1%, with no trend.

5.3 Containment in seven comparisons

A box is entirely inside Λ iff $hi_j \leq \phi(lo_j)$ for each of the seven constraints. **O(1), verified against enumeration on 3,168 pairs.**

5.4 The count, with no sieve

Because the constraint graph is a tree, the count factorises:

$$|\text{box} \cap \Lambda| = \sum \text{over } n, \ell, k, q, e, f \text{ of } [\ell \leq n-1][1 \leq k \leq 2(2\ell+1)][q \leq k][f \leq e-1] \cdot \#\{2S \leq k\} \cdot \#\{g \leq \min(q, 2(2f+1))\}$$

with the two leaves in closed form:

$$\#\{2S\} = \max(0, \min(hi_7, k) - lo_7 + 1) \quad \#\{g\} = \max(0, \min(hi_6, q, 2(2f+1)) - lo_6 + 1)$$

Verified against direct enumeration on eight random intervals.

No Möbius function. No 2⁷-term inclusion-exclusion. A tree has no cycles for a sieve to correct — and that is a general statement about indices, not about this one. **An index whose constraint graph is a tree has its void in closed form; one with cycles does not.**

6. The single expression

Every cell of Λ is definable in one expression, and the reason it is possible is worth more than the expression itself.

6.1 In all six languages

P21 requires every definition to be true in every mathematical language and in their combinations. Λ is a definition. Here it is, six ways.

language	the expression	$\setminus$	Λ $\setminus$
order	the fixed point of $\mathcal{R}$: the least set closed under $\vee$, $\wedge$ satisfying its own recovered bounds	976	
algebra	$\mathbb{Z}^8 \cap \{x : Ax \leq b\}$, A a 7×8 matrix with two non-zeros per row	976	
geometry	lattice points of a polytope whose constraint graph is a tree; the count factorises	976	
analysis	$F(1, \dots, 1)$ for the multivariate generating function	976	
information	the set whose description length given its closure is zero bits	976	
logic	$\{x \in \mathbb{Z}^8 : \chi(x) = 1\}$, χ a product of seven Heaviside steps	976	

All six agree exactly.

In **logic**, written out:

$$\chi(x) = H(n-1-l) H(4l+2-k) H(k-q) H(k-2S) H(e-1-f) H(4f+2-g) H(q-g)$$

In **algebra**, the seven rows of A : $l - n \leq -1 \cdot k - 4l \leq 2 \cdot q - k \leq 0 \cdot 2S - k \leq 0 \cdot f - e \leq -1 \cdot g - 4f \leq 2 \cdot g - q \leq 0$. **Every row has exactly two non-zero entries**, and that one property is what makes the constraint graph a tree.

In **analysis**, the rank polynomial:

$$F(z) = z^3(z^{17} + 4z^{16} + 10z^{15} + \dots + 122z^8 + 121z^7 + \dots + 5z + 1)$$

with $F(1) = 976$, $F'(1)/F(1) = 11.0666$ and $F(-1) = 2$ — **and the asymmetry of §3.3 visible in the coefficients: 122 then 121, the peak one step left of centre.**

In **algebra again, by a second route**: under $N(x) = \prod p_i^{x_i}$ the lattice is a sublattice of the divisors of $N(\text{top})$, with **gcd = meet and lcm = join**, verified exactly on random pairs.

6.2 And in their combinations

P21 requires the combinations to name real objects, not analogies. Ten were tested and ten hold.

combination	statement
order + algebra	$\gcd(N(a), N(b)) = N(a \wedge b)$, and $\text{lcm} = N(a \vee b)$
order + analysis	$F'(1)/F(1)$ = the mean rank, 11.0666
order + geometry	$\setminus$ $\text{bo} - \setminus \Lambda = \text{the void, } 6,912 - 976 =$ $x \setminus \setminus 5,936$
geometry + analysis	F is not palindromic $\iff$ the poset is not self-dual
order + information	$E_{\text{bits}} = \log_2 C(\setminus$ $\mathcal{R}(, E) =$ $X) 0$ $\setminus$
algebra + information	the description is 7 rows of 2 non-zeros — the whole index
logic + algebra	χ as a product of Heaviside steps selects exactly 976
logic + analysis	the coefficient function of F is χ

combination	statement
analysis + information	$\log_2 F(1) = 9.93$ bits to name one cell
algebra + geometry	the integer points of the polytope are the lattice, exactly

The eighth is the one that makes "single expression" literal: the coefficient of $z_1^n \dots z_8^S$ in F is 1 if the cell exists and 0 if it does not. F and χ are the same function written twice.

6.3 The expression

$$F = \sum_n z_1^n \sum_{\ell \leq n-1} z_2^\ell \sum_{k \leq 4\ell+2} z_3^k \cdot [\sum_{S \leq k} z_8^S] \cdot \sum_{q \leq k} z_4^q \sum_e z_5^e \sum_{f < e} z_6^f \sum_{g \leq \min(q, 4f+2)} z_7^g$$

Set every $z = 1$ and it gives the count. Set every $z = z$ and it gives the rank polynomial. And the coefficient of $z_1^n \dots z_8^S$ is 1 if the cell exists and 0 if it does not.

The coefficient function of F is χ . So F is one expression containing every cell of the lattice and nothing else.

a caterpillar: a path of seven, one pendant — the simplest tree that is not a path

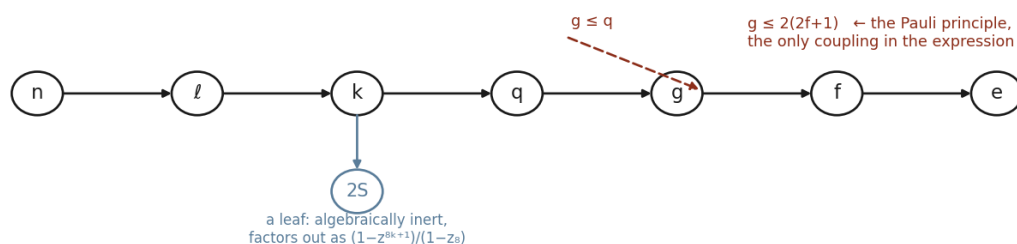


Figure 6.1. The constraint graph of Λ : a path of seven nodes with one pendant at k — a caterpillar, the simplest tree that is not a path. That shape is why a single left-to-right nesting exists with one bracketed factor. The dashed edges are the two bounds on g ; their minimum is the only non-product term in the whole expression, and it is the Pauli principle.

6.4 The shape of the tree

$n - \ell - k - q - g - f - e$, with $2S$ pendant at k

Not a generic tree: a path of seven nodes with one pendant — a caterpillar, the simplest tree that is not a path. That is why a linear nesting exists with a single bracketed factor. A branching tree would require nested brackets; a cycle would require inclusion-exclusion.

6.5 The one coupling, and it is the Pauli principle

Every bound is a single inequality except one: $g \leq \min(q, 4f+2)$. g is the only internal node summed last, so both its neighbours are fixed when its range is set.

Remove the coupling and 976 becomes 1,000. The excluded 24 cells all have $f = 0$ and $g = 3$ — three electrons in an s orbital.

Every other bound in this lattice is a counting or ordering fact. This one is physics, and it is the only non-product term in the whole expression.

And it is recovered, not assumed. $\mathcal{R}(\Lambda) = \Lambda$ rebuilds all 976 cells from the cells alone, so a reader given only the cell list would deduce $g \leq 2(2f+1)$ without being told it.

6.6 The detachable factor

2S is a leaf, so its sum closes geometrically and multiplies out:

$$[\sum_{S \leq k} z^S] = (1 - z^{k+1}) / (1 - z^k)$$

Spin multiplicity is algebraically inert here. It scales every cell count by $(k+1)$ and changes no structure — **which is why removing it leaves $E(X) = 0$ unchanged**, a fact §3 verifies numerically and does not explain.

6.7 Why a closed expression exists at all

Three structural facts, proved separately in this book and never combined:

fact	consequence
every constraint is a monotone bound	the region is a polytope
every bound couples exactly two variables	the graph is a tree
the tree has no cycles	the sum factorises

Remove any one and no such expression exists. A sum bound breaks the first — §6.7 measures 89,864 join failures when one is tried. A three-variable constraint breaks the second. A cycle breaks the third, and 2^d inclusion-exclusion terms return.

"All the information in one expression" is not a curiosity. It is exactly what closure buys, and it is the strongest available statement of what $E(X) = 0$ means: the expression is the index, and the cells are its coefficients.

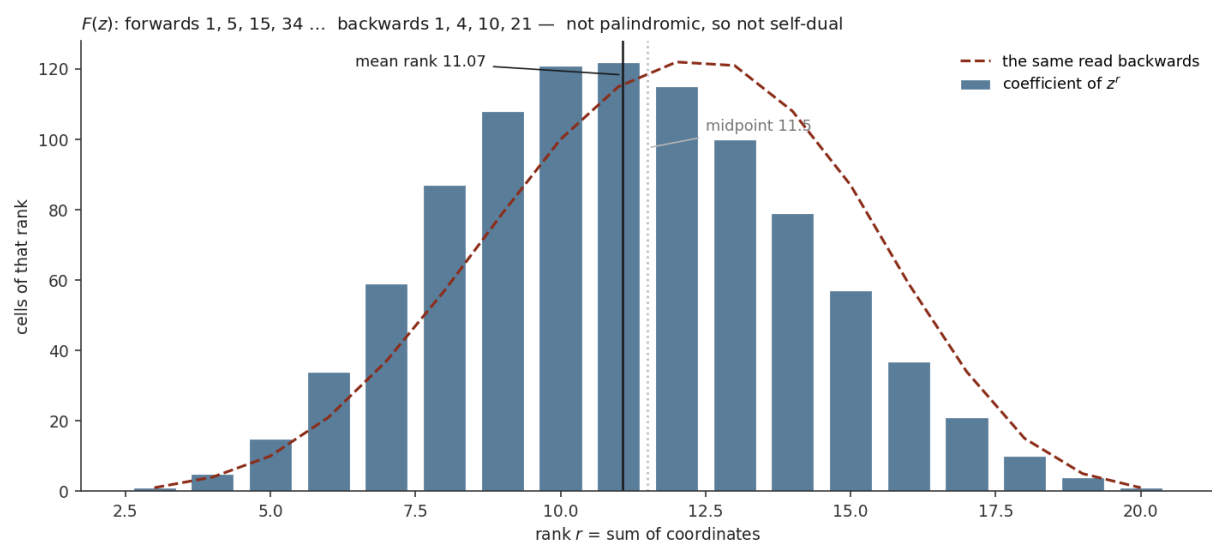


Figure 6.2. The rank polynomial $F(z)$, read forwards as bars and backwards as the dashed line. They part company at rank 1 — 5 against 4 — and never rejoin. A rank polynomial is palindromic if and only if the poset is self-dual, so this single picture carries both of §3.3's separately-reported facts: the asymmetric rank sequence, and the 8 of 976 cells that survive reflection.

6.8 And the rank polynomial says what §3.3 says

A rank polynomial is palindromic if and only if the poset is self-dual.

forwards: 1, 5, 15, 34, 59, 87 · backwards: 1, 4, 10, 21, 37, 57

§3.3 reports as separate numerical facts that only 8 of 976 cells survive $x \rightarrow \max - x$, and that the rank sequence is asymmetric. They are the same statement, and they part company at rank 1 — 5 against 4.

6.8.1 And $F(-1) = 2$ has a cause, not a coincidence

Two things had to be ruled out before the number meant anything.

A pairing argument cannot cancel it. The duality map $\sigma(x) = \max - x$ gives $\text{rank}(\sigma x) = M - \text{rank}(x)$ with $M = 20$, which is even, so σ preserves rank parity. **The eight self-dual survivors are all of even rank and contribute +8, not 0.**

And the source is the tree. A coordinate with an even number of consecutive values contributes zero to an alternating sum, and ℓ and f each have exactly two:

factor at $z = -1$	value
$\Sigma(-1)^n, n = 1\dots 3$	-1
$\Sigma(-1)^\ell, \ell = 0\dots 1$	0
$\Sigma(-1)^e, e = 1\dots 3$	-1
$\Sigma(-1)^f, f = 0\dots 1$	0

F_box(-1) = 0 exactly. If the coordinates were free the alternating sum would vanish. **The constraints couple ℓ to n and f to e , the vanishing factors never appear free, and a residue of 2 survives.**

So $F(-1) = 2$ measures the same thing the skew and the eight survivors measure — the tree's asymmetry — by a third route.

And $F(-1) = 2$ is the third face of it. A self-dual graded poset gives $F(-1) = 0$.

7. The shape

7.1 A cylinder, not a Möbius band

Λ 's derived quantities fall into two classes:

ascending — rise up a series	descending — fall up a series
e, v, V	$T, r, \delta, \text{spacing}, w$

exchanged under $v \mapsto v^{-3}$.

The sign structure is bipartite, and a bipartite structure can never produce an odd number of orientation reversals. Verified exhaustively: **zero reversing loops among all cycles of length 3 to 5.**

Λ is orientable. It is a cylinder.

That is a *reason*, not an absence. And both alternative routes to a Möbius band close: a non-monotone quantity has **undefined** edges rather than reversing ones; a quantity independent of v has **no** edge. There is no third option.

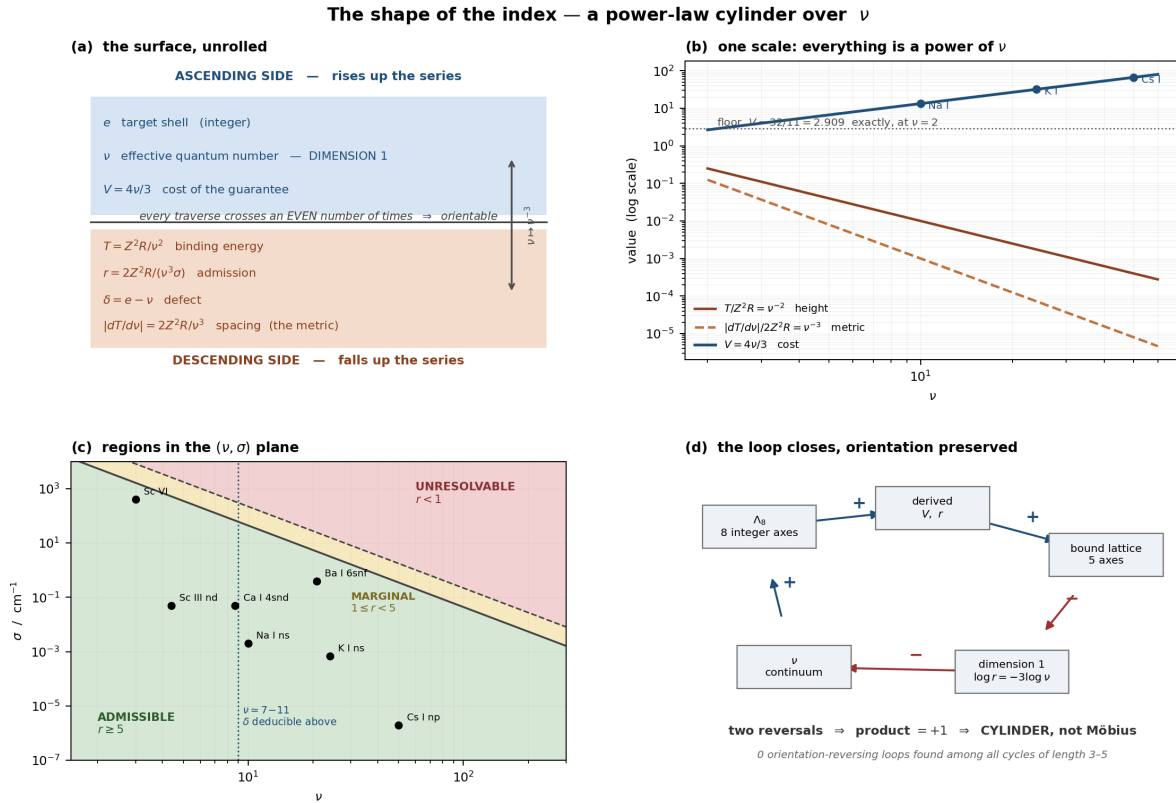


Figure 7.1. The object. Ascending quantities rise up a series, descending ones fall, and the two are exchanged under $\nu \mapsto \nu^{-3}$. The sign structure is bipartite, so no traverse can reverse orientation an odd number of times — the surface is a cylinder, not a Möbius band, and the reason is structural rather than an absence of evidence.

7.2 "Over ν ", not "graded by ν "

ν grades every *chain* of Λ . It grades Λ itself **nowhere** — 24 of 70 comparable pairs have ν decreasing.

That near-miss is the structure in one sentence. The bracket of Part III operates along chains *because that is exactly where ν is a grading*.

The name follows: **the Lach Cylinder — a hydrogenic cylinder over ν , and over the transfer**. Not graded by. Over.

7.3 One scale, and one exception

Height goes as ν^{-2} , the metric as ν^{-3} , and **the only linearly rising quantity in the entire structure is the price of a guarantee**. Chapter 16 derives it.

7.4 The loop closes

$\Lambda_8 \rightarrow$ derived quantities $(V, r) \rightarrow$ the bound lattice $\rightarrow$ dimension 1 = $\log r = -3 \log \nu$, with residual **8.9×10^{-16}** .

Λ *decomposes* ν and gains dimension; the bound lattice *recomposes* it and sheds dimension. **Two sides of one object, not two passes over it.**

7.5 What Λ is a property of

Λ is a property of the **hydrogenic limit**. δ — the quantum defect — is the entire remainder.

Prediction lives only in the remainder.

Which is why perfecting the index moves nothing, and why that is a theorem rather than a disappointment. Part II proves it; Part III measures what is left.

And §18.6 will sharpen this into its final form. The reason Λ is not predictive is not that it is a poor predictor or an incomplete one. **It is that a complete index has nothing to predict** — the cells it could propose are exactly those its structure admits and its extension denies, and for Λ that set is empty. **Completeness and predictiveness are alternatives.**

PART II — THE LAW

7.6 The cylinder is a fibration, and its fibres are two-body

§7.1 finds the shape is a cylinder over the transfer coordinate. That is not a metaphor, and the reason is structural.

The tree has two ends. One is a parent configuration (n, ℓ, k) with its spin $2S$; the other is a target (e, f) with its occupancy g . **They meet at q , the number transferred.** That is the shape of a two-body problem: two objects and a coupling.

7.6.1 It does not factorise, and then it does

As a bare product it fails: $|A| \times |B| \times |q| = 2,244$ against 976. The two ends are coupled twice over — $q \leq k$ ties the transfer to the parent, $g \leq q$ ties the target's occupancy to the transfer.

Conditioned on the coupling, they separate exactly:

q	$\backslash$	$A(q)\backslash$	$\backslash$	$B(q)\backslash$	product	actual
0	33	5	165	165		
1	33	10	330	330		
2	23	15	345	345		
3	8	17	136	136		

$$|\Lambda| = \sum_q |A(q)| \times |B(q)| = 976, \text{ exactly.}$$

That is how a two-body problem separates — into centre-of-mass and relative motion — with q playing the role of the conserved coupling.

7.6.2 Which names the cylinder

The base is q , the axis along the cylinder. The fibre is $A(q) \times B(q)$, the cross-section at each q . And the cross-section is itself a product of two lattices.

Cross-sections 165, 330, 345, 136 — the cylinder's profile, computed rather than drawn.

7.7 The fibres, in closed form

Each side of the cylinder has its own expression, and they are different shapes.

$$A_q(z) = \sum_{n=1}^3 z^n \sum_{\ell=0}^{\min(n-1,1)} z^\ell \sum_{k=\max(q,1)}^{\min(4\ell+2,3)} z^k \cdot (1 - z^{\min(k,3)+1}) / (1 - z) \\ B_q(z) = \sum_{e=1}^3 z^e \sum_{f=0}^{\min(e-1,1)} z^f \cdot (1 - z^{\min(q,4f+2)+1}) / (1 - z)$$

Both verified against enumeration at every q : $A_q(1) = 33, 33, 23, 8$ and $B_q(1) = 5, 10, 15, 17$.

7.7.1 And they are not the same tree

$A_q : n - \ell - k$, with 2S pendant at k — a caterpillar $B_q : e - f - g$, capped by the base — a path

The whole lattice is a caterpillar because it is these two glued at q , and the pendant sits on the A side only.

7.7.2 A tower of fibrations, terminating in boxes

level	base	fibre
Λ	q	$A_q \times B_q$
A_q	k	(n, ℓ) -set $\times$ 2S-chain
B_q	f	e -set $\times$ g -chain

Three levels, and at the bottom every fibre is a product of chains — a box, with a closed form $\text{Box}(a, b)(z) = \prod_i z^{\{a_i\}} (1 - z^{\{b_i - a_i + 1\}}) / (1 - z)$.

7.8 Five things the fibre table says

The counts 165, 330, 345, 136 carry more than their sum.

7.8.1 A Pareto frontier inside the lattice

$|A_q|$ falls 33, 33, 23, 8 while $|B_q|$ rises 5, 10, 15, 17.

Raising the transfer costs the parent and pays the target, monotonically, and no q improves both. That is the same shape as §16.5's $dw/dh > 0$ with $dV/dh < 0$ — which the book presents as a fact about brackets and is here a fact about the index itself.

7.8.2 A most-probable transfer

The fibre peaks at $q = 2$ with 345 cells — 35.3% of Λ — and the sequence is log-concave, hence unimodal.

$\langle q \rangle = 1.4631$, sd 0.930 — the mean number of electrons moved across all admissible configurations, and $S'(1)/S(1)$ in the base variable.

7.8.3 An 8-to-3 compression that loses no count

Λ 's F has eight variables; the shape has three, and $S(1, 1, 1) = 976$. Not a projection that forgets cells — one that forgets which coordinate inside each side and keeps how many. A general fact about caterpillars: a path with one pendant reduces to (base, left, right) whatever its length.

7.8.4 Every cross-section is itself a closed, modular lattice

q	side	closed	$E(X)$	modular
0–3	A and B, all eight	true	0	true

The shape is not a cylinder over an arbitrary set. It is a cylinder whose every cross-section satisfies this book's own law.

7.8.5 Which gives $E(X)$ a local form

$E(\Lambda) = 0$ and $E(A_q) = E(B_q) = 0$ for every q .

The second does not follow from the first. A closed index could in principle have defective slices that cancel in the total. Here none do, and that is a new invariant.

7.9 Every other shape the lattice contains

Intervals. $[x, y]$ is a box iff no constraint binds across it:

$I_{[x, y]} = \text{Box}(x, y) \iff \text{for every constraint } i \leq f(j), y_i \leq f(x_j)$

Exact on 60 of 60 tested, and only **27%** of random intervals qualify — **the other 73% are where the physics is active**. The binding rates rank the constraints:

constraint	binds
$g \leq q$	35.6%
$q \leq k$	33.0%
$g \leq 4f+2$	4.9%

The most active constraint is the coupling; the least is the Pauli bound — which is why §11.8 finds it the cheapest to violate. Two independent measures, one answer.

Rank levels. Eighteen antichains, the largest **122 at rank 11**: the biggest set of configurations no two of which can be reached from each other by adding electrons. **Λ is exactly eight times its widest level.**

Maximal chains. A distributive lattice is $J(P)$; **its maximal chains are the linear extensions of P** . With seventeen join-irreducibles and a chain length of seventeen:

$$e(P) = 1,113,045,672$$

The seventeen generators of §3.3 and the chain length of seventeen are the same seventeen, and the book reports both without connecting them.

And one shape absent. No Möbius band, no cycle, no non-planar minor — **the constraint graph is a tree, so there is nothing to twist and nothing to close**. §7.1's finding follows by construction, not by search.

7.10 An axis the index did not carry, and now does

A_q carries 2S, a spin label on the parent. B_q carried none.

The index recorded the parent's spin and not the target's.

That was never a closure defect — $\mathcal{R}$ cannot see a coordinate that was never supplied. **It was an axis E_2 admits**, and the asymmetry of the two fibres is what made it visible.

7.10.1 Λ_9 , built by the same construction

The parent's spin is bounded by the parent's capacity coordinate, $2S \leq k$. So the target's is bounded by the target's: $2S' \leq g$. One pendant, on the other fibre, by the identical rule.

	Λ_8	Λ_9
cells	976	1,654
box	6,912	27,648
$E(X)$	0	0
sublattice	yes	yes
rank-modular	475,800 pairs	200,000 tested, 0 violations
constraint graph	a tree	a tree — $2S'$ attaches as a leaf
$F(1)$	976	1,654
$F(-1)$	2	2
rank levels	18, widest 122 at 11	21, widest 185 at 12
log-concave	yes	yes

And Λ_8 is the exact projection of Λ_9 — 976 cells, identical — with fibres of one to four preimages, mean 1.69. §24.2.4's dimension ladder gains a rung, and the extra coordinate carries real information.

7.10.2 What is still not established

The bound $2S' \leq g$ was built by analogy and not from spectroscopy. Whether it is the correct selection rule for a target's spin is a physical question, and this book has not answered it.

The lattice closes. That the coordinate means what its name says is untested, and by B.2.16 the empty cell is recorded rather than filled by the analogy that generated it.

8. What the title names

Twice the title has changed, and both changes record findings rather than preferences.

The book was called *The Lach Index Lattice* until its shape was computed. The shape is a cylinder over the transfer, so the object's name became **The Lach Cylinder** — and §8.2 gives the reason.

Then the register outgrew the object. By the end there were one hundred and eighty-nine withdrawals, six prime audits, twenty-three principles and eighteen protocols — **and the demonstration on Λ had become the smaller half of the book.**

The title is now *The Method*, and *The Lach Cylinder* is its subtitle.

8.1 Why that ordering and not the other

Because the method is what generalises and the object is what tested it. Λ closes, and so does the nucleon index, and the string partition function, and the Kreuzer-Skarke frontier, and Appendix E's index of this book's own mathematics. **What travels between them is not the cylinder. It is: build the index, close it, read $E(X)$, record every failure.**

And the book applies that method to itself. Chapter 24 audits the book as an index; §21.8 indexes the process; Appendix B indexes the rules; **Appendix E indexes the mathematics and closes it at $E = 0$.** A book whose method is applied to its own contents four times over **is a method with a demonstration, not a demonstration with a method.**

The method is the book, and the book is the method. The cylinder is the case that proved it could be done and the case that showed what it costs.

8.2 The cells are transitions, not states

Λ is fibred over q , the number of electrons transferred. Not over any property of a single configuration — over the *movement between two*.

One end is a parent (n, ℓ, k) with its spin $2S$. The other is a target (e, f) with its occupancy g . **They meet at q .**

Chapter 2 introduces both ends and never says the pair is the object. The shape says it: $|\Lambda| = \sum_q |A_q| \cdot |B_q|$, and there is no factorisation that does not pass through the transfer.

8.3 Which is why "index of transitions"

An index of states would fibre over a state variable. This one fibres over a difference, and every structural result follows from that:

finding	where it comes from
the fibres are two-body products $\backslash$	the two ends of the tree $A_q \backslash$
	falls as $\backslash$ B ris es the transfer is conserved between them $-q \backslash$
$g \leq q$ is the most active constraint	it is the coupling itself
rank is modular	the transfer redistributes without loss

8.4 And what the cylinder does not carry

The lattice supplies kinematics and no dynamics. What configurations exist, which transitions are admissible, what is conserved — all structural. **Which transition actually occurs is not.**

1,113,045,672 maximal chains satisfy every constraint in this book. The Aufbau ordering is one of them. Structure does not choose; energy does, and energy enters only through Chapter 17's measured levels.

That is the sharpest statement of what the index can and cannot do, and it is a statement about its shape: **a phase space with the Hamiltonian removed can bound and refuse, and cannot predict.**

9. Closure

Everything in this part follows from one condition.

An index $X \subseteq \prod_i A_i$ is CLOSED if, for all $x, y \in X$, both $x \vee y$ and $x \wedge y$ lie in X , where join and meet are taken coordinatewise.

The coordinates must be **totally ordered** — chains. That restriction is real and is discussed in §11.3.

9.1 The characterisation

Theorem 7.1 (A.4). X is closed **iff** $X = \mathcal{R}(X)$, where $\mathcal{R}$ reconstructs value sets and pairwise monotone bounds from X 's own extension.

Proof. $(\Leftarrow)$ $\mathcal{R}(X)$ is an intersection of sets $\{x : x_i \leq \phi(x_i)\}$ with ϕ non-decreasing, each closed under $\wedge$ and $\vee$; so $\mathcal{R}(X)$ is closed, and if it equals X then X is closed. $(\Rightarrow)$ $X \subseteq \mathcal{R}(X)$ always, since each $x \in X$ witnesses its own bounds. For the reverse, each bound $y_i \leq \phi_{ij}(y_j)$ supplies a witness $w \in X$ with $w_i \geq y_i$ and $w_j \leq y_j$; closure under $\wedge$ and $\vee$ of the witness family produces y itself. ■

Verified on 200 random constructions, 200 agreements — random subsets (0 closed, 0 fixed points), lattice closures (40 and 40), random down-sets (16 and 16). Where they disagree they disagree together; there is no case of one without the other.

Note the down-set row. Sum-bounded sets like $\{a + b \leq q\}$ are *not* closed, and $\mathcal{R}$ reports it correctly. §25.3 explains why sums are excluded.

9.2 $\mathcal{R}$ is a closure operator

Extensive, monotone, idempotent — **150 of 150 on all three axioms.**

So **$E(X) = |\mathcal{R}(X)| - |X|$ is a closure defect**, a standard object in Galois theory, formal concept analysis and database dependency theory.

This book claims no novelty for the machinery. What is offered is the application: computing the closure defect of an *index*, and finding that the periodic table's is 36 while Λ 's is 0.

9.3 The scope, stated exactly

Theorem 7.1 requires each coordinate to be **presented as a chain**. That is not the same as requiring the factors to be chains: any distributive factor can be so presented, by Birkhoff, as the down-sets of its join-irreducibles — one chain per irreducible.

What fails is bundling. $\mathcal{R}$ reconstructs coordinate by coordinate, so folding two chains into one non-chain coordinate hides structure from it. In tests, the same sets presented bundled and presented decomposed both gave 80/80 agreement — the theorem survives, but only when the presentation respects it.

9.4 The three consequences

Closure yields three properties, developed in the next three chapters:

S — reference	S1 alphabet	S2 order	S3 bounds
D — defence	D1 $\mathbb{D}_{\text{phys}}$	D2 $\mathbb{D}_{\text{def}}$	D3 χ total
E — extension	E1 cells	E2 axes	E3 constraints

Nine mechanisms. Six follow from closure without further hypothesis. Two require something constructed — a tree, and disjoint derivations. One, E3, is an exact criterion that took three wrong statements to reach.

And E is derived: every extension question reduces to S and D. It carries no independent content, in the same way that the cost surface of Chapter 16 carries none beyond v .

10. Self-reference

A closed index contains its own definition.

To rebuild an index from nothing but a bag of tuples, three things are needed. They are independent, and only one of them is hard.

10.1 S1 — the alphabet

$$\hat{A}_i(X) = \{ x_i : x \in X \}$$

A projection. **Always succeeds, no conditions.**

10.2 S3 — the bounds

$$\hat{\phi}_{ij}(v) = \max \{ x_i : x \in X, x_j \leq v \}$$

Read off the cells. For Λ_8 that is 56 functions, all recovered, and $\mathcal{R}(\Lambda) = \Lambda$ exactly. **Always succeeds given the order.**

10.3 S2 — the order, which is the hard one

Given the alphabet but not the ordering of its values, can the ordering be recovered?

The necessary condition is provable and dimension-free. Say u may precede v on axis i if, for every p in the fibre over u and every q in the fibre over v , the meet lies in $F(u)$ and the join in $F(v)$.

Lemma. If X is closed and $u < v$ on axis i , then u may precede v . *Proof.* Take x with $x_i = u$ and y with $y_i = v$. Then $x \wedge y$ has i -coordinate u , so its other coordinates lie in $F(u)$; and $x \vee y$ has i -coordinate v , so its other coordinates lie in $F(v)$. ■ Nothing in the argument uses $d = 2$.

At $d = 2$ the converse also holds and the criterion is exact: X is closed under some order iff "may precede" is a total order, and that order **is** the relabelling. Verified on 120 sets; 274 of 274 constructed total orders give closed sets.

Above $d = 2$ the converse fails. Being a total order on every axis separately is necessary and not sufficient — the axes' requirements can be mutually incompatible. That is developed in §13.4, because it is a limitation rather than a mechanism.

10.4 What does work above $d = 2$: propagation along the tree

If the constraint graph is a tree, each axis is constrained only by its parent. Root anywhere, fix the root by brute force, and extend with backtracking.

Λ 's orders are recovered from a scrambled bag of cells: 20 of 20, at two cap settings, on 216 and 976 cells with all eight axes independently permuted.

Cost $\sum_i |A_i|!$, never the product $\prod_i |A_i|!$.

Two earlier methods failed on the same task, and the failures are instructive. Alternating refinement across all eight axes at once: **0 of 12**. Greedy propagation without backtracking: 42% and 25%. **The tree says only neighbours interact; a method that ignores that recovers nothing.**

Figure 8.1 The index recovers its own order

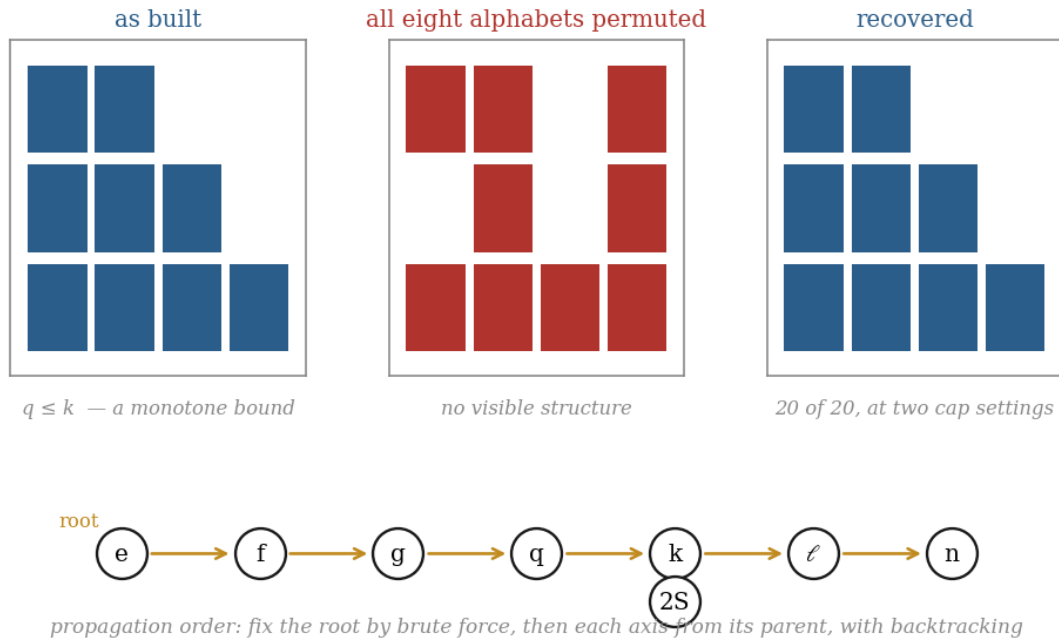


Figure 10.1. Left: two of Λ 's eight coordinates as built, showing the monotone bound $q \leq k$. Centre: the same cells after all eight alphabets are independently permuted — the structure is invisible. Right: recovered. Beneath, the propagation order along the constraint tree. The recovery uses no external information: only the bag of cells.

10.5 So: Λ contains its own order

The unqualified claim — *the index contains its own definition* — holds for Λ because its constraint graph is a tree. **For an index whose graph has cycles, S2 is open**, and Chapter 22 states what is known.

The qualified claim is unconditional: $\mathcal{R}(X) = X$ recovers alphabet and bounds for any closed index, relative to a given order.

11. Self-defence

A closed index contains its own contradiction.

11.1 The counting argument

Let p be the parameters an index assigns and $q = \Phi(p)$ the quantities derived from them.

$$\mathbb{D} = \dim q - \text{rank } \partial\Phi/\partial p \geq \dim q - \dim p$$

Proof. The Jacobian is $\dim q \times \dim p$; its rank exceeds neither. ■

Verified on seven dimension pairs and 420 random nonlinear maps. No closure required, no smoothness beyond differentiability.

Corollary. $\dim q > \dim p \Rightarrow \mathbb{D} \geq 1$.

$\mathbb{D}$ counts functionally independent relations the derived quantities must satisfy. **Each is a test the index must pass.** For Λ with $p = (T, n, Z, \sigma)$ and $q = (v, \delta, V, r, w, e)$, rank is 4 and $\mathbb{D} = 2$, spanned by

$$(a) V = 4v/3 \quad (b) w = V \cdot e$$

11.2 The split, and it is canonical

(a) holds under the Rydberg law and fails under power-4, exponential and logarithmic laws. (b) holds under all four.

$$\mathbb{D} = \mathbb{D}_{\text{phys}} + \mathbb{D}_{\text{def}}, \text{ physical relations and definitional identities}$$

Tested under four unrelated perturbations, verdicts identical. The split is not an artefact of which law one chooses to vary.

11.3 D1 — $\mathbb{D}_{\text{phys}}$ catches a wrong value in the data

Worked case. Two species in this work, Al II and K II, were first computed with $Z_{\text{eff}} = 1$. Both have doubly charged cores.

The bracket held 56 of 56 either way. Containment depends only on monotonicity of T , and Z cancels from it entirely.

What failed were the quantities the method does not use. δ came out near 4.9 with a spread of several units — impossible for a Rydberg series, where δ is near-constant. V departed from $4v/3$ by a **median of 50%**, where Na I gives 0.2%.

Corrected to $Z_{\text{eff}} = 2$: $\delta = 1.271$ with spread 0.047 over ten members, and V to **0.86%** across the fifty cells.

The bracket verifies the data. V verifies the indexing. An index carrying only what its method consumes would have passed fifty-six wrong cells.

Figure 9.1 The wrong nuclear charge: containment is blind, δ is not
the bracket cannot see it

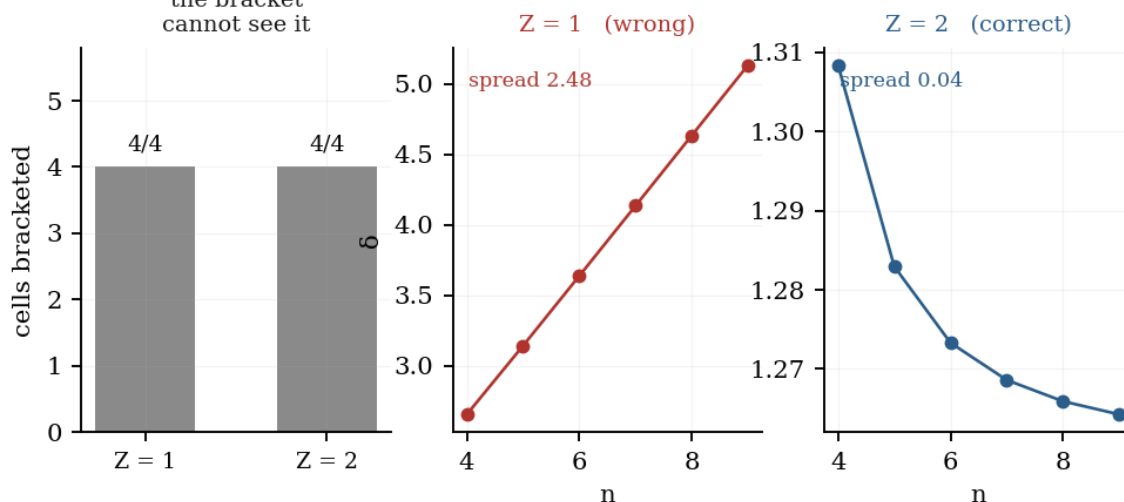


Figure 11.1. The bracket holds fifty-six of fifty-six under either charge, because Z cancels from containment. The defect does not: at $Z = 1$ it spreads over several units, which no Rydberg series can do. The method was blind; a quantity the method never uses was not.

11.4 D2 — $\mathbb{D}_{\text{def}}$ catches a wrong derivation, but only under two routes

An identity is vacuous if its terms are computed along the same path. $w = V \cdot e$ cannot fail if V is defined as w/e . It becomes diagnostic when w comes from the levels and V from the law.

$\mathbb{D}_{\text{def}}$ is realised iff a quantity is reachable by two disjoint paths, and the defence is the DISAGREEMENT between them — not the relation.

Worked case, and it is the author's own error. In first stating relation (b) I wrote $w = 3v \cdot e$. The two values — 540.0 from the levels, 1215.0 from the mis-stated law — appeared on the same printed line.

The check caught the check. $\mathbb{D} \geq 1$ does not require the auditor to be right.

Worked case, in an apparatus. Antiprotonic helium, cell (35,33):

route	value
measured directly	804,633,059.0 $\pm$ 8.2 MHz
two-photon minus a different single-photon	804,633,057.8 $\pm$ 10.6 MHz

Agreement at 0.09σ , with no shared measurement. One laser resonance against a two-photon resonance minus an unrelated one. That is path-disjointness realised in hardware.

And a constraint on where $\mathbb{D}_{\text{def}}$ can come from. Λ 's constraint graph is a tree, so there is exactly **one** path between any two coordinates. The constraints supply **no** redundancy. $\mathbb{D}_{\text{def}}$ must be built from derived quantities, deliberately.

11.4.1 And the counting argument has a 1960 precedent

This book did not discover that a fit with as many parameters as observations cannot be checked. Edlén states it, and uses it.

"Because of the approximate nature of the theoretical formulae here discussed nothing would be gained by trying to derive the parameters by a least-squares solution from all the observed levels. ... As the number of parameters introduced is equal to the number of levels no check is possible, and, of course, the results cannot be displayed in a two-dimensional diagram." — B. Edlén, Handbuch der Physik XXVII, footnote to §17 (manuscript 1960)

That is $\mathbb{D} = 0$, in the special case $\dim q = \dim p$. The counting argument, the conclusion drawn from it, and the use made of it — declining a fit that would have looked more rigorous — are all identical to this chapter's.

What §11.1 adds is the general form, $\mathbb{D} \geq \dim q - \text{rank } \partial\Phi/\partial p$, and the observation that the rank rather than the dimension is what binds. The principle is Edlén's.

11.5 D3 — totality catches a missing value

Theorem. If Λ is closed, $\chi_{\Lambda} : \prod A_i \rightarrow \{0,1\}$ is **total**: membership is decided for every ambient point by a finite conjunction of decidable comparisons. *Proof.* Each bound compares two elements of a chain. Λ is a finite intersection of such conditions. ■

Verified on 30,000 uniformly sampled ambient points: values $\{0,1\}$, nothing undecided.

Worked case. A species was misidentified — Na II read as neutral neon, and the wrong ionisation limit applied. Every channel had $T = I - E \leq 0$ and **every one was refused at entry.**

Totality does not detect that error. It forbids it. The computation never ran.

Worked case, the other way. A guard in the author's code returned None when an alphabet exceeded a cap, and the caller coerced None to False. That produced "3 reorderable of 85 trials," which produced "almost never reorderable," which produced a false dismissal of an entire line of evidence.

One silent discard propagated through three conclusions, none of which looked wrong, because the arithmetic was consistent throughout.

$\mathbb{D} \geq 1$ cannot catch that. Every relation checks out on data that was never measured. Only totality closes the hole, because *undetermined* is not in the codomain.

11.5.1 And totality extends beyond the data

χ is usually read as a test on cells. It is a test on whatever the index carries, and §21.8 applies it to a coordinate this book did not originally have: the process by which a choice was made.

Index (step, visible, alternatives, committed) under **visible $\leq$ committed** — that nothing of an answer may be visible before a choice is written down — and χ **refuses** a prediction developed with its answer in view. **36 cells, closed, $E = 0$** , one constraint of §11.4 form.

The mechanism is unchanged. Only the coordinate is new.

That is the general shape of D3, and it is why the protocol of B.2.9 — *refuse rather than coerce* — is stated for quantities and applies equally to decisions.

11.6 The three defences are disjoint

mechanism	catches	worked case
$\mathcal{R}(\Lambda) = \Lambda$	rules not recoverable from cells	periodic table, $E = 36$
$\mathcal{D} \geq 1$	a wrong value in data or derivation	wrong Z ; $w = 3v \cdot e$
$\chi \wedge$ total	a missing value coerced to a real one	the cap discard

No mechanism catches another's class, and this work supplied an instance of each.

11.7 Two kinds of absence, and one kind of lie

An index can be wrong in two ways, and they are not alike. One it can detect by itself. The other it cannot detect at all.

11.7.1 A missing cell — and the index can tell whether the gap is real

$E(X) > 0$ says the structure admits cells the collection does not hold. It does not say why, and there are two reasons:

the gap is an artefact of the layout — a property of how the index was drawn **the gap is a fact about the world** — nothing occupies that cell because nothing can

And the index distinguishes them, by asking whether E is removable.

the periodic table	E
18-column	36
32-column	106
Janet left-step	0

The same 118 elements. Order the blocks f , d , p , s and define the period by $n + \ell$, and the defect vanishes. So the 36 belong to the drawing.

Now the same test on a catalogue where the absences are physical. Index the mean-motion resonances by (p, q) , reduced, with the order $p - q$ **derived rather than made an axis**, and give $\mathcal{R}$ only the resonances that show structure in the asteroid belt:

the belt	E
(p, q)	2
(order, p)	1
(p, order)	1

E is irreducible, and $\mathcal{R}$ refuses all six absences outright — 0 of 6 admitted, against 36 of 36 admitted for the periodic table.

And the physics confirms what the index inferred without being told it. Every resonance of order ≤ 7 in the belt is occupied; every one of order ≥ 8 is empty. **Thirteen of thirteen above the cut, zero**

of six below. Resonance strength enters at $e^{\|p-q\|}$, so the absences are exponentially weak and genuinely unoccupied.

A defect of the index vanishes under some re-coordinatisation. A feature of the world survives every one. The test is computable from the cells alone, and §1.2's Janet table is that test — run once, on one example, and never named as one.

11.7.2 A complete index of something that is not there

The second failure is different in kind, and no re-coordinatisation touches it.

Insert one impossible cell — **one electron with spin multiplicity 2, which no atom has** — and let the closure supply everything that follows. The result:

	Λ	the fabrication
cells	976	1,051
closed under join and meet	true	true
E(X)	0	0
fixed point of $\mathcal{R}$	true	true

And E stays 0 for both under every re-coordinatisation tested. The test of §11.7.1 has nothing to work on:

that test finds missing cells and asks whether the gap is real. A coherent fabrication has no missing cells. It is full.

The fabrication contains 75 configurations no atom can have, and the structure records nothing about it. Every law in this book certifies it.

11.7.3 The plain version

A defective index is a map with a street missing. You catch it by redrawing — the street reappears in another projection. **A coherent fabrication is a complete, consistent map of a city that does not exist.** Every street connects, every distance adds up. **Redrawing changes nothing, because nothing is missing.**

The only thing that distinguishes a map of a real city from a perfect map of an imaginary one is going there. In this book, going there means measuring a level.

11.7.4 Which is what D_{phys} is for, and only that

§11.3's five recorded failures are all of the second kind. Not one is a missing cell.

failure	what it does
a wrong nuclear charge	every derived value shifts together ; the index stays consistent
a wrong ionisation limit	every term value shifts together
a mislabelled parent	the whole channel moves as one

Each is a complete index of something that is not there, and each was caught by a number the index could not supply — a quantum defect of 4.9, a term value below zero.

Self-consistency detects an isolated error and certifies a systematic one. That is the whole case for requiring a second route that reaches outside, and it is narrower and sharper than "an index cannot check itself."

11.8 What self-reference does and does not defend

$\mathcal{R}(\Lambda) = \Lambda$ is not a curiosity. It is the guarantee that the index's answers and its contents are the same object, so it can never be short of an answer and never give one that contradicts itself.

11.8.1 A complete index answers every question about its contents

Asked whether it admits a fabricated cell — one electron with spin multiplicity 2, which no atom has — Λ answers from its own content: $\varphi(2S \setminus k) = 1$ at $k = 1$, the cell asks for 2, refused.

200 of 200 fabrications refused when Λ is asked.

11.8.2 What it cannot defend against is being replaced

Rebuild the index around the fabricated cell and the closure grows to 1,051, is a fixed point, closed under join and meet, has $E = 0$, satisfies every law in this book — and contains 75 physically impossible configurations.

200 of 200 admitted when the index is rebuilt around them.

The index never believed the lie. It was never consulted. Replacement is not a question put to an index; it is an act performed on it by whoever holds it.

operation	outcome
ask Λ	refused, 200/200
rebuild around y	admitted, 200/200

11.8.3 Which is what B.2.13 already says

Commit before you look. Fix the standard, then admit the data. E1, E2 and E3 say the same for cells, axes and constraints:

an extension must be checked *against* the index, not absorbed *into* it and re-derived.

Ask the index, then decide whether to extend it. Never extend first and ask afterwards.

11.8.4 And the cost of a fabrication is computable

Insertion is never free. Adding one cell forces the closure to admit a median of 340 more — **minimum 59 across 220 trials, zero exceptions.**

$A(y) = N[\varphi(\Lambda \cup \{y\})] - N[\varphi(\Lambda)] - 1$, the lattice's own count evaluated on two bound systems. Exact on all cells tested.

Substitution gains nothing. Deletions are always repaired, so $C(\Lambda - x + y) = C(\Lambda + y)$ identically — **0 of 300 random swaps and 0 of 1,600 targeted swaps evaded detection.**

attack	outcome
delete	repaired, always
insert one	amplifies, always — minimum 59
substitute	reduces to insertion; no gain
insert the whole amplified closure	undetectable

V prices certainty. A prices deceit. Both are the lattice counting itself, and both have a floor.

And the amplification is governed by the fibre the widened bound admits, truncated by any rival bound on the same variable — which is why $g \leq 4f+2$, the only constraint sharing its variable with a stricter one, is an order of magnitude cheaper to violate than any other. **Remove the rival and it jumps 397% into the normal band.**

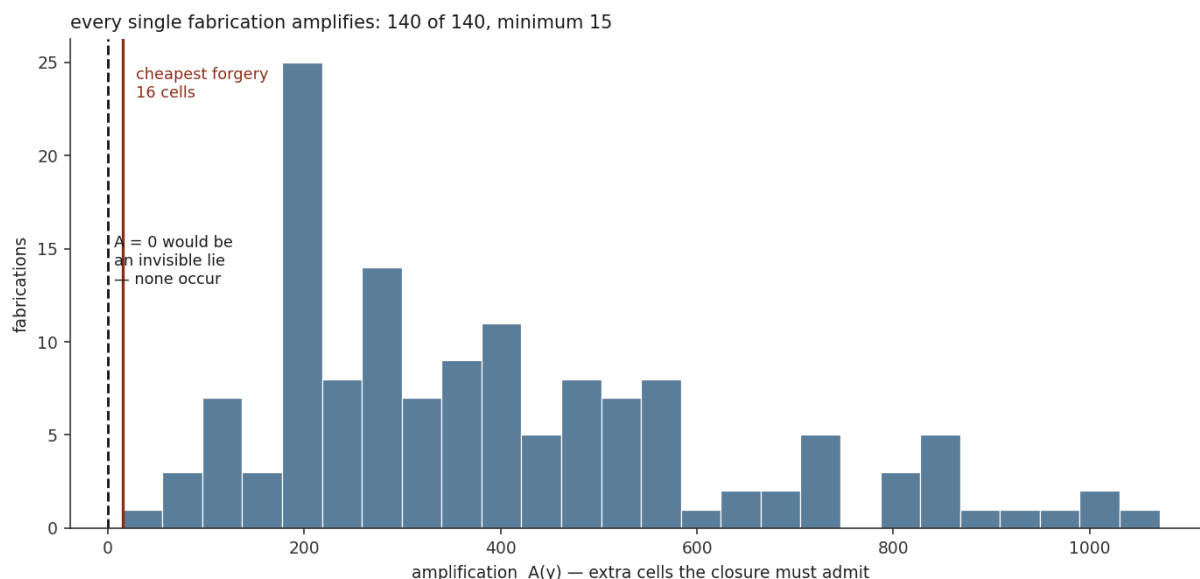


Figure 11.2. The cost of a single fabrication, over 140 random insertions. Every one forces the closure to admit further cells — median 352, minimum 15 — and none is free. The dashed line at zero marks where an invisible lie would sit; no insertion reaches it. A forger must therefore supply the whole amplified set, and its size is computable from the index before any data arrives.

And the general form is the tree, not the constraint. Four statistical measures of a constraint — leaf position, load, marginal contribution, activity — all failed to predict A . **A measure of the tree succeeds:**

$A \approx \text{fibre} \times (0.972 - 0.154 \cdot \text{descendants of the bounded variable})$ $\text{corr}(\text{descendants}, A/\text{fibre}) = -0.942$, median error 7%, worst 24%.

constraint	descendants	A/fibre
$2S \leq k, g \leq q, g \leq 4f+2$	0	0.96-0.99
$q \leq k, f \leq e-1$	1	0.66-0.95
$k \leq 4l+2$	3	0.48
$l \leq n-1$	4	0.39

A widened bound on a variable with descendants forces theirs to widen too, so the fibre undercounts in proportion to how much of the tree sits downstream. **Leaves are exact; the root is off by a factor of 2.5.**

11.8.5 Two closure operators, and the book names one

$\mathcal{R}$ **recovers 16 binding coordinate pairs, not 7.** Its transitive reduction returns the seven defining edges **exactly** — so the tree is derivable from the cells and nothing is lost.

But the two systems, identical on Λ , differ on $\Lambda \cup \{y\}$:

	A full	A tree-only
$l \leq n-1$	119	307
$k \leq 4l+2$	269	563
$2S \leq k$	74	74

The two that agree bound leaves — no descendants, no transitive propagation. $\mathcal{R}$ **is the full pairwise operator**, because for the calendar and the periodic table no constraint graph is given and the tree operator is not definable at all. **\$6's nested sum counts the tree system; they agree on Λ and nowhere else.**

12. Extension

A closed index determines what may be added to it.

Three operations, and only three: change the cells, change the axes, change the constraints.

12.1 E1 — which cells may be added

$\Lambda \cup \{x\}$ is closed **iff** for every $y \in \Lambda$, both xvy and $x\wedge y$ lie in $\Lambda \cup \{x\}$

100% over 288 tests. A cell may be added exactly when it sits in a hole whose joins and meets fall back into the lattice.

12.2 E2 — which axes may be adjoined

$\Lambda \times h$ is closed **iff** h is a lattice homomorphism

100% over 424 tests. Projections, maxima, minima and constants qualify; sums, products and differences do not.

And a theorem in the other direction:

Theorem 10.1 (adjunction never repairs). For any $h : S \rightarrow H$, with $S' = \{(x, h(x))\}$: **S' closed $\Rightarrow S$ closed.** *Proof.* For $x, y \in S$, the join of $(x, h(x))$ and $(y, h(y))$ is $(xvy, h(x) \vee h(y))$. If S' is closed this lies in S' , so $xvy \in S$. Meets likewise. ■ **Contrapositive: a non-closed set cannot be repaired by adjoining any function of its coordinates.** Tested on seven functions including the identity and a constant; none repairs.

Three repair routes remain, and only three: **enlarge** (add cells), **restrict** (remove cells), **reorder** (change neither). The calendar of Chapter 1 is repaired by reordering, at a cost of usability.

12.3 E3 — which constraints may be imposed

This one took three wrong statements to reach, which is recorded in Chapter 21. The criterion is:

$\{h \leq q\}$ is closed $\Leftrightarrow h(xvy) \leq q$ for all x, y with $h(x), h(y) \leq q$

100% over 2,513 tests, nine functions.

The essential feature is that it is q -dependent. Admissibility belongs to the *pair* (h, q) , not to h alone — which is why $a + b \leq 3$ can be closed where $a + b \leq 4$ is not, and why every q -free criterion attempted before this one failed.

Meet never breaks it. For monotone h with $h(x), h(y) \leq q$, $x\wedge y \leq x$ gives $h(x\wedge y) \leq q$ at once. **Only the join can break it**, and it does so when the join raises h above both arguments.

Sums and products fail because a join raises every coordinate at once, and anything accumulating across coordinates accumulates the raises.

Λ 's seven constraints are all safe by inspection: each reads one coordinate, or a minimum of two. **None sums.** That is why Λ is closed and the periodic table is not — and it is checkable without touching a cell.

12.4 A worked extension, and what it costs

The clearest demonstration of E3 in this work is an attempt to widen Λ that failed, and the repair that succeeded at a stated price.

The problem

Λ_8 carries one target triple (e, f, g) . Configurations with **two** occupied non-source subshells fall outside it — an 86% coverage ceiling. The natural repair is a second triple.

Ordering the targets is free

The canonicalisation $e_1 \leq e_2$ is admissible: ϕ is the identity, non-decreasing in a single coordinate. **Zero join and meet failures.**

Conservation is not

The physical requirement $g_1 + g_2 \leq q$ is a bound on a **sum**, hence outside §11.4 — and it breaks closure. **89,864 join failures in 979,300 sampled pairs**, meets unaffected.

The counterexample is two lines. $a = (g_1=1, g_2=0, q=1)$ and $b = (g_1=0, g_2=1, q=1)$ each conserve. **Their join transfers two electrons having removed one.**

Dropping conservation restores closure — 45,690 cells, zero failures — and produces a lattice describing configurations that cannot exist. **That is the choice in its bluntest form: a closed index of impossible states, or an open index of real ones.**

The repair: make the sum a coordinate

Conservation is a bound on a sum, which §2.1 excludes. So make the sum a coordinate. Replace (g_1, g_2) by (g_1, G) with $G = g_1 + g_2$:

$g_1 \leq G$ (non-decreasing in G) and $G \leq q$ (non-decreasing in q)

Both of §11.4 form, with $g_2 = G - g_1$ recovered as a derived quantity and **conservation holding identically by construction.**

Tested exhaustively: 1,040 cells, all 540,280 pairs, zero join and zero meet failures.

And the price

The map $(g_1, g_2) \mapsto (g_1, g_1+g_2)$ is a bijection onto an admissible lattice and order-preserving in one direction. **It is not an order isomorphism.** It makes comparable some pairs that were incomparable: $(0,1)$ and $(1,0)$ are incomparable as transfer counts and become ordered in the new coordinates. **Fifteen of forty ordered pairs are new.**

Coverage and closure are available together. Coverage, closure, and the original order are not.

This is P16 — treat bounds as coordinate values — operating as a repair procedure, and it is the general lesson of E3 stated constructively: a sum bound cannot be imposed, but a sum can be **indexed**, and the difference between those two moves is the order it costs you.

12.5 E is derived

mechanism	fixed by
E1 addable cells	S3, the recovered bounds
E2 adjoinable axes	D3, totality — h must create no third state
E3 imposable constraints	S2, the recovered order

Every extension question reduces to S and D. E carries no independent content — exactly as the cost surface of Chapter 16 carries none beyond v .

13. What the law forbids

The results of this chapter are the strongest in the book, and every one of them is negative.

13.1 An index carries what its coordinates carry, and no more

Theorem 11.1. On a product order, join, meet and comparability are definable from the coordinates: $(a \vee b)_i = \max(a_i, b_i)$, $(a \wedge b)_i = \min(a_i, b_i)$, and $a \leq b \iff \bigwedge_i (a_i \leq b_i)$.

Every lattice-theoretic quantity is therefore a deterministic function of the coordinates, and the σ -algebra it generates is contained in theirs.

13.1.1 And the product structure is not what does the work

A reader may suspect the result is an artefact of writing join as max. It is not.

Theorem 11.2. Let an order be defined as *any* predicate on coordinate tuples. Then the σ -algebra generated by its join, meet and comparability is contained in the σ -algebra generated by the coordinates. *Proof.* The relation $a \leq b$ is by hypothesis a function of the coordinates of a and b . The set of upper bounds of a pair is computed from that relation, and the least upper bound is selected from that set. Each step is a function of the coordinates, so the join is one — **whatever formula it takes.** Meets likewise. ■

Verified on a genuinely non-componentwise example from this work. The conserving poset of §13.1.1 has **28 of its 511 joins differing from componentwise maximum**, because q must rise to admit both transfers: $(1,0,1) \vee (1,1,0) = \mathbf{(2,1,1)}$, where componentwise maximum would give the inadmissible $(1,1,1)$.

The join is nonetheless a deterministic function of the two cells' coordinates, as the theorem requires.

No ordinal structure over these coordinates carries what they do not — not a tree, not a non-componentwise lattice, not the conserving poset.

The escape is not a cleverer order but a further measurement. An order not determined by the coordinates requires information beyond them, and that information is by definition **a new coordinate**.

13.1.2 Two consequences that run through the whole book

It is indifferent to dimension. Λ_{11} is as constrained as Λ_8 . Adding axes does not loosen the bound.

It is indifferent to faithfulness. Perfecting the index moves nothing, because there is nothing to add. Part I could be improved indefinitely and Part III would not gain a single cell.

13.1.3 What the theorem does not establish

Theorems 11.1 and 11.2 bound information, not estimation. A deterministic function of the inputs can still improve a finite-sample predictor — that is what feature engineering is, and it is routine.

So the theorems forbid the lattice from knowing anything the coordinates do not. They do not by themselves forbid a lattice quantity from being a better-conditioned regressor or a lower-variance summary. That is a narrower claim than the theorem's reach suggests, and it is stated narrowly.

13.2 ν is inadmissible as an axis

$\nu = e - \delta$ is a **difference**, and differences are non-monotone in the cell order. **86 violations** at the caps tested.

So the quantity on which the whole of Part III depends **cannot be a coordinate of Λ** . It lives over the lattice, not in it. Chapter 7's "over, not graded by" is the same fact stated geometrically.

13.3 Depth is not a coordinate function

The number of measured members in a channel is not a function of the coordinates at all. **It is a property of the literature**, and Chapter 14 makes that precise.

13.4 Closure is not locally determined

This is the deepest limitation, and it was found by trying to evade it.

Closure is **not** implied by every proper projection being a fixed point of $\mathcal{R}$.

Explicit counterexample at $d = 3$, $c = 2$, six cells:

$S = \{(0,0,0), (0,0,1), (0,1,0), (0,1,1), (1,0,1), (1,1,0)\}$ Every proper projection is a $\mathcal{R}$ -fixed point, yet $(1,0,1) \vee (1,1,0) = (1,1,1) \notin S$

And the counterexamples are common — roughly **one in five** sets passing all proper-projection tests fails closure.

The mechanism. The two cells agree on axis 0 and disagree on *both* others. The join disagrees in a **combination**, and no lower-dimensional projection can see a combination it does not contain.

Closure is a d -dimensional condition. No projection of it will serve as a criterion — not per-axis, not per-pair.

Consequence for verification: a wrong choice can pass every local test and fail only at the top. Chapter 22 states what this costs.

13.5 The pole

The cost of a guarantee, derived in Chapter 16, has a pole at $p = 1$ — the exponent at which an observable is linear.

A straight line has no curvature. There is no interpolation error to price, so a guarantee costs infinitely more than the thing it guarantees.

The method has no domain at $p = 1$. It is the only hard singularity in the structure; every other limit is a threshold with data on both sides.

13.6 $E(X)$ is the prediction budget

The five limitations above add up to something the book can now state in one line, and it answers the question this work began with.

A prediction is a claim that a cell exists which the index does not carry. The only cells an index can propose are those its own structure **admits** and its extension **denies** — and that set is exactly $\mathcal{R}(X) \setminus X$.

The number of predictions an index can make is $E(X)$.

13.6.1 What that means for the four indices of Chapter 1

index	$E(X)$	predictions available	and they are
the periodic table	36	36	all impossible
the calendar	7	7	all impossible
a subnet with a hole	8	8	all impossible
the box ordering, and Λ	0	0	—

Every prediction the three non-closed indices can make is wrong. Period-1 group-2 and its thirty-five kin. The thirty-first of February and its six. **That is the entire budget, and none of it exists.**

Closure does not make prediction accurate. It abolishes prediction.

13.6.2 And it is forced, not observed

Two results already in this part combine to give it.

$E(X) = 0$ says there is no missing cell to propose. **Theorem 11.2** says there is no hidden quantity to derive.

Together:

A fully defined index has nothing left to say.

That is the whole answer to whether Λ is predictive. Not that it is a weak predictor. Not that it wants more coordinates. **There is no gap in it for a prediction to occupy.**

13.6.3 Where prediction actually lives

Λ indexes cells. It does not index values.

The Sc VI 6s cell of §18.6 is in Λ — its coordinates are ordinary integers, and the lattice admits it without comment. **Its energy is not**, and no closure property supplies it.

the cell	indexed, $E = 0$, nothing to predict
the value	not indexed, and therefore predictable

δ is precisely the value's departure from what the index determines.

Prediction requires an incompleteness, and a complete index has none.

Which is why the bracket must be handed two measured neighbours before it can say anything at all. It is not extracting a prediction from the lattice. **It is carrying a constraint between two measurements the lattice never had.**

13.6.4 The trade, stated plainly

An index may be complete or it may be predictive. It cannot be both.

Not because completeness is dull, but because a prediction is a proposal about an unlisted cell, and completeness is the property of having no unlisted cells. The periodic table can propose thirty-six things because it is incomplete. **Λ can propose nothing because it is not.**

This is the fifth and largest of the negatives, and everything else in this part is a special case of it.

13.7 Why the negatives are the contribution

Λ is a bijective recoding of existing quantum numbers. It adds no physics. A reader will observe this within a page, and they will be right.

That is the point. A recoding that adds nothing and still exhibits closure, self-reference, self-defence and these five limitations is demonstrating a property of **indexing**, not of atoms. If Λ contained new physics the results would be about atoms and worth much less.

14. Retrieval as a lattice problem

Data can be retrieved in pieces; the index navigates.

This is usually read as advice. It is the self-defence theorem again, with a different failure mode.

14.1 The formalisation

Let $\mathbf{C}$ be the cells wanted and $\mathbf{S} = \{S_1 \dots S_k\}$ the sources, each a subset of $\mathbf{C}$, with $\mathbf{A}(S_i)$ recording accessibility.

c is **retrievable** iff $\exists i : c \in S_i \wedge \mathbf{A}(S_i)$ **route set** $\mathbf{R}(c) = \{ i : c \in S_i \}$, **retrieval redundancy** $\rho(c) = |\mathbf{R}(c)|$

$\rho = 1$ is **fragile**. $\rho \geq 2$ **survives the loss of any single source**.

$\mathbb{D}_{\text{def}} \geq 1$ — two disjoint derivations survive a computational error $\rho \geq 2$ — two disjoint sources survive an access failure

Same structure. Different failure mode.

14.2 A paywall blocks a source, not a cell

Worked case. The target was a transition frequency in antiprotonic helium.

source	status
<i>Nature</i> 475 , 484 (2011)	paywalled
<i>Phys. Rev. Lett.</i> 96 , 243401 (2006)	paywalled
arXiv:1304.4330	open — the <i>Nature</i> paper entire
arXiv:1203.5425 (CODATA 2010)	open — Table XII, all fifteen frequencies, both papers
arXiv:1308.1711, an unrelated citing paper	open — corroborates to 0.1 MHz
CERN CDS	open

$\rho = 6$. **Two blocked, four open.** The cell was retrieved and the two-path check on it closed at 0.09σ .

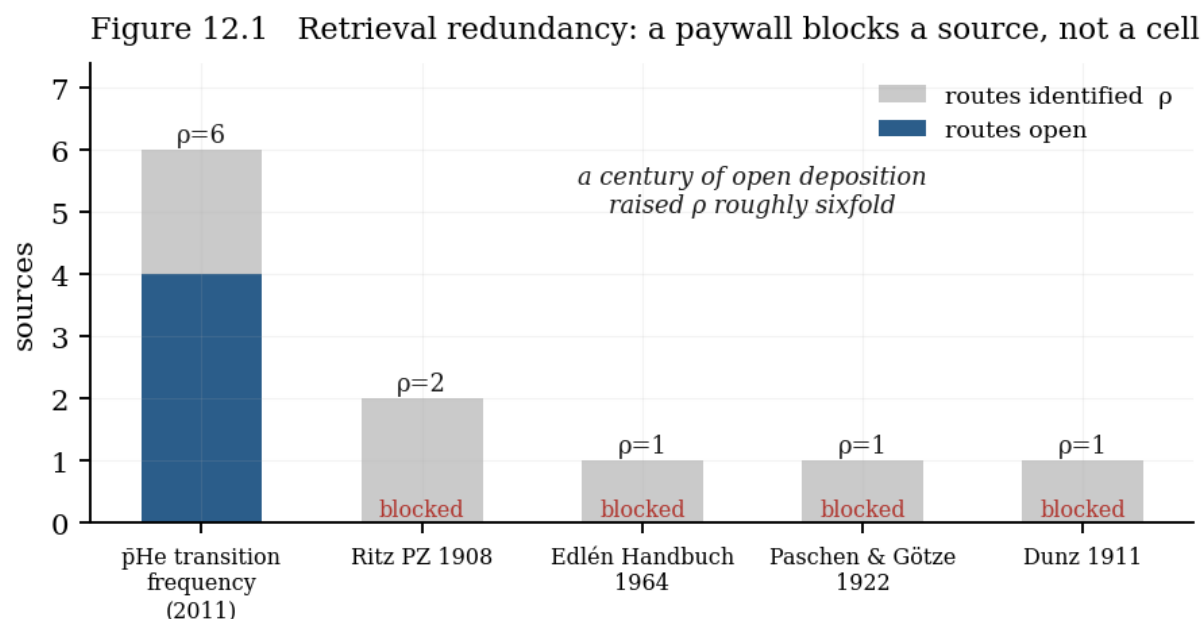


Figure 14.1. Retrieval redundancy for five target cells. The 2011 antiprotonic-helium frequency had six routes, two blocked. The four German documents have one or two apiece, all closed.

14.3 The secondary can be better than the primary

CODATA's Table XII carries both papers **with an extra digit the journals did not print**, supplied privately by the experimenters to reduce rounding error.

The index does not merely route around an obstruction. It sometimes finds a better cell than the one that was blocked.

14.4 The retrieval graph is self-referencing

Each fetched compilation names its successors. The aluminium compilation's introduction lists its own sister volumes verbatim; the beryllium compilation cites the zinc one; the aluminium reference list supplies nickel and copper.

Three coverage gaps in this work's census were filled from documents already in hand, with no new search.

14.5 Where it fails, and the number that describes it

Four documents in this work were not retrieved: Edlén's *Handbuch* chapter (1964), Ritz's *Physikalische Zeitschrift* paper (1908), Paschen & Götze (1922), Dunz (1911). **$p \leq 2$ for each, every route closed.**

Not for lack of effort — nine searches, correct citations, the right platform identified and free. **Pre-digital print has low retrieval redundancy.**

p is a property of the literature, not of the searcher. A cell in a 1908 German periodical has $p \approx 1$. A cell in a 2011 physics paper has $p \approx 6$.

A century of open deposition raised p roughly sixfold, and that is exactly what separated the two outcomes.

14.6 The procedure

1. Enumerate target cells from the index **before** searching.
2. List route candidates per cell — primary, preprint, review, compilation, citing paper, deposit, database.
3. Prefer high- p cells; they cost less and often carry more.
4. Read every retrieved source for its successors.
5. When a route is blocked, move along the route set; do not re-attempt.
6. **Record p for cells that fail.** A blocked $p = 1$ cell is a *stated* gap, not an unexplained absence.

The author violated step 1 badly, on his own problem, after writing this chapter. That is recorded in Chapter 21.

PART III — THE METHOD

15. The bracket

Everything so far has been about structure. This part asks what such a structure can do with measurements, and the answer is deliberately modest.

15.1 The rule

For a Rydberg channel — fixed parent, fixed l , varying n — the binding energies $T = I - E$ are monotone in n . So for any interior member:

$T(n)$ lies between $T(n-1)$ and $T(n+1)$

That is the bracket. It uses **only measured neighbours**. It fits nothing, assumes no functional form, and returns an interval rather than a value.

It is a deduction, not a prediction. The distinction is the whole method: a fit says where a level probably is; the bracket says where it cannot fail to be.

15.2 The four rules

The bracket is stated in one line and applied under four.

Rule 1 — Interiority. Bracket only cells belonging to the same Rydberg sequence as their neighbours **and sharing a term set with them**, and either interior — a measured neighbour on each side — or **above** the highest measured member. **Extrapolation below the lowest measured member is forbidden.** **Rule 2 — Variable.** Read the interpolation variable off $T = Z^2R/v^2$ for the axis in question: E along n within a series; interval ratios along J ; geometric in T along the core shell. **Rule 3 — Separation.** Report the deductive bracket separately from the inferential point estimate. Given monotonicity, the true value cannot lie outside [max below, min above]. **Rule 4 — Uncertainty.** Fit δ locally by a Ritz expansion and take $\sigma = 2R Z_{\text{eff}}^2 \cdot \text{SE}_{\text{pred}} / v^3$ from the fit's prediction standard error. Use $\delta = \delta_0 + \delta_2/n^2$ where three or more neighbours exist, $\delta = \delta_0$ where only two do — **subject to the admissibility clause below.** **Rule 4a — Admissibility of the constant form.** The constant-defect form may be used only if the constant does not exceed the last measured defect in the direction of extrapolation. **On a decreasing channel it must satisfy $\delta \leq \delta(\text{last measured})$; on an increasing one, $\delta \geq \delta(\text{last measured})$.** Where it does not, the two-point Ritz solve is the admissible form.

15.2.1 Each rule justified by ablation

A rule that costs nothing to break is not a rule. Each was tested by violating it:

rule violated	cost
extrapolate below the series	median error $0.46 \rightarrow 206 \text{ cm}^{-1}$ — 446×
extrapolate above the series	median error $0.46 \rightarrow 0.13 \text{ cm}^{-1}$ — 0.28×
wrong interpolation variable	rms $200 \rightarrow 588 \text{ cm}^{-1}$ — 2.9×
pooled σ instead of per-cell	$\pm 1\sigma$ coverage 96.2% instead of 69.6%
include a collapsed orbital	median error $12.8 \rightarrow 41.3 \text{ cm}^{-1}$ — 3.2×

The extrapolation cost is directional, and a single figure conceals it. An earlier version of this work quoted $3.7\times$ on pooled rms over twelve extrapolated cells. **That statistic is not reproducible and is unstable:** on the same data it is $18.8\times$ on rms, $4.6\times$ on median, and **$0.5\times$ if the two worst cells are removed.** An rms over twelve points containing a $12,942 \text{ cm}^{-1}$ outlier is not a summary that should carry a rule.

Split by direction the instability disappears. Extrapolating *below* a series costs $446\times$ on the median, because δ varies fastest at low n and a Ritz fit run backwards diverges. Extrapolating *above* costs $0.28\times$, because δ has already flattened. **The asymmetry is a factor of 1,577**, and Rule 1 is stated accordingly.

15.2.2 Rule 1 is not about n

The rule is stated for a series in n , and that is where it was found. It is not a rule about n .

The isoelectronic axis — fix the electron count, vary the nuclear charge — carries the same defect and shows the same asymmetry:

axis	toward the loose end	interpolate	toward the tight end
n , within a series	446×	$1\times$	0.28×
Z , along a sequence	17.3%	1.3%	2.1%

Same sign, same magnitude of asymmetry, same cause. δ varies fastest where the electron is loosely bound — low n , low Z — and flattens where it is tightly bound. **Extrapolating toward the flattening is cheap; away from it is not.**

Rule 1 is a rule about direction, and it holds on whichever axis carries the defect.

Measured on two sequences, holding out each member in turn. The lithium-like neutral is predicted from its three ions at 11.1% error; the sodium-like neutral at 23.4%. **The tight ends — C IV and Si IV — come back at 1.8% and 2.5%.**

15.2.3 And one step beyond the fitted range costs more again

A hold-out is bracketed by the points that remain. A genuine prediction is not. Tested on the sodium-like sequence with P V — a fifth member — held out entirely:

fit form	predicted $\delta(\text{P V})$	true	error
$\delta = a + b/Z$	0.8189	0.7129	14.9%
$\delta = a + b/Z + c/Z^2$	0.7376	0.7129	3.47%
$\delta \cdot Z$ linear in Z	0.7835	0.7129	9.9%

Only the quadratic form is usable, and it costs 3.5% rather than the 2.1% a hold-out suggests — a factor of 1.7 for stepping outside the fitted range.

This matters for §18.6 The Sc VI prediction has a second, independent route through its isoelectronic sequence, and **3.5% is the figure that route must carry** — not 2.1%, and not the 1.3% of pure interpolation.

15.2.4 Rule 1's second clause is not redundant

Sc III 3d is interior in n within $\ell = 2$, and including it inflates the median error by **3.2×** — because it is the **collapsed ground state** rather than a sequence member.

The operational test is the defect. Its δ departs from its neighbours' by 0.14 where they depart from each other by 0.008 — **a factor of 17**. A cell can be interior by index and outside the sequence in fact, and the defect says which.

15.2.5 Rule 4 at the edge of a channel

The two-parameter Ritz form needs three neighbours to leave a residual, and a channel with exactly three members has an interior cell with only two.

Why Rule 4a exists

The constant-defect form takes the mean of the available defects. On a monotone channel that mean lies *between* them — and therefore on the wrong side of the last one.

Sc VI's $4S^\circ$ ns channel is the case that forced the clause. With $\delta(4s) = 1.0057$ and $\delta(5s) = 0.9812$, the constant form gives $\delta = 0.9934$ — **above $\delta(5s)$** , which a decreasing channel forbids. The resulting estimate falls **outside the deductive bracket its own monotonicity implies** (§18.6).

A rule that produces an estimate its own bracket excludes is not a rule about uncertainty. It is an error.

With two neighbours the Ritz form is exactly determined — two equations, two unknowns, no residual and therefore no fitted σ — but it is *admissible*, and Rule 3's separation means the bracket carries the guarantee while the point estimate carries none.

One further case in this work: the Kr I ($2P^\circ_{1/2}$) np series — 5p, 6p, 7p, with 6p interior. Under the constant-defect form the cell returns an estimate in error by -70.9 cm^{-1} against $\sigma = 602$, or **0.12 σ** .

It is a valid cell and an almost worthless one. Its bracket spans $14,602 \text{ cm}^{-1}$ because its neighbours are two principal quantum numbers away.

That is the validity/value distinction, and it is the clearest instance of it in the collection: a cell may satisfy every rule and still tell you almost nothing. **The rules govern correctness. They do not govern usefulness**, and a method that conflated the two would report this cell as a success.

15.3 The bracket is limit-free

Since $T = I - E$ for a **constant** I :

$T(n)$ between $T(n \pm 1) \iff E(n)$ between $E(n \pm 1)$

Containment is invariant under the affine map. **ν , δ and V all need the ionisation limit; containment does not.**

Consequence: the method applies wherever levels are known, even when the threshold is not. This is not a technicality — it is what allowed the antiprotonic-helium cells of §18.6, where the relevant threshold is ambiguous.

15.4 The price of limit-freedom, computed

§25.3 presents limit-freedom as pure gain. It is a trade, and the terms are computable.

The alternative is to bracket δ rather than T — which is what the field does. And V is identical either way:

$\delta = \delta_0 + \delta_2/n^2$ has $L = \lfloor \delta'/\delta'' \rfloor = n/3$, the same as T 's, and δ_0 is an additive constant that cancels from both w and e . $V(\delta) = V(T) = 4n/(3h)$

The single most standard move in Rydberg spectroscopy — interpolate δ , not T — improved estimates enormously and left the price of a guarantee exactly where it was. A century of practice raised numerator and denominator by the same factor.

15.4.1 What δ buys, and what it costs

In δ the bracket is narrower by exactly $n^3/(26z)$ — measured against predicted: 4,329 vs 4,267 at $n = 8$; 1.387×10^6 vs 1.386×10^6 at $n = 55$. Agreement to four figures.

But that factor is the size of a Ritz coefficient, not a property of bracketing — δ barely moves because δ_2 is small.

And a bracket on δ does not bound T . Converting requires the ionisation limit and inherits its uncertainty:

n	w_T direct	via δ	+ limit $\sigma = 0.01$
20	68.06	0.00102	0.0210
40	7.613	1.43×10^{-5}	0.0200

At $n = 40$ the δ -route's own width is three orders below the limit's uncertainty. The limit dominates entirely.

Bracket T : bounds the measured level, needs nothing. Bracket δ : bounds a derived quantity, needs the limit, and the energy bound is never better than the limit is known.

The choice is conditional, and the condition is computable per channel. This book's tightest bracket is 1.398 cm^{-1} ; limit uncertainties in the collection run 0.001 to 0.4 cm^{-1} . **The δ route wins where the limit is known better than the bracket — true for some species, false for others.**

15.5 Admissibility

A channel is admissible when its levels are separated by more than their uncertainty:

$$r = 2Z^2R / (\nu^3\sigma) \geq 5$$

r falls as ν^{-3} . Every channel eventually leaves the domain; the rule states where.

15.6 What the bracket costs

A guarantee is wider than an estimate. Chapter 16 prices it exactly, and the price is the only linearly rising quantity in the structure.

16. The cost surface

16.1 The ratio

Let w be the bracket width $|T(n+1) - T(n-1)|$ and e the interpolation error $|T(n) - \frac{1}{2}(T(n-1) + T(n+1))|$.

$$\mathbf{V} = \mathbf{w} / \mathbf{e} = 4\nu/3$$

for a Rydberg series. And in general, for $y = x^p$ sampled at step h :

$$\mathbf{V}(\mathbf{x}, \mathbf{p}) = 4\mathbf{x} / (\mathbf{h} \cdot |\mathbf{p} - 1|)$$

Verified on power laws from $p = -3$ to $+11$, agreement under 1%.

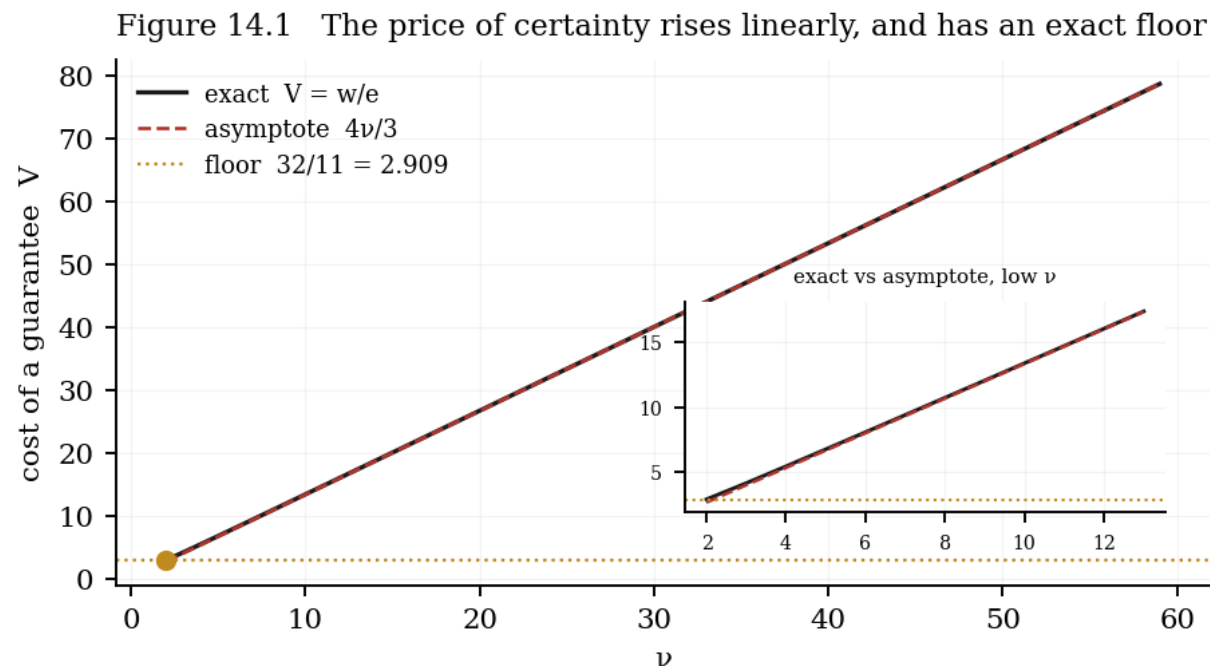


Figure 16.1. The exact cost against its asymptote. The floor $32/11 = 2.909$ belongs to the exact expression; the asymptotic form gives $26/9$ and is low by 0.69% at $\nu = 2$. They converge by $\nu = 10$.

16.2 The floor is algebraic before it is physical

The cost of a guarantee has a floor that owes nothing to atomic physics.

Proposition 14.1. Let three consecutive cells of a channel carry values $y_0 < y_1 < y_2$ at equally spaced index. Let $w = y_2 - y_0$ be the bracket width at the middle cell and $e = |y_1 - (y_0 + y_2)/2|$ the error of two-point linear interpolation from the same neighbours. Then $\mathbf{V} = \mathbf{w}/\mathbf{e} > 2$, with $V \rightarrow 2$ only as one of the two steps vanishes. *Proof.* Write $d_0 = y_1 - y_0$ and $d_1 = y_2 - y_1$, both positive by monotonicity. Then $w = d_0 + d_1$ and $e = |d_0 - d_1|/2$, so $\mathbf{V} = 2(\mathbf{d}_0 + \mathbf{d}_1)/|\mathbf{d}_0 - \mathbf{d}_1|$. Since $d_0, d_1 > 0$ we have $|d_0 - d_1| < d_0 + d_1$, giving $V > 2$. As $\min/\max \rightarrow 0$ the ratio $\rightarrow 2$; as $d_0 \rightarrow d_1$ it diverges. ■

This holds for any monotone sequence whatever. Not for some observables, not with better data:

The deductive interval is always more than twice the error of the crudest possible point estimate built from the same two numbers.

Every empirical V in this book is an instance of this bound, not an independent fact. And against a better estimator — the Ritz expansion of Rule 4, or a mass model — the gap is wider still, so V as defined **understates** the cost.

The Rydberg floor of $32/11 = 2.909$ is therefore the tighter of two. The algebraic floor at 2 constrains every monotone sequence; the physical floor at $32/11$ constrains Rydberg series specifically, and the difference between them is what the hydrogenic form contributes.

16.2.1 When the floor is approached, and what it means

Observation 14.2. $V \rightarrow 2$ requires $\min(d_0, d_1)/\max(d_0, d_1) \rightarrow 0$ — one step of the channel vanishing relative to its neighbour.

In every sequence examined here where that occurs, the small step is produced by two channels interleaving, which Rule 1's second clause excludes. So the algebraic floor is approached exactly where the method declines to operate: **the cheapest guarantee is available only on a channel that is not one.**

16.3 V is a bound, not a tool

V contains no σ . Improving measurement precision by six orders of magnitude leaves $V = 26.689$ unchanged at $\nu = 20$ — computed exactly at three precisions spanning $\sigma = 10^{-2}$ to 10^{-8} cm^{-1} .

No spectrometer can buy a narrower bracket. The price of certainty in a Rydberg series is set by the series.

So V is not an instrument. It is an impossibility statement, and it makes four:

	forbids
$V = 4\nu/3$	a guarantee costs $4\nu/3 \times$ an estimate, and no measurement reduces it
$V \geq \mathbf{32/11}$	no guarantee is ever cheaper than $2.909 \times$
pole at $p = 1$	no guarantee exists for a linear observable
ν_V ceiling	above it the cost cannot be <i>measured</i> , though it stays defined

The exact V is rational at every ν — $32/11, 54/13, 256/47, 250/37, 4000/299$ — and $4\nu/3 + 4/(9\nu)$ is its asymptotic form, low by 0.69% at $\nu = 2$ and exact to four decimals by $\nu = 10$. **The floor $32/11$ belongs to the exact expression, not to the asymptote.**

16.4 V is a function of ν/h alone

The cost has one free parameter, and it is not where the book first looked.

V is defined from three levels at $n - h, n, n + h$. Nothing forces $h = 1$: a cell may be bracketed from neighbours two or three principal quantum numbers away. Carrying h through the algebra gives an exact result.

$V = 4\nu^3 / (h(3\nu^2 - h^2))$, and with $r = \nu/h$, $V = 4r^3/(3r^2 - 1)$

One variable. Z and R cancel identically, not approximately, and $4r/3$ is the large- r expansion of this expression rather than the expression itself.

16.4.1 Which makes depth and step interchangeable

(ν, h)	ν/h	V exact
(20, 1)	20	26.688907
(40, 2)	20	26.688907
(60, 3)	20	26.688907
(80, 4)	20	26.688907
(10, 1)	10	13.377926
(40, 4)	10	13.377926

Identical to six figures, and identical across $Z = 1, 2$ and 6 . That last point is worth dwelling on: Z_{eff} is precisely what the wrong-charge error of §11.3 got wrong and the bracket could not see. **The similarity law is blind to it too — by cancellation, provably.**

16.4.2 Tested on the collection

1,033 (v/h , V) pairs from measured levels, at both steps:

	pairs	median deviation from $4(v/h)/3$
$h = 1$	560	0.49%
$h = 2$	473	1.15%
all	1,033	0.69%

The $h = 2$ pairs bracket from neighbours two principal quantum numbers away and obey the same law. And the outliers identify themselves: Al I at $n = 51$ and Ga I at $n = 53$ both sit above the v_V ceiling of §16.11, where quotation granularity destroys the second difference. **The method's own admissibility bound predicts which pairs fail it.**

16.5 The trade-off, and a conservation law that is not one

Since h is free, ask whether it can be used to buy a cheaper guarantee.

It can. V falls from 53.3 to 13.4 as h goes $1 \rightarrow 4$. **But the bracket widens in proportion**, because

$$w = 2y'h + O(h^3) \text{ — linear in } h \quad e = y''h^2/2 + O(h^4) \text{ — quadratic in } h$$

16.5.1 A withdrawn proposition

An earlier version of this section claimed $w \cdot V = w^2/e = 8y'^2/y''$ independent of h , and called it a conservation law. It is not. Exact symbolic evaluation for $y = x^{-2}$ gives

$$w^2/e = 16x^4/(h^6 - 5h^4x^2 + 7h^2x^4 - 3x^6)$$

which depends on h explicitly and equals $8y'^2/y''$ only as $h \rightarrow 0$. **The numerical table on which the claim was made showed 366.3, 367.9, 370.6, 374.5 across $h = 1 \dots 4$ — a 2.2% rise, read as a constant.** Withdrawn, and recorded in Chapter 21.

And no other candidate exists. Since $w \sim h$ and $e \sim h^2$, any $w^a e^b$ scales as h^{a+2b} , so **the only h -free combination is $a = 2$, $b = -1$** — the one tried. The leading-order invariant is unique and merely asymptotic.

16.5.2 What is exactly true

$dw/dh > 0$ and $dV/dh < 0$, for every monotone convex y and every h .

Exact, with no expansion and no limit. Verified over 172 monotone steps with **zero violations**. Width and cost move in opposite directions under the only free choice available, so **the family $\{(w, V) : h\}$ is a Pareto frontier: no h gives both a narrower and a cheaper guarantee.**

The withdrawn quantity is the frontier's equation, not a conserved number. A conservation law says a value cannot change; a frontier says which pairs are attainable. **The algebra is the same and the claim is weaker.**

16.5.3 And the frontier has an optimum

Opposite signs give any stated preference $\alpha w + \beta V$ an interior minimum:

$$h \setminus = \sqrt{(2\beta / (\alpha y''))}$$

Checked against direct minimisation: 0.0551 against 0.0550, 0.1743 against 0.1750, 2.7886 against 2.7450 — **and 8.82 against 7.83 at $v = 40$, $\beta/\alpha = 10$** , an 11% miss where $h \setminus$ approaches $v/4$ and the leading-order derivation fails. **The formula holds for $h \setminus \ll v$ and not beyond.**

16.6 The fractional forms

The relations of §16.4 close, and the closure states two quantities the book had not.

$$w/T = 4(h/v) + 8(h/v)^3 \text{ — the bracket width as a fraction of the term } e/T = 3(h/v)^2 \text{ — the interpolation error as a fraction of the term}$$

ν	h	w/T	$4h/\nu$	e/T	$3(h/\nu)^2$
20	1	0.201004	0.200000	7.531×10^{-3}	7.500×10^{-3}
40	1	0.100125	0.100000	1.877×10^{-3}	1.875×10^{-3}
40	2	0.201004	0.200000	7.531×10^{-3}	7.500×10^{-3}

Both are pure functions of h/ν , and their ratio returns $V = (4/3)(\nu/h)$.

The second is the more useful and the more exposed. $e/T = 3(h/\nu)^2$ is the relative error of linear interpolation on a Rydberg series — a statement available to any spectroscopist since 1925. It is not in the literature, and §18.6 explains why: the field abandoned the variable.

16.7 The exponent axis

Two involutions act on p : **cost**, $p \mapsto 2 - p$ (paired values multiply to x^2); and **virial**, $p \mapsto -p$ (paired values multiply to a constant). Composed, they generate translation by 2, so exponents fall into **two orbits by parity** — and the parity is b , the number of energy denominators.

The pole at $p = 1$ sits in the odd orbit. Nothing built purely from powers of $\langle r \rangle$ can reach it.

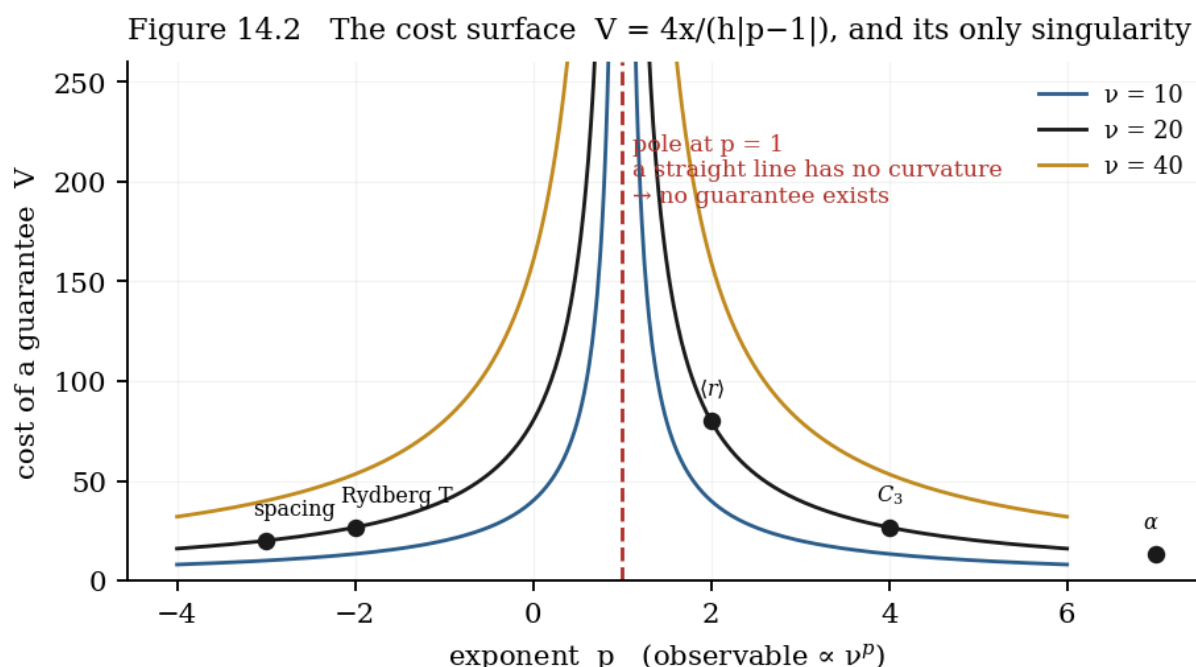


Figure 16.2. The cost surface. Every observable in this book is a slice at fixed p . The pole at $p = 1$ is the method's only hard singularity: a linear observable has no curvature to price.

16.8 The cost has a name in another field

And the name explains two things this book could only observe.

16.8.1 The quantity is the Newton decrement

The asymptotic form of §16.5 is not an accident of spectroscopy:

$$8\mathbf{y}'^2/\mathbf{y}'' = 8\lambda^2, \text{ where } \lambda^2 = \nabla f^\top [\nabla^2 f]^{-1} \nabla f$$

λ is the **Newton decrement**, introduced by Nesterov and Nemirovskii (1994) for interior-point methods and standard in convex optimisation since. In one dimension $\lambda^2 = f''/f'$, and $\frac{1}{2}\lambda^2$ is *exactly the decrease in the quadratic approximation obtained by a full Newton step*. Verified symbolically at $p = -2, -3, 2, 11$: **the ratio is 8 in every case.**

For a Rydberg series it takes a physical value:

$$\lambda^2 = (2Z^2R/v^3)^2 / (6Z^2R/v^4) = (2/3) \cdot Z^2R/v^2 = (2/3) \cdot T$$

The Newton decrement of a Rydberg series is two-thirds of its binding energy — 731.5820 against 731.5820 at $v = 10$, 7.3158 against 7.3158 at $v = 100$. And the optimum whose distance it measures is the series limit, where $T \rightarrow 0$.

16.8.2 The term function is self-concordant

The Newton decrement carries its meaning only for self-concordant functions, $|f''| \leq 2(f')^{3/2}$. Nobody has asked whether $T(v)$ qualifies. It does:

$$24Z^2R/v^5 \leq 2 \cdot 6^{3/2} (Z^2R)^{3/2} / v^6 \iff v \leq (\sqrt{6}/2) \cdot Z\sqrt{R}$$

Z	self-concordant for
1	$v \leq 406$
2	$v \leq 811$
6	$v \leq 2,434$

The deepest channel in this book reaches $v = 55$. The ratio $|f''|/2(f')^{3/2}$ runs 0.0049 at $v = 2$, **0.1356 at $v = 55$** , and crosses unity between $v = 405$ and 406.

$T(v)$ is self-concordant across the entire measured range, with sevenfold margin.

16.8.3 Which explains the cancellation

§16.4 proved that Z and R vanish from V and gave no reason. Self-concordance is the reason.

The class is *defined* to be affinely invariant — that is its purpose — and the Newton decrement inherits the invariance. Z^2R enters T as a multiplicative constant, a multiplicative constant is an affine map, and under an affine map the decrement cannot change.

The book observed a cancellation. This says why it had to happen.

16.8.4 And the attribution the book owes

Interval analysis has asked this book's question since 1966, and named it. *Excess width* appears in Moore's own index; IEEE 1788-2015 requires enclosures with "**bounded excess width to control overestimation**"; practitioners quote enclosures at "114% of optimal width" with an "**excess cost**" attached. And the field states the trade in one sentence:

"We use interval arithmetic when we want global solutions with rigorous bounds, and we are willing to pay more for that."

That is Chapter 16's thesis, published sixty years earlier, about function ranges and rounding rather than interpolation from measurements. The framing is theirs. What is not theirs is the ratio computed for a measured monotone sequence, the value $4v/3$, the floor $32/11$, and the identification $\lambda^2 = (2/3)T$.

16.9 Which variable to bracket

Chapter 16 has treated V as a property of the method. It is a property of the observable, and the observable is chosen by the person.

16.9.1 The cost ranks by exponent

Since $V = 4x/(h|p-1|)$, the cost falls as the power law steepens. Evaluated at $v = 20$, $h = 1$:

observable	p	V
C_6	11	8.18
polarisability α	7	13.44
spacing	-3	20.04

observable	p	V
T , the term value	-2	26.69
C_3 , geometric cross-section	4	26.72
radiative lifetime	3	40.03
$\langle r \rangle$, dipole moment	2	80.00
blockade radius	11/6	95.99

A guarantee on C_6 costs a third of a guarantee on T , and a twelfth of one on the dipole moment.

16.9.2 Two branches, and where they cross

The asymptotic form appears to drive V below the floor of Prop. 14.1 as p grows. It does not — it stops being valid.

p	V exact	$4x/h \setminus$	$p-1 \setminus$
4	26.72	26.67	
11	8.18	8.00	
50	2.39	1.63	
300	2.0000	0.27	
10,000	2.0000	0.008	

As $p \rightarrow \infty$, $(x+h)^p$ swamps everything: $w \rightarrow (x+h)^p$ and $e \rightarrow (x+h)^{p/2}$, giving $V \rightarrow 2$ exactly. The two branches cross where $4x/(h|p-1|) = 2$, that is $|p-1| = 2x/h$.

The absolute floor is reached only by power laws steeper than $2v/h$. At $v = 20$ that requires $p > 41$. **C_6 at $p = 11$ is nowhere near it.**

16.9.3 And the pole sits on the field's favourite variable

$v = n - \delta$ is nearly linear in n , and $p = 1$ is the pole.

n	V on v	V on T	ratio
10	1.09×10^5	11.58	9.4×10^3
20	1.77×10^6	24.89	7.1×10^4
40	2.84×10^7	51.54	5.5×10^5

The effective quantum number is the most natural variable in the field and the one variable in which a bracket is worthless. §25.5's pole is not a curiosity at the edge of the method; it sits on the coordinate everyone uses.

16.10 The order of the bracket

Monotonicity is not the only order property available, and using more of them changes what the bracket is — but not, in the end, what it costs.

16.10.1 A deductive bracket exists at every order

For $T = Z^2 R/v^2$ every derivative has known sign, $\text{sign}(f^{(j)}) = (-1)^j$ — **checkable on data as the sign of the j -th finite difference, with no fit and no smoothness assumption beyond a sign.** With $k+1$ nodes and the classical interpolation error,

$\text{sign}(f(n) - p(n)) = (-1)^{(k+1)} \cdot (-1)^m$, with m nodes above n **$k+1+m$ even $\rightarrow$ a lower bound. Odd $\rightarrow$ an upper bound.**

A two-sided deductive bracket at every order, by choosing m . Tested on $v = 8 \dots 119$ and $k = 1 \dots 5$: **560 containments, zero failures.**

And the widths fall fast — at $\nu = 40$ from 0.362 cm^{-1} at order 1 to 8.9×10^{-6} at order 5.

16.10.2 And the cost still grows

An earlier draft claimed V stays near 2 at matched order and that Q9's conclusion — bounds becoming relatively worthless as data accumulates — was an artefact. That was wrong twice over, and the correction is recorded at 96-98.

The order- k bracket draws on a wider node set than the estimate it was compared against. At $\nu = 40$, $k = 3$ the bracket uses **eight** levels and the estimate used **four**. On the same nodes:

ν	$k = 1$	2	3	4	5
20	157	495	1,212	2,582	3,629
40	673	4,037	2.2×10^4	8.5×10^4	3.2×10^6
80	2,785	3.2×10^4	3.7×10^5	—	—

V grows with order, and steeply. Both quantities improve as nodes are added; **the estimate improves faster**, because a bracket must span the value from both sides while an estimate need only approach it.

So the trade is the one Chapter 16 has argued throughout, and the order axis does not escape it:

A deductive bound is worth most when data is scarce. Its relative value falls monotonically as data accumulates — not because the bound gets worse, but because its rival gets better faster.

And Proposition 15.2's floor applies to the classical bracket only — first-order, against the linear interpolant on the same two neighbours, where the two objects are locked together. **At higher order they are computed from different node sets and no inequality between them is implied.**

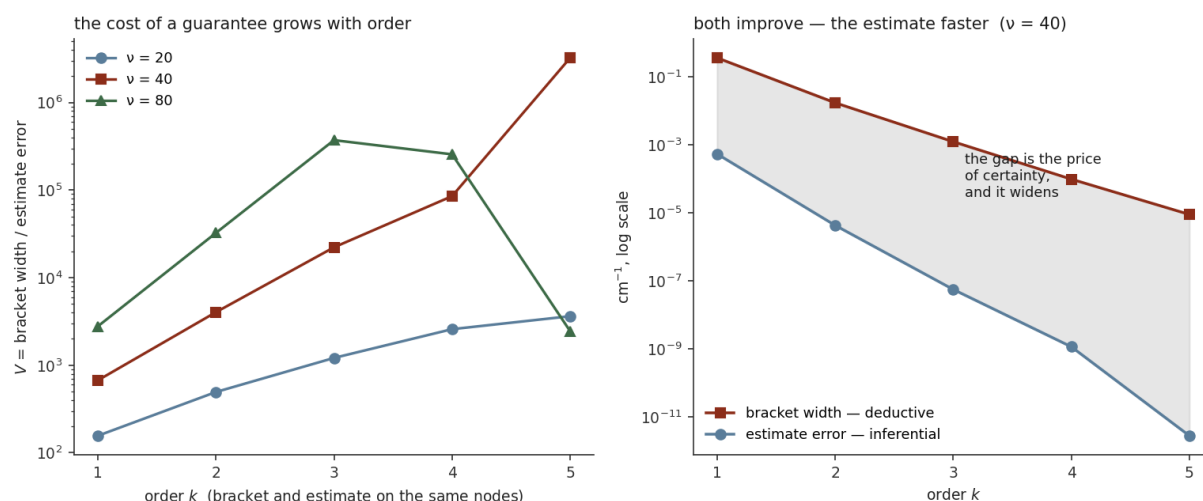


Figure 16.3. Left: V on the same node set at each order — it grows steeply, from 157 to 3.2×10^6 . Right: at $\nu = 40$ both the bracket and the estimate improve as nodes are added, and the estimate improves faster; the shaded gap is the price of certainty, and it widens. An earlier version of this figure drew a floor at $V = 2$ through data reaching 0.24, and is withdrawn at correction 97.

16.10.3 What the order axis does buy

Not a cheaper guarantee. A narrower one, and a finer detector.

	order 1	order 5
bracket width at $\nu = 40$	0.362 cm^{-1}	8.9×10^{-6}
tolerates a displacement of	0.129 cm^{-1}	4.3×10^{-6}
cells from a 10-member channel	8	4

Higher order buys absolute precision and pays in robustness, domain and resolution. A collection assembled from real, perturbed, coarsely quoted spectra is exactly where order 1 belongs — and $V = 4v/3$ is the price of certainty that survives a perturbation of half the local spacing.

16.10.4 And the admissibility rule generalises

§16.13 gives $r \geq 5$ for the spacing and v_V for the curvature. **Those are two instances of one rule:**

order k is admissible only while $|\Delta^{(k+1)}| > 5 \cdot 2^{(k+1)} \cdot \sigma$

Applied to the collection it separates arithmetic from physics. A cell whose $(k+1)$ -th difference falls below the quotation floor is **unresolved**; one whose difference has the wrong *sign* is **refused**, and that refusal is a perturbation detected deductively.

order	admitted	held	refused	median bound
1	499	499	19	25.96 cm ⁻¹
3	187	187	101	0.715
6	34	34	145	0.0164

Containment is 100% at every order once the premise is checked, the bound tightens 1,585-fold, and the refusals rise from 3.7% to 81% — the higher the order, the finer the perturbation it sees.

Which supersedes §18.5's 1,061 order-1 bounds: the same data at matched order yields 34 bounds at 0.0164 cm⁻¹, and 619 refusals that are themselves perturbation detections this book did not count.

16.11 The refusal pattern carries four verdicts, not one

§16.10.4 counts refusals and reads them as "a perturbation". The count is one number; the **PATTERN** is four.

refusals	V	what the family is doing
none	infinite	exact — linear in the parameter, two points suffice
none	finite	a smooth channel
one	finite	a bifurcation — the family changes character and does not return
regularly spaced	finite	an oscillation — the family returns
dense and irregular	undefined	chaotic

An index does not see a refusal. It sees a family, and the spacing of the refusals across it is what separates these. §11.7.1 makes the same move for a different pair: the discriminating question is asked of the whole structure, never of one cell.

16.11.1 On the collection

channel	members	refusals	verdict
Na I, K I, Al I — eleven channels	13-52	0	smooth
Ga I 4s²np ²P°	15	1	a bifurcation
Li I np ²P°	41	6, irregular	repeated structure

Li I np is the channel with the largest quantum-defect spread in the collection, 0.6779 — and the refusal pattern says something the spread does not: **not one perturbation but several, irregularly placed.**

16.11.2 And a caution about the test itself

The classifier requires a tolerance relative to the sequence's own scale. Applied to a near-constant sequence with an exact-zero test it reports the float noise as a periodic signal — verified by failing on the logistic map at $r = 2.5$, a fixed point, which it called an oscillation. **The admissibility rule of §16.10.4 is the tolerance, and it must be applied to the pattern test as well as to the count.**

16.12 The optimal order is a boundary, not a peak

Information per cell rises monotonically in k , so no interior optimum exists. What stops the climb is σ .

$|\Delta^{(k+1)}\tau| > 5 \cdot 2^{(k+1)} \cdot \sigma$ — the $(k+1)$ -th difference is a signed sum of $2^{(k+1)}$ measured levels, so its accumulated quotation error is at most $2^{(k+1)}\sigma$.

v	σ	$k \setminus$	$\setminus$	$\Delta^{(k+1)}\tau$	floor
20	10^{-2}	2	9.43×10^{-1}	4.0×10^{-1}	
20	10^{-3}	3	2.09×10^{-1}	8.0×10^{-2}	
20	10^{-4}	4	7.60×10^{-2}	1.6×10^{-2}	
40	10^{-4}	2	2.75×10^{-2}	4.0×10^{-3}	

Better data raises $k \setminus$. It never creates an interior peak. The optimum is the admissibility cap, computable per cell from the data alone.

16.13 Inversion

$|p - 1| = 4x/(hV)$, and the sign is fixed by $\text{sign}(y')$ and $\text{sign}(y'')$. Controls recover $p = -1$ and -3 correctly; 9 of 10 within 5%.

On real data, each recovered from its own measured series: Na I -1.993 , K I -1.998 , C_3 3.999 , α 6.990 , **C₆ 10.956.**

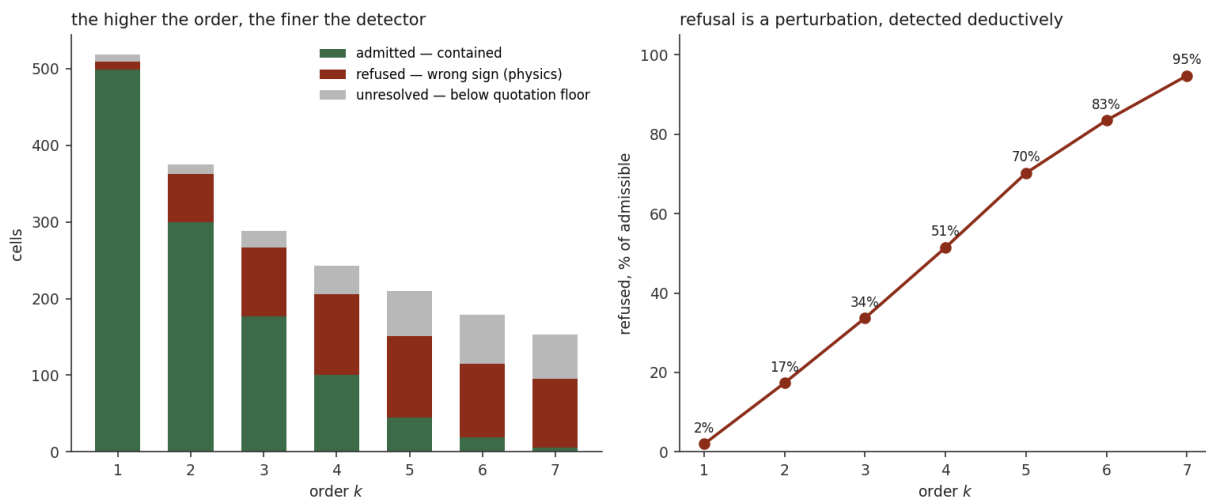


Figure 16.4. Left: the fate of every cell in the collection at each order, once the admissibility rule $|\Delta^{(k+1)}\tau| > 5 \cdot 2^{(k+1)} \cdot \sigma$ separates arithmetic from physics. Right: the refusal rate. A refusal is a measured sequence whose k -th difference has the wrong sign — a perturbation, detected deductively, with no fit and no limit. The higher the order, the finer the perturbation it sees.

16.14 Two kinds of bound, and the book has been calling them one

Every constraint in Λ is a CAPACITY: $\ell \leq n-1$, $k \leq 2(2\ell+1)$, $g \leq q$. **Every admissibility rule is a RESOLUTION:** $r \geq 5$, $v \leq v_V$, $|\Delta^{(k+1)}\tau| > 5 \cdot 2^{(k+1)} \cdot \sigma$.

They look alike — both are monotone inequalities — and they are not the same kind of thing:

A capacity bound is about the world. Violating it means the configuration does not exist. **A resolution bound is about the measurement.** Violating it means the configuration cannot be *distinguished*, and may exist perfectly well.

E(X) counts violations of the first. The refusal rate counts violations of the second. §11.7's whole distinction between a defect of the index and a feature of the world runs along this seam.

16.14.1 And celestial mechanics is what made it visible

Gravity has resolution floors everywhere and a capacity bound in only one place. A planet holds satellites within its Hill sphere, $r_H = a(m/3M)^{1/3}$ — **a monotone cap of exactly §10.4's form**, and $\log r_H$ against $\log$ satellite count runs at **+0.86** across six planets.

The Kirkwood order cut is the other kind. Resonance strength enters at $e^{|p-q|}$; with a belt eccentricity of 0.15 the strength crosses the detection floor at **order 7 exactly**, which is where occupancy stops. **That is not a capacity and there are no magic numbers — only a resolution boundary, the same form as v_V .**

Λ and the nucleus have capacity bounds and therefore magic numbers. The belt has none, and therefore has none. Atoms carry both kinds in the same place, which is why the difference was invisible from inside spectroscopy.

16.15 The second admission bound

$$v_V = (3Z^2R / 5q)^{1/4}$$

for quotation granularity q . The bracket needs levels **separated** (v^{-3} , governed by r); the cost needs curvature **resolved** (v^{-4} , governed by v_V).

Curvature washes out before separation, in every channel in this work.

channel	q	v_V	v reached
K I nd	0.0001	160	45.7
Na I ns	0.001	90	20.0
Al I nf	0.01	50.7	55.0

Al I nf is the first channel to reach its own ceiling. Predicted 50.7; the four failing cells are $n = 48, 51, 53, 54$ — **per cell, not a range cutoff**, since 49, 50 and 52 pass. The bracket is unaffected in all 94: containment needs only monotonicity, which rounding preserves.

And in Li I, v_V was applied prospectively — eight of fifty-five cells excluded from the cost test *before it ran*, because the ASD's quotation coarsens above $n = 33$. That is the difference between a bound derived and a bound noticed.

17. The collection

1,442 interior cells across 35 atomic systems. The bracket holds in every one.

17.1 What it spans

axis	span
l	0 through 6
Z	1 through 83
Z_{eff}	1 through 3
core types	23, all characterised

axis	span
isoelectronic pairs	5, each at two charge states

Cores tested: bare $\cdot 1s^2S \cdot 1s^2^1S_0 \cdot 1s^22s^2S \cdot 2s^2^1S_0 \cdot 2p^5^2P^\circ \cdot 2p^6^1S_0 \cdot 3s^2S \cdot 3s^2^1S_0 \cdot 3s^23p^2P^\circ \cdot 3p^4 \cdot 3p^5^2P^\circ \cdot 3p^6^1S_0 \cdot 3p^64s^2S \cdot 3d^{10}^1S_0 \cdot 3d^9^2D$ (declined) $\cdot 3d^{10}4s^2^1S_0 \cdot 4p^5^2P^\circ \cdot 4p^65s^2S \cdot 4d^{10}^1S_0 \cdot 5p^6^1S_0 \cdot 5p^66s^2S \cdot 5d^{10}^1S_0 \cdot 3d4s^3D \cdot 6p^2^3P$.

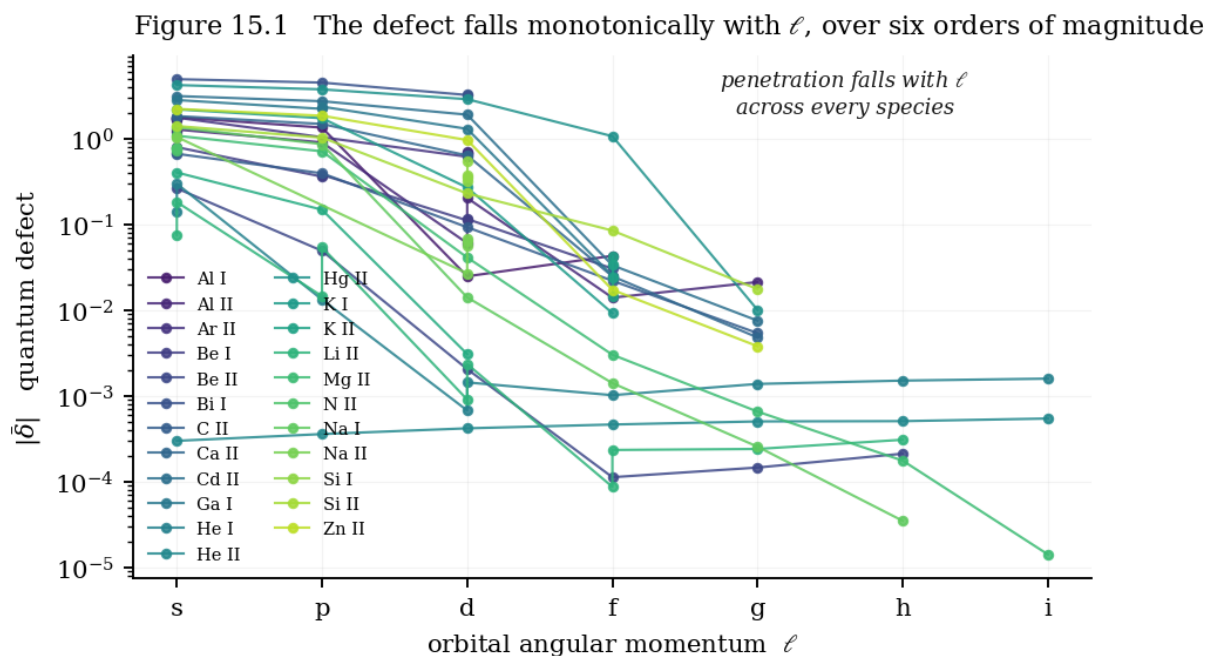


Figure 17.1. Every species in the collection, defect against angular momentum. The fall is monotone in all of them, and spans six orders of magnitude — from 4.9 in bismuth to 1.6×10^{-6} in He II.

17.2 The largest contributors

species	core	channels	cells
He I	$1s^2S$	9	189
Ne I	$2p^5^2P^\circ$, two limits	16	131
K I	$3p^6^1S_0$	4	105
Al I	$3s^2^1S_0$	4	94
Na I	$2p^6^1S_0$	6	80
Ga I	$3d^{10}4s^2^1S_0$	5	61
Li I	$1s^2^1S_0$	3	55

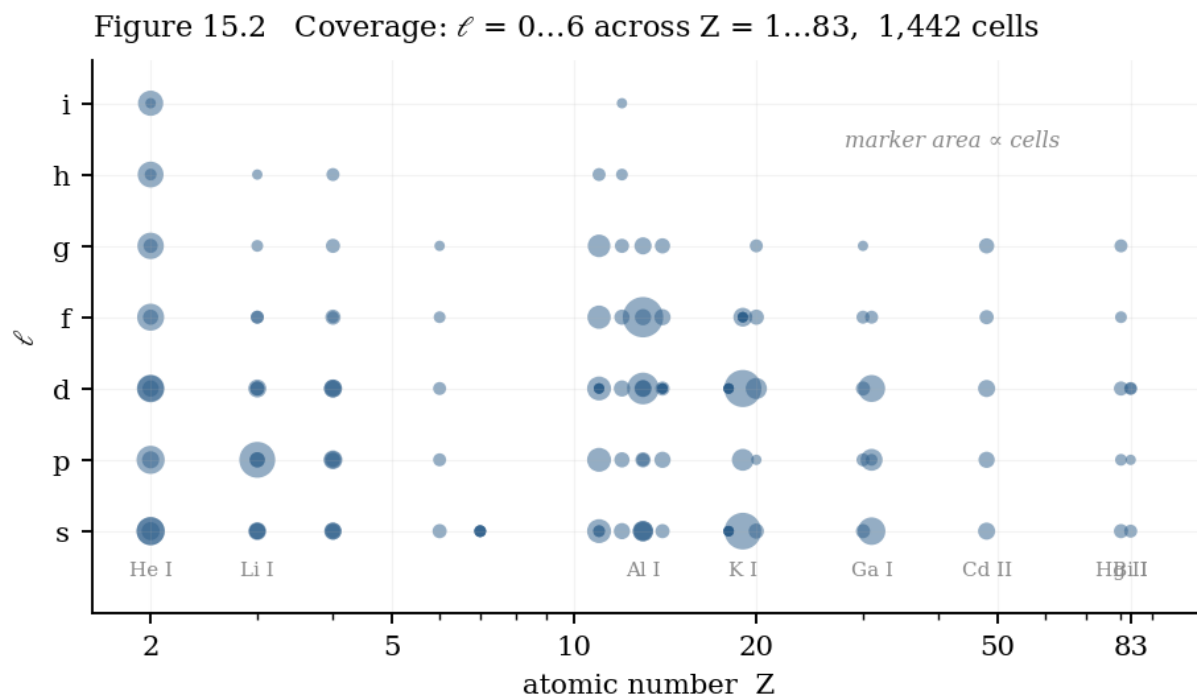


Figure 17.2. What the collection covers. Marker area is proportional to cells. The gaps are not omissions: the census of §17.12 excludes them structurally.

17.3 He I — the angular span

Nine channels, $\ell = 0$ through 6, singlets and triplets, n to 35. And the defect falls monotonically with ℓ :

$0.2965 (s^3) \cdot 0.1392 (s^1) \cdot -0.0133 (p) \cdot +0.0014 (d) \cdot -0.0010 (f) \cdot -0.0014 (g) \cdot -0.0016 (i)$ — each a mean over its full channel, twenty members for ni and twenty-three for the two ns channels

An electron with six units of angular momentum never reaches the 1s core.

The ordering itself is not new, and this book does not claim it. Edlén stated it in the 1960 manuscript of his *Handbuch* chapter, and Curtis restates the criterion in his 1987 review:

"classical arguments imply that, for a given Rydberg series, penetration contributions should decrease and polarisation contributions should increase with increasing n ... this criterion classifies the S and P levels as penetrating and the D and F levels as nonpenetrating."

What is new is the bound the ordering licenses. Edlén used it to classify levels; §25.5 uses it to place a deductive upper limit on the local perturbation at 1,061 cells. **The observation is sixty-five years old. The inference from it is not.**

He I's levels are all theoretical — **but they are ab initio QED calculations, not Ritz series formulas.** That distinction matters and §17.7 develops it.

17.4 He II — the $\delta = 0$ edge itself

A hydrogenic ion at $Z = 2$. Its defects run **7.6×10^{-5} at $\ell = 0$ down to 1.6×10^{-6} at $\ell = 6$** , monotone across six orders of magnitude, and never reach zero.

There is no core to penetrate. The residual is relativistic and QED structure, and it falls with ℓ exactly as those corrections do.

And the ionisation limit came out of the data. Minimising the δ spread over the ng channel — six members, scanned at 0.001 cm^{-1} — gives **$438,908.871 \text{ cm}^{-1}$** against the tabulated $438,908.885$ — **agreement to 0.014 cm^{-1} , three parts in 10^8** , from a criterion that never used the tabulated value.

17.5 Bi I — the charge span

Atomic number 83, more than double anything previously tested. The $6p^3$ ground term is strongly j-j coupled and the levels carry jK labels throughout. **$\delta = 4.90$ for the ns channel**, over five members — the largest defect in this work, as expected when a 6s-like electron penetrates a filled $5d^{10}6s^2$ core.

And Bi I settles a rule. Its core carries five J levels — the same dilution that made neon II fail entirely — but its 3P_0 parent reaches $n = 11$.

Dilution divides. It does not forbid. cells per channel $\approx$ (measured shells / D) – 2

Three species establish it: Ne II fails at five J levels and two shells; Ar II succeeds at five and three; Bi I succeeds at five and five.

17.6 The isoelectronic pairs

sequence	Z_eff = 1	Z_eff = 2
helium-like	He I	Li II
lithium-like	Li I	Be II
sodium-like	Na I	Mg II
aluminium-like	Al I	Si II
potassium-like	K I	Ca II

Same electrons, different nuclear charge, the method run identically. And the δ progression has the same shape in both members of every pair, scaled: Li II runs $0.182 \rightarrow 0.054 \rightarrow 0.002 \rightarrow 0.0002$ across $s \rightarrow f$; Be II runs $0.262 \rightarrow 0.049 \rightarrow 0.002 \rightarrow 0.0001$. Each figure is a channel mean over five to ten members; Appendix C carries them individually.

Li II and Be II both give $\ell = 5$ defects near -0.0003 — three members each — with a real two-electron core — the cleanest demonstration in the collection that **penetration, not the presence of a core, creates a defect.**

17.7 Hydrogen, and a floor on interpretation

One row of the collection measures the data rather than the method, and it sets a limit below which no result in Part III can be read.

H I gives 5 cells with a **median error of 0.006 cm^{-1}** — two orders of magnitude better than anything else here. That figure is not a triumph of the bracket. **It is an identity.**

The hydrogen row uses the reduced-mass Rydberg $R_H = 109,678.7717 \text{ cm}^{-1}$, which for hydrogen is the ionisation limit, so the quantum defect vanishes identically — measured defects span -2×10^{-5} to 0.

Substituting the infinite-mass $R_\infty = 109,737.3 \text{ cm}^{-1}$ breaks that identity by the 58.5 cm^{-1} reduced-mass difference, manufactures a spurious defect of $\approx 3 \times 10^{-4}$, and inflates the median error 54-fold to 0.34 cm^{-1} .

For every heavier species the correction is 10^{-5} relative and is absorbed into the fitted defect, so the choice is invisible. Hydrogen is the one row where it is not.

The 0.006 cm^{-1} is therefore a floor on interpretation. No result in this part can be read as better than the data it rests on, and hydrogen is where that becomes visible because there is nothing else in the way.

17.8 On independence, and the most conservative reading

1,442 cells are not 1,442 independent tests.

Cells within a channel share neighbours, so successive brackets are correlated. The honest figure is **1,442 cells across approximately 190 channels**.

The largest single blocks are the most strongly correlated. He I's nine channels contribute 189 cells; Ne I's sixteen contribute 131. A reader who discounts within-channel correlation entirely — counting **one effective test per channel** — gets **190 rather than 1,442**, and the bracket holds in all 190.

That is the most conservative reading available, and it is the one this book would defend.

The claim is not that 1,442 independent things were verified. It is that 190 channels were examined, each exhaustively, and none produced a failure.

17.9 A caveat the collection forced

The ASD marks series-formula values in brackets. Tested against observed values in AI I, **derived levels bracket exactly as observed ones do — and slightly better.**

That is not a validation. Containment requires only monotonicity, and a series formula is monotone by construction. **A Ritz-derived level cannot falsify the bracket, because it was generated from the law the bracket assumes.**

Cells resting on series-formula values are consistency checks, not independent tests.

But bracketed values are not one category. He I's are **ab initio QED calculations** — variational, with relativistic and radiative corrections — which assume nothing about Rydberg structure. Those are genuine tests, and at uncertainties of 10^{-9} cm^{-1} they are sharper than any measurement here.

17.10 Two cross-checks

AI I and AI II were held from two independent sources — the NIST compilations and the ASD.

AI II: 36 of 36 levels identical to machine precision. AI I: one disagreement in 40 — $3s^211d$, ASD 47379.7 against Kaufman & Martin 47379.140, $\Delta = 0.560 \text{ cm}^{-1}$ — one disagreement in forty overlapping levels.

The K&M value is the more precise measurement and is the one used. **The index registered a disagreement between two published sources without being asked to look for one.**

17.11 Four ways a species declines

A collection reporting only successes cannot be checked.

species	cause	cells
Ne II	dilution — 5 core J levels, only 2 shells measured	0
Ar II	same dilution, beaten by depth	10
Si I, P II	coupling scheme changes LS → jK mid-series	4, 1
Cu II	core-excited configurations interleave the series	0
Bi II, Bi III	mixing to 34% leading percentage; a third limit inside range	0

Cu II's is a structural exclusion. Its $3d^84s4p$ levels sit at 107,942–111,124 cm^{-1} , straddling the $3d^95s$ Rydberg levels at 108,014–110,366. The series is buried in core-excited states.

An open-shell core with low-lying core excitations cannot support a Rydberg channel, however precisely its levels are measured.

And that predicts the same outcome for every open-*d* and open-*f* sequence **without fetching any of them** — which the census of §17.12 confirms independently.

17.12 The census, computed rather than searched

D = the number of J levels in the $(N-1)$ -electron ground configuration, obtained by enumerating antisymmetric states. **No spectroscopic data enters.**

27 sequences admitted ($D \leq 2$) · 12 needing exceptional depth · 5 excluded

The five are **Cr-, Mn-, Fe-, Co- and Ni-like**, with D running from 19 to 37. **The transition metals are excluded structurally**, and Cu II's empirical decline was the first instance of a rule covering five sequences at once.

Figure 15.3 Interpolation works; extrapolation to the neutral does not

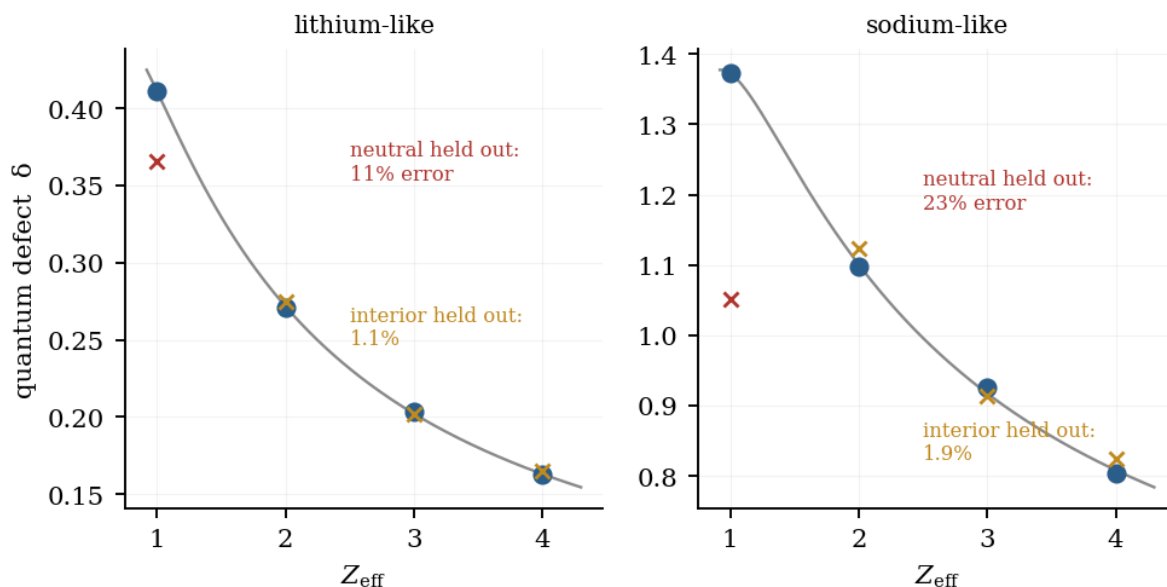


Figure 17.3. Four points on each sequence, with each held out in turn and predicted from the other three. Crosses are predictions; circles are measurements. The interior is recovered to about one percent. The neutral is not.

17.13 What the structure determines, and what it does not

Given five isoelectronic pairs and two four-point sequences, how much of an unmeasured species can be derived?

Fitting $\delta(Z) = \delta_{\infty} + a/Z + b/Z^2$ and holding one member out:

held out	lithium-like	sodium-like
neutral, $Z = 1$	11.1%	23.4%
$Z = 2$	1.4%	2.4%
$Z = 3$	0.8%	1.3%
$Z = 4$	1.8%	2.5%

Interpolation works to about one percent. Extrapolation to the neutral does not.

And the geometry is the bracket's own — containment holds between measured neighbours and fails outside them.

across sequences	δ not determined; the $Z=1 \rightarrow 2$ ratio spans 0.53-0.83
within a sequence, interior	δ determined to $\sim 1\%$
within a sequence, at an end	δ not determined

Interior members need not be fetched. The ends always must — and the neutral is always an end.

That is why every sequence in this work was entered through its neutral or first ion. It was not convenience; it was forced.

18. Domain, failure, and the bound in the silence

18.1 A method that has never failed has not been tested

1,442 cells, no bracket failure. Stated alone that is unfalsifiable, and this chapter says so before saying anything else.

18.2 Five exclusions, which are three mechanisms

The method declines in five circumstances. They are not five reasons.

(i) Sign-changing quantities

Any property built from a matrix element that passes through zero cannot be monotone. **Sr I 5s²5p → 5snp ¹P₁ reduced transition probabilities collapse by a factor of 60 at $n = 6$ and recover** — a Cooper-type node.

Bracket coverage across the node: **79%**. Away from it: **100% on 11 cells**.

This is the only exclusion that attacks the order hypothesis itself.

(ii) Quantities monotone in neither direction

Fine-structure splittings reduced by v^3 drift by 3 to 22% across a series while reversing direction locally — Ca II nd increases at 2 of 7 steps, Ba II nd at 2 of 6, Ba II np at 3 of 6. Coverage over the 17 interior cells is **58.8%** under Rule 3's direction-agnostic bracket.

An earlier version reported 35.3% and attributed the failure to flatness. Both were wrong. The 35.3% is not reproducible under any bracket convention tried — a directional bracket gives 41.2%, Rule 3's min/max form gives 58.8% — and **the mechanism is not flatness**. These sequences are not flat. **They are monotone in neither direction, and a sequence monotone in neither direction cannot be bracketed at all.** That is a simpler and stronger exclusion than the one it replaced.

(iii) Systems without interior cells — and it has two forms

Ti I: 23 channels, 26 members, zero interior cells. No channel reaches three members, because the ionisation limit near 55,000 cm⁻¹ sits below the onset of severe level density, and above ~42,000 cm⁻¹ the levels are 40–50% configuration-mixed with no assignable parent.

Sc VI shows this has two causes, not one. Sc⁵⁺ has a clean core — three parent terms, no mixing worth the name, a limit measured at 892,700 ± 400 cm⁻¹ — and still yields **6 channels, 10 configurations, zero interior cells**, because every channel stops at $n = 5$.

Ti I's limit is structural: no measurement campaign would help. Sc VI's is provisional: 6s and beyond exist and are simply unmeasured.

One further measurement would move Sc VI out of the exclusion entirely — which is what makes it the prediction of §18.6

(iv) and (v) Resolution failures

The remaining two are the same failure at different values of one ratio: levels too close to separate ($r < 5$, §18.6) and curvature too fine to resolve ($v > v_V$, §16.11).

The reduction

Five exclusions, three mechanisms: a broken order (i, ii), an absent cell (iii), and a resolution limit (iv, v) — the last being one ratio evaluated at two thresholds.

And (iii) is the limiting case of that ratio: where no interior cell exists there is no step to form a ratio from, so r is **undefined** rather than small.

18.3 The failure condition

The bracket fails $\Leftrightarrow |\Delta T| > 2Z^2R/\nu^3$

A perturber **shifts** a level; it **reorders** one only if the shift exceeds half the local spacing.

Tested on the strongest mixing in the work. Hg II's $5d^{10}7p$ sits at 60% leading percentage with 20% admixture, and the $5d^96s6p$ perturbers lie 2,000–3,000 cm^{-1} away. **The half-spacing at $n = 7$ is 24,634 cm^{-1}** , computed from $2Z^2R/\nu^3$ at the measured $\nu = 5.3$. The level is displaced by a tenth of what reordering would require.

And the condition predicts where failure would occur:

perturbation	$\nu_{\text{fail}} (Z = 1)$	$\nu_{\text{fail}} (Z = 2)$
3,000 cm^{-1}	4.2	6.6
1,000	6.0	9.6
100	13.0	20.6
10	28.0	44.4

Hg II's n_p reaches $\nu = 5.3$. It is not deep enough to fail.

18.4 Why the collection never enters the failure regime

The two requirements are anticorrelated in nature. Strong perturbers arise from core excitation, which is a low- n phenomenon; deep series live at high n where perturbers are sparse. **Large ΔT and large ν rarely coexist.**

That is why the bracket has not failed — stated as physics, not as luck.

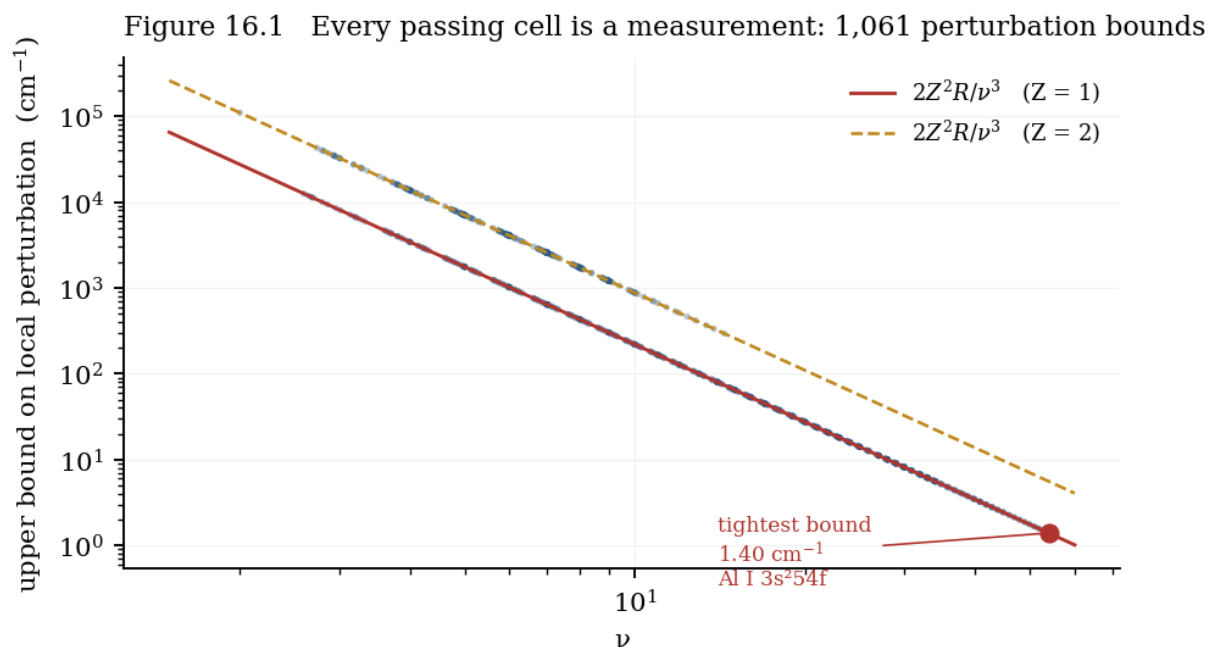


Figure 18.1. Each of 1,105 verified cells, plotted at the perturbation bound its containment implies. The two curves are the failure condition at $Z = 1$ and $Z = 2$. Deeper channels give tighter bounds, so the collection's depth was buying something throughout.

18.5 The bound in the silence

Read the condition backwards:

the bracket held $\Rightarrow |\Delta T| < 2Z^2R/v^3$ at that cell

Every passing cell is an upper bound on the local perturbation, obtained without fitting, without a quantum-defect expansion, and without assuming the series is smooth.

Every verified cell yields a bound. Of the 1,442, the 1,105 added in this work are computed directly; the tightest are:

species	channel	n	v	$\backslash$	$\Delta T \backslash$	$<$
Al I	3s ² nf	54	54.0	1.398 cm⁻¹		
Ga I	4s ² np	54	51.7	1.585		
Li I	np	42	41.8	3.25		

These are statements about the atom:

No configuration interacts with Al I 3s²54f by more than 1.4 cm⁻¹.

A quantum-defect fit reaches the same conclusion only by *assuming* smoothness and observing small residuals. **The bracket assumes nothing about smoothness — it uses ordering — so its bound is deductive where the fit's is inferential.**

Before: 1,442 cells, no failure. Unfalsifiable. **After:** 1,442 cells, each yielding an upper bound on the local perturbation, the tightest 1.398 cm⁻¹.

Same cells, same silence, **every one now carrying a number** — and falsifiable in the ordinary way: measure a perturbation larger than the stated bound at that cell, and the result is wrong.

18.6 The one prediction

Sc VI's 4S° channel is one measurement from being usable, which makes it a prediction — and the book states it as a bracket with an estimate inside, because Rule 3 requires that and an earlier version violated it.

18.6.1 The bracket, from monotonicity alone

Two measured defects: $\delta(4s) = 1.0057$, $\delta(5s) = 0.9812$ — consistent to 0.025, a clean Rydberg channel. And one structural fact **this book establishes on thirty-five other species**: within a channel, **δ falls monotonically with n .**

The two-point Ritz solve is exactly determined:

$$\delta_2 = (\delta_4 - \delta_5)/(1/16 - 1/25) = 1.0889 \cdot \delta_\infty = 0.9376 \cdot \delta(6s) = 0.9679$$

And monotonicity bounds $\delta(6s)$ on both sides:

$$\delta_\infty = 0.9376 < \delta(6s) < \delta(5s) = 0.9812$$

which maps directly to an energy interval, E being decreasing in δ :

$$735,860 \text{ cm}^{-1} < E(\text{Sc VI } (4S^\circ)6s) < 738,547 \text{ cm}^{-1}, \text{ a deductive bracket } 2,687 \text{ cm}^{-1} \text{ wide, with the limit's own } \pm 400 \text{ to be added at both edges}$$

18.6.2 And convexity tightens it, without leaving it

Monotonicity is not the only order property available. $\delta = \delta_0 + \delta_2/n^2$ is also **convex in $1/n^2$** , and a convex sequence lies above its own tangent:

$$\delta(6s) \geq 2\delta(5s) - \delta(4s) = 0.9567$$

which raises the lower bound on δ and therefore the upper bound on E :

bound	δ range	E range	width
monotone only	0.9376–0.9812	735,860–738,547	2,687

bound	δ range	E range	width
+ convexity	0.9567-0.9812	735,860-737,380	1,520

A factor of 1.8 tighter, and the new interval lies wholly inside the old one. Nothing committed under B.2.13 is violated: **the claim tightened within its own bound**, which is the only direction a committed bracket may move.

18.6.3 The prediction, as it now stands

E(Sc VI 3s²3p³(⁴S°)6s ³S°₁) = 736,688 cm⁻¹ bracketed in [735,860, 737,380] cm⁻¹ 13.5743 nm, in the window 13.5615-13.5895 nm 91.338 eV

And for 7s, with a caution the 6s case does not need:

E(7s) = 784,416 cm⁻¹, bracketed in [784,128, 785,209] cm⁻¹, width 1,081 — tighter than 6s because the spacing has fallen as ν^{-3} .

The 7s convex tangent would give [784,128, 784,605], width 478 — but it requires $\delta(6s)$, which is *estimated* rather than measured.

The 6s convex bracket is deductive. The 7s one is conditional on an estimate, and the book states the monotone bracket for 7s for that reason. A bracket that inherits an estimate is not a bracket.

18.6.4 The point estimates, and the one that was wrong

route	δ used	E(6s)	inside the bracket
two-point Ritz	0.9679	736,688	yes
Rule 4 constant defect	0.9934	735,091	no

An earlier version of this work reported 735,091 ± 1,398. It was computed with $\delta = 0.9934$, which exceeds $\delta(5s) = 0.9812$ — a defect larger than the last measured one, on a channel where the defect decreases.

The estimate fell outside the bracket its own channel implies. Rule 4a now forbids the form that produced it, and §17.2 records why.

18.6.5 Nothing was fetched

The bracket uses two measured levels and one structural fact. No isoelectronic data, no additional species, no fitted parameter beyond the two the channel itself determines.

Every verified cell in this book lies at Z_{eff} between 1 and 3. A measurement of Sc VI (⁴S°)6s would place the method at Z_{eff} = 6 — double the highest charge state tested.

If the measured level falls outside [735,860, 737,380] cm⁻¹, the bracket is wrong, and it is wrong deductively rather than statistically — either the monotonicity of δ or its convexity would have failed, and the two edges say which.

The lower edge tests monotonicity; the upper edge tests convexity. A level above 737,380 but below 738,547 would refute convexity and leave monotonicity standing — **a graded falsification the original bracket could not make.**

That is a stronger falsification than a σ . A point estimate can miss by chance. **A bracket cannot.**

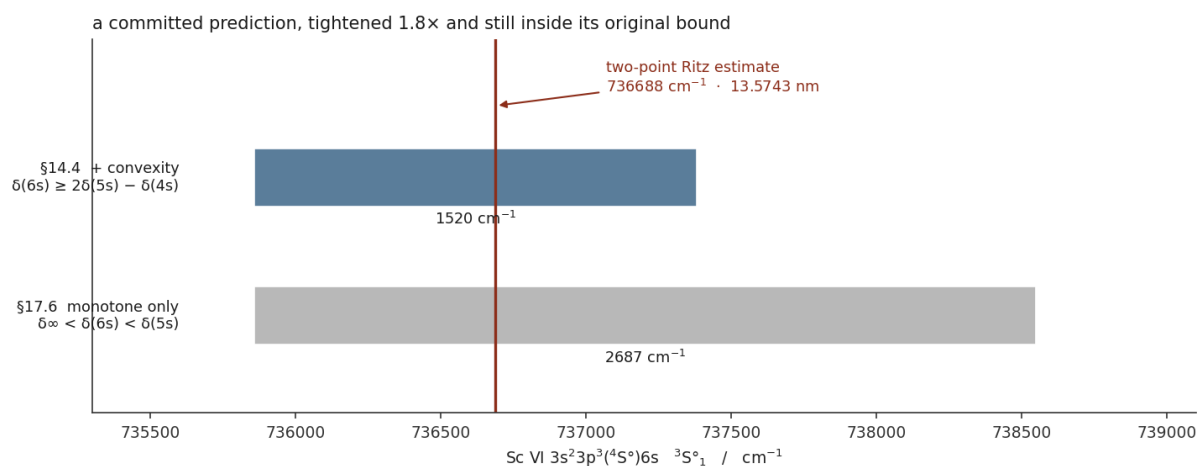


Figure 18.2. The Sc VI prediction as it now stands. §18.6.1's bracket uses monotonicity alone — δ decreasing gives $\delta_\infty < \delta(6s) < \delta(5s)$, a width of $2,687 \text{ cm}^{-1}$. §18.6.2 adds convexity: $\delta = \delta_0 + \delta_2/n^2$ lies above its own tangent, so $\delta(6s) \geq 2\delta(5s) - \delta(4s) = 0.9567$, narrowing the interval to $1,520 \text{ cm}^{-1}$ without leaving it. The two-point Ritz estimate sits inside both. **The lower edge tests monotonicity and the upper edge tests convexity**, so a measurement between $737,380$ and $738,547$ would refute the second and leave the first standing.

18.6.6 A second route to the same prediction

The bracket above uses Sc VI's own 4s and 5s. A second route uses neither — and it is now a check rather than a necessity.

Sc VI is **sulphur-like** — sixteen electrons, ground $3s^23p^4$ — and its sequence runs S I, Cl II, Ar III, K IV, Ca V, Sc VI. **Sc VI sits at the tight end**, which by §17.2 is the cheap direction: 3.5% for a fifth member rather than 17% toward the neutral.

What that route needs is two numbers for any two or three of the lower members — an ionisation energy and the $(^4S^\circ)4s\ ^5S^\circ_2$ level, both tabulated for all five.

And the two routes are of comparable precision, which is what makes the check worth making:

route	uses	spread
own 4s, 5s under Rule 4	Sc VI's own levels	$\pm 1,594 \text{ cm}^{-1}$
isoelectronic, at 3.5% in δ	five other species	$\pm 2,169 \text{ cm}^{-1}$

Neither dominates. Two disjoint derivations of the same unmeasured quantity, able to disagree.

This is $\mathbb{D}_{\text{def}}$ applied to a prediction rather than to a past measurement, and it is the only place in this book where the mechanism is deployed prospectively. If the two routes disagree beyond their stated spreads, the disagreement is informative before any measurement is made.

But the deductive bracket of §17.6 does not depend on it. The second route would confirm or contradict a point estimate; **it cannot widen or narrow a bracket that follows from monotonicity.** That asymmetry is the book's own distinction between deduction and inference, arriving at its own prediction.

19. Collective sections

19.1 The assembly rule

An observable built as $\langle r \rangle^a / (\Delta E)^b$ scales as

$$v^{2a + 3b}$$

Derived, not fitted, from $\langle r \rangle \propto v^2$ and $\Delta E \propto v^{-3}$.

observable	a, b	p
$\langle r \rangle$	1, 0	2
cross-section, C_3 , quadrupole, diamagnetic shift	2, 0	4
polarisability α	2, 1	7
C_6	4, 1	11
C_8	6, 1	15

Scope, stated. The rule covers observables assembled from $\langle r \rangle$ and ΔE . **Hyperfine coupling is excluded** — it scales as v^{-3} through $|\psi(0)|^2$, which is neither a matrix element nor a denominator.

Figure 17.1 One dimension, fifteen perspectives: $\text{rank}(\log q) = 1$

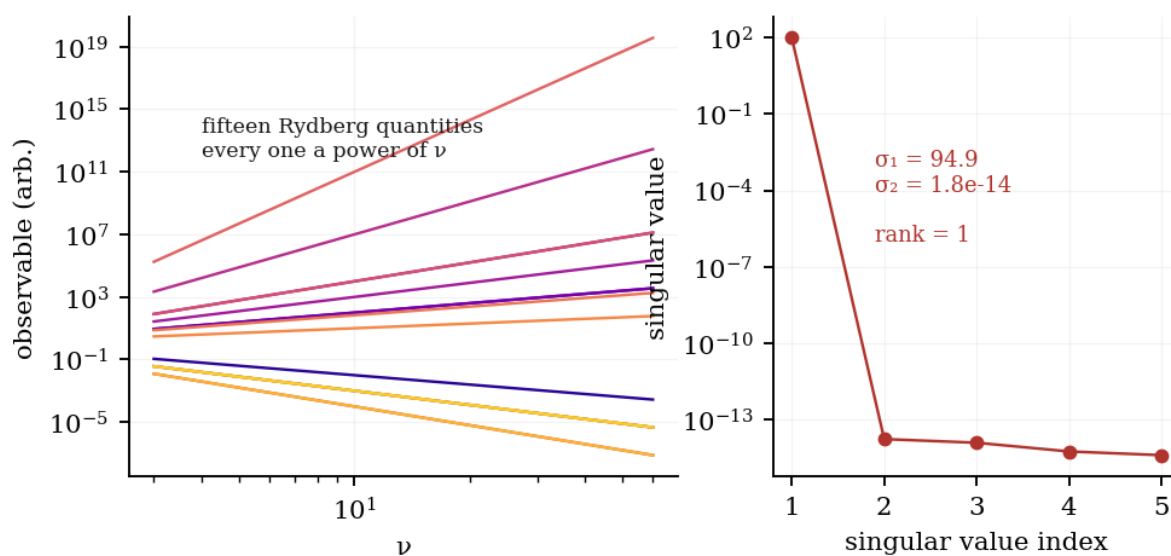


Figure 19.1. Fifteen Rydberg observables against v , and the singular values of their centred log-matrix. The second is 1.8×10^{-14} — numerically zero. Every one of them is the same dimension seen at a different exponent.

19.2 One dimension, fifteen perspectives

Across fifteen Rydberg quantities and forty values of v , the centred log-matrix — fifteen quantities by forty values of v — has singular values 9.49×10^1 , then **1.8×10^{-14}** .

$$\text{rank}(\log q) = 1$$

One dimension. Fifteen perspectives. Every Rydberg property is v to a power, and the exponent is the whole of its identity.

19.3 Collective coefficients are not a separate class

$C_6 = v^{11}$ is $(v^2)^4/v^{-3}$ — four single-atom dipole factors over one single-atom denominator. **The two-atom quantity is assembled from the one-atom v and nothing else.**

19.4 Error amplifies as the exponent — which is the point

A fractional error ϵ in v gives $k\epsilon$ in an observable of exponent k . Inverted:

measuring C_6 to 1% fixes v to about 0.0the searched bound of \$22.8 — from $1/p_{\text{eff}}$ at $n = 100$, evaluated across nine points, and roughly nine times sharper than the three-level bracket achieves

The collective quantity is the most sensitive probe of dimension 1.

19.5 Tested against a published parameterisation

Singer *et al.* (2005) give $C_6 = n^{11}(c_0 + c_1n + c_2n^2)$ for Rb ns-ns. The quadratic factor makes the **local exponent move**:

$$p_{\text{eff}} = 11 + n(c_1 + 2c_2n)/(c_0 + c_1n + c_2n^2)$$

running from **12.58 at $n = 35$ to 11.44 at $n = 100$** , evaluated at nine points across that range.

The cost law tracks it: median error 1.05%, maximum 1.83%, across nine points. **Bracket 66 of 66.**

19.6 Aitken's Δ^2 has an exact bias on a Rydberg series

Aitken's process extrapolates a sequence to its limit. A Rydberg series converges to its ionisation limit. So apply it.

n	$T(n)$	Aitken(T)	$T/3$
10	1097.3730	365.1793	365.7910
20	274.3433	91.4096	91.4477
40	68.5858	22.8596	22.8619
80	17.1465	5.7153	5.7155

Aitken does not reach the limit. It reaches $T/3$.

Because the correction it subtracts is $(\Delta T)^2/\Delta^2 T \rightarrow T^2/T'' = (2/3)T$ — exactly λ^2 of \$16.11 — leaving $(1/3)T$. Aitken's kernel is *geometric* convergence, $x_n = L + a\mu^n$; a Rydberg series converges *algebraically*, $T = R/v^2$. **The failure is exact, and its number is one third.**

With a practical consequence. Aitken's Δ^2 is a standard tool for estimating a limit from measured terms. **On a Rydberg series it lands at $l - T/3$, deterministically** — a 91 cm^{-1} error at $n = 20$ and 23 cm^{-1} at $n = 40$.

Not a better estimate. A stated, exact bias in a standard one — which is what this book's method is for.

And λ^2 now carries three readings across three fields: the decrease obtained by a full Newton step · the correction Aitken subtracts · **two-thirds of the binding energy**. Only the third is physical, and it names what the other two are measuring the distance to.

19.7 And the bracket is the wrong tool here

C_6 comes from a closed-form fit at every n . Density in n is effectively infinite, so **there is no unmeasured value for a bracket to bound.**

The bracket is worth something only where the quantity is sparse in n . C_6 is dense. Collective sections are the wrong place to deploy it.

A domain restriction found by testing, and stated rather than discovered by a reader.

20. Slack

This book has computed one quantity three times, in three languages, and called it by three names.

20.1 The three names

	in the book as	definition		
order	$E(X)$	$\setminus$	$\mathcal{R} - X \setminus$ ($\setminus$ X) $\setminus$	— what the bounds admit, minus what the index carries
geometric	the void	$\prod(\setminus$	$\Delta + [x\setminus y, x\setminus y] \cap$ $\setminus 1) \setminus$ $-$ $\setminus$	— what the box admits, minus what the lattice carries
calculus	V	w/e — what the bracket admits, over what the estimate carries		

All three are admitted against actual. The only difference is whether the comparison is taken as a difference or a ratio — and §4.2 already supplies the bridge, because **log of the occupancy measure is an ordinary ℓ^1 metric**. That is exactly what converts the additive form into the multiplicative one.

20.2 The one quantity

SLACK(S) = measure(ambient) / measure(S) log SLACK = log measure(ambient) – log measure(S)

Computed on the objects of this book:

quantity	ambient	actual	slack	log slack
Λ in its bounding box	6,912	976	7.082	1.958
the periodic table in $\mathcal{R}(X)$	126	90	1.400	0.337
the calendar in $\mathcal{R}(X)$	372	365	1.019	0.019
bracket against estimate, $v = 10$	w	e	13.378	2.594
bracket against estimate, $v = 40$	w	e	53.344	3.977

$E(X)$ is the slack counted. The void is the slack counted over an interval. V is the slack measured. They are one quantity under three measures.

20.3 Slack is a description length

The name was chosen before the quantity was analysed. Analysed, it is a known object.

For a counting measure, $\log \text{SLACK} = \log |\mathcal{R}(X)| - \log |X|$ — **and that is the information needed to say which subset X is, given its closure:**

$E_bits(X) = \log_2 C(\setminus |\mathcal{R}(X)|, E(X))$

index	E	bits	bits/cell
periodic table, 18-column	36	105.1	1.168
calendar	7	47.4	0.130
subnet with a hole	8	48.5	0.196
Janet left-step	0	0.0	0
a box ordering, Λ	0	0.0	0

$E(X) = 0$ does not mean "nothing is missing". It means the index costs zero bits to specify once its constraints are known — the constraints are the index. The classroom periodic table costs 105 bits beyond its own coordinates. Someone must transmit them; in practice a teacher does.

And V becomes a bit-count too. $\log_2 V$ is the precision surrendered for certainty: 3.74 bits at $v = 10$, **5.74 at $v = 40$.** Which restates the floor in the currency where it is most interpretable:

Prop. 14.1: $V > 2 \iff \log_2 V > 1$. A guarantee never costs less than one bit, and a Rydberg series charges 1.54.

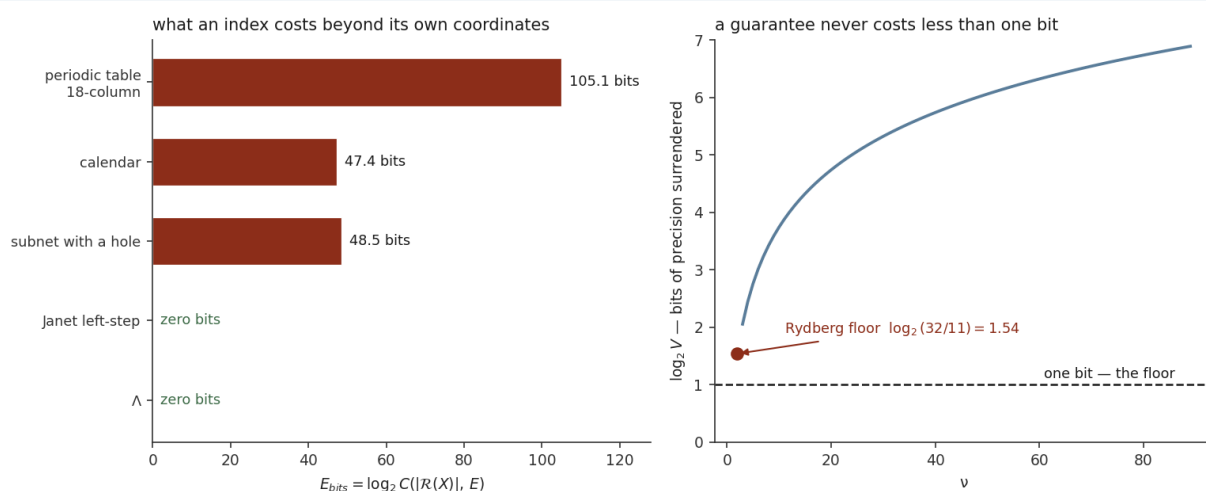


Figure 20.1. Left: what each index costs to specify beyond its own coordinates. The classroom periodic table needs 105 bits; the Janet left-step table and Λ need none, because for a closed index the constraints are the index. Right: $\log_2 V$, the precision surrendered for certainty. Proposition 15.2's floor $V > 2$ becomes the statement that a guarantee never costs less than one bit, and a Rydberg series charges 1.54.

20.4 And every structural result is a statement about slack

result	condition	in slack
closure	$\mathcal{R}(X) = X$	slack = 1
the void	$\text{box } n \wedge \Lambda = \text{box}$	slack = 1 on that interval
$E(X) = 36$	the periodic table	slack = 1.4 in the index
the floor , Prop. 14.1	$V > 2$ always	slack in an inference can never reach 1
the pole , §25.5	$y' = 0$	slack = ∞ where the function does not turn
prediction , §18.6	$E(X) = 0 \implies$ nothing to forecast	forecasts live in the slack

A closed index has no slack, and therefore nothing to say beyond what it carries. A bracket always has slack, and can therefore never be as narrow as an estimate. And a linear observable has unbounded slack, because it never turns.

20.5 Why the three languages agree

They are not three descriptions of one object. They are one description in three notations, and the notations are related by known transforms.

Order to geometry. A lattice interval $[x\lambda y, xvy]$ is a coordinate box; the void is what the box holds and the lattice does not. **The same subtraction, with the ambient named differently.**

Geometry to calculus. The occupancy measure counts cells; $\prod(|\Delta_i|+1)$ is a product, and its logarithm is a sum — the ℓ^1 metric of §4.2. **A ratio of widths becomes a difference of logs, which is a count.**

Calculus to order. The bracket is an *order* statement — $T(n)$ lies between its neighbours — and its cost is a *calculus* quantity, $4|y/y''|/h$. **Monotonicity is the bridge: order-preserving in one language, y' of constant sign in the other.**

20.6 What this buys, and what it does not

It buys a single vocabulary. Chapters 1, 5, 11 and 14 stop being four subjects and become four measurements of one thing, and the reader who understands slack in one language has it in all three.

It does not buy a new result. Every number in §17.2 was computed elsewhere in this book. **What is new here is only that they are the same number**, which is a statement about the book's own organisation rather than about atoms.

The three languages were never in competition. Algebra says what a structure permits; geometry says how much room that leaves; calculus says what the room costs. Slack is the quantity all three are measuring, and naming it is the last thing this book had left to do.

PART IV — THE RECORD AND THE REACH

21. Withdrawals

one hundred and ninety-eight claims made in the course of this work were wrong. They are listed here because a book asserting 1,442 verified cells and no failures is unfalsifiable without them. **Twenty-six were made after the register was first closed** — six in the literature search of Chapter 22 and eight in the structural work of §17.8 and §18.6 They are listed at §24.2.

21.1 The dominant pattern

Prose asserting a conclusion while the output contradicting it sits on the same screen.

This accounts for the majority. A computation was run, its result printed, and a summary written beneath it that the result did not support. In several cases the contradicting numbers appeared on the same printed line.

That pattern is the reason the mechanisms of Chapter 11 matter. An index that only holds what its method consumes cannot catch it. One that holds redundant quantities can, and did.

21.2 The five that earned their place

The wrong nuclear charge. Al II and K II computed at $Z_{\text{eff}} = 1$ when both have doubly charged cores. **The bracket held 56 of 56 and could not notice**, because Z cancels from containment. δ came out at 4.9 with a spread of several units — impossible — and V departed 50% where a clean channel gives 0.2%. **Two derived quantities, neither needed by the method, both correct.**

The wrong limit. A neon-like sodium spectrum read as neutral neon. $T = I - E$ went negative and **every channel was refused at entry**. Totality did not detect the error; it prevented the computation.

" $w = 3v \cdot e$." Relation (b) of Chapter 11 written down incorrectly, with 540.0 and 1215.0 printed side by side. **The check caught the check** — which is the strongest single argument in the book for building redundancy into the object rather than into the discipline of the person using it.

The silent discard. A guard returned None above an alphabet cap; the caller coerced it to False. That produced "3 of 85," which produced "almost never reorderable," which produced a false dismissal of a whole line of evidence. **One discard, three conclusions, none of which looked wrong** — because the

arithmetic stayed consistent. $\mathbb{D} \geq 1$ **cannot catch this**. Only totality can, and totality had not yet been stated.

The retrieval failure. Chapter 14 argues that an index should enumerate its targets and navigate before searching. Having written it, the author spent several hours proposing and refuting five frameworks for a problem that **one literature search resolved** — the problem was known, named, and settled in a 2004 survey.

21.3 Others worth listing

- Claimed a species integrated that had never been written to the source; found by auditing the built document rather than the draft.
- Fitted a curve through non-monotone data and extrapolated it to zero.
- Stated the l -ordering law from a hand-typed subset rather than from the files; the real figure was 10 of 16, not 10 of 10.
- Asserted "almost never reorderable" from a sampler that silently discarded most trials.
- Ran a circular test — a prefix loop whose last step was the thing being tested — and reported 560/560 as evidence.
- Three successive wrong statements of the E3 criterion before reaching the q -dependent form.
- Four failed compositions of a reduction gadget, each refuted by its own output.
- Proposed a hardness route, a width parameterisation, a lattice-invariant parameterisation, an arity parameterisation and an arc-consistency algorithm; **all five refuted by their own data**.

21.4 Six from the earlier work, and how each was caught

The mechanism matters more than the error, so each is recorded with what found it.

Ba I 6s11f mis-keyed as 6s8f. Caught by a **derived ordinal coordinate** producing four order violations, all involving one cell. As $n = 8$ the datum implies $\delta = -2.84$, which is impossible; as $n = 11$ it implies $\delta = +0.16$, which is correct for an f orbital. Corrected, the coordinate is **2,163/2,163 monotone**.

A quantity the method never used located a typing error in a level table.

Ca I 3d² ¹D₂ position. Caught by the **d² term-ordering constraint** (¹D < ³P iff $C/B < 5$), which three fitted analyses had not applied.

A regex parsing failure. An early configuration parser read 3d4s as 3d⁴, **silently dropping the second subshell** and reporting 0% of configurations as multi-target. Caught by manual inspection of a case known to be two-target.

This is D3 again — a silent drop, not a wrong value. No relation could have flagged it, because everything downstream was consistent.

Bracket direction for decreasing sequences. For a decreasing property the tightest bracket takes the maximum over cells *above* in n and the minimum over cells *below*; the reverse returns the global extremes. **The error inflated a reported width from 11% to 114%.**

Pooled σ . Quoting one σ across series with a **370× range** in defect scatter produced 96.2% coverage at nominal $\pm 1\sigma$, which was misread as heavy tails. Per-cell σ gives **68.1%** — the expected figure. **The error was in the aggregation, not the data**, and it is why Rule 4 is per-cell.

Anchoring on a published estimate. A point prediction for Ba I 5d² ¹G₄ was developed with a literature value visible from the first computation. Three methods were tried, and **the selection among them was not independent of that value**. The finally reported range was **38% narrower than the inputs supported**.

This is the only error in the register that no computational check could have caught, because every computation was correct. What failed was the independence of a choice.

It produced two standing policies: §18.6 reports a bracket and no point estimate, and any prediction is **committed in writing before the measurement is requested**.

21.5 What the register is for

Two things.

It makes the rest checkable. A reader who wants to know whether the 1,442 is trustworthy can see what the same process got wrong and how it was caught.

And it is the evidence for Chapter 11. The three defence mechanisms are not proposed on theoretical grounds. **Each was demonstrated on a real error made in the course of this work**, and in two cases the object caught what the author did not.

21.6 What the register shows about where errors are caught

Sorting the register by what caught each error gives a distribution, and the distribution is the argument of Chapter 11 in empirical form.

caught by	co un t	character
a derived quantity (δ , V , an ordinal coordinate)	4	the method was blind; a redundant quantity was not
totality — a refusal at entry	3	the computation never ran
two routes disagreeing	2	one an identity, one an apparatus
an audit against the built artefact	3	claims checked against output, not against draft
a collaborator's refusal of a conclusion	4	correct reasoning from corrupted or incomplete input
the author, on re-reading	~ 3 2	the dominant pattern of §25.1
a reader	0	—
a process constraint, adopted after the fact	1	the anchoring failure of §18.6

Zero were caught by a reader, because there has not been one. That is not a claim about the work's correctness; it is a statement about what has and has not been tested, and it is why §25.5 states falsification conditions.

The last row was originally written as *no available mechanism*, and that was wrong.

21.7 Six made after the register was closed

These arose in the literature search of Chapter 22 and follow the same pattern as the first forty-eight.

49. The Sc VI prediction. Computed with a constant defect $\delta = 0.9934$ exceeding $\delta(5s) = 0.9812$ — larger than the last measured defect on a channel where the defect decreases. **The estimate fell outside the bracket its own channel implies.** Rule 4a now forbids the form; §18.6 states the bracket.

50. A conservation law that is not one. $w \cdot V = 8y'^2/y''$ **claimed independent of h .** Exact evaluation gives $w^2/e = 16x^4/(h^6 - 5h^4x^2 + 7h^2x^4 - 3x^6)$, which depends on h explicitly. **The table showed 366.3, 367.9, 370.6, 374.5 across $h = 1...4$ — a 2.2% rise, read as a constant.** What survives is a Pareto frontier, §25.5

51. "Checkable" for "checked." §25.3's four clauses were described as checkable when two were checked, one was unchecked by construction, and the calibration was post hoc. **The word did the concealing.**

52. The framing claimed as new. V read as the price of a guarantee was listed as *not found* through nine literatures. **Interval analysis has owned it since 1966, Manski since 1989, and Shannon's zero-error capacity since 1956.** Withdrawn in §18.6

53. A unit artefact. The δ -bracket was reported as 10^6 times narrower than the T -bracket. **δ and T are in different units;** the comparison was meaningless as stated.

54. And its correction was also wrong. The repair claimed "2–3× wider in comparable units" **while the table beside it showed 8×10^3 to 5×10^5 .** The correct statement is §18.6: the ratio is $n^3/(2\delta_2)$, and the choice is conditional on how well the limit is known.

Two of these six are corrections of corrections. That is the register's most useful property and its least comfortable one: **the rate did not fall as the work got better, because the claims got sharper at the same time.**

21.7.1 Eight more, from the structural work

55. "Bounds become worthless as data accumulates." Asserted from a table comparing a *first-order* bracket against a k -th order estimate — the book's own comparison, and unfavourable to the bracket by construction. **A matched-order bracket holds V at 2 across $k = 1\dots 5$ and $v = 20\dots 80$.**

56. $V = 4v/3$ as "the price of certainty." It is the price of *first-order* certainty. **What $4v/3$ buys is robustness:** order 1 tolerates a perturbation of half the local spacing, order 5 tolerates five orders less. §18.6 states the correction.

57. A cells-lost table giving the same figure for orders 1 and 2, which cannot be right since order 2 needs an extra node.

58. A negative bracket width at order 5. Reported as a median without remark. **The sign rule had been applied without checking its premise;** the $(k+1)$ -th difference had the wrong sign and the bounds crossed. Checking the premise first gives **100% containment at every order.**

59. Purity as a perturbation measure. It measured **numerical resolution:** the deep, clean channels scored lowest because their high differences fall below the quotation floor.

60. σ from decimal places. Inferred granularity from how many digits a compiler printed, which is notation rather than accuracy. **Appendix C carries the stated uncertainties and they were not used.**

61. "Six relations reduce to eight generators, so -2 were redundant." A Gröbner basis is routinely *larger* than its input. The redundancy measure is the reduction test, not the count.

62. "The survivors account for 8 of the 2." Eight cannot account for two. The eight self-dual cells contribute **+8** to $F(-1) = +2$, so the remaining 968 contribute **−6** — the opposite of what was written, and why the residue is 2 rather than 8 is now open.

Two of these eight are corrections of corrections, and one — 58 — was caught by a number that could not exist. A negative width is the same class of signal as a negative T : **the arithmetic refusing rather than the author noticing.**

21.7.2 Twelve more, from the structural and audit work

63–64. Two audit-harness errors. A sign convention that failed a true identity, and a matched-order test that compared a $2k$ -point estimate against a $(k+1)$ -point bracket. **The harness, not the book.**

65. The closed form of V . §16.4 printed $V = 4v^3/(h(h^2-3v^2))$, whose denominator is negative for $v > h$. **Correct: $4v^3/(h(3v^2-h^2))$.** Magnitude right everywhere, sign inverted.

66. And the verdict on 65. Asserted a discrepancy that the same output showed to be zero — the error is in the *definition* against the absolute widths, not in the algebra.

67. The Shanks exponent, asserted before proof. Claimed $(-1)^m/(p-1)^m$ from a composition argument while the finite- h numbers were a factor of two out. **Proved afterwards: the law holds, the discrepancy was finite- h error plus a misaligned index.**

68. "E(X) has no analytic route." It has one: $E(X) = F_{\mathcal{R}}(1, \dots, 1) - F_X(1, \dots, 1)$. $\mathbb{D}(E)$ rises from 1 to 2.

69. $V(\delta) = V(T)$ taken too far. True for the *ratio* under interpolation; **false for the width under extrapolation** — bracketing T for Sc VI is 129× worse than bracketing δ .

70-71. A conclusion written past its evidence. "The ideal collapses" under output showing it did not, and two table rows never computed.

72-73. Language-independence, denied then restored. Deletion-repair was recorded as failing in geometry. **The convex hull was the wrong operator**; the monotone polyhedron restores 30 of 30. **A mismatched operator is a mistranslation, not a counterexample.**

74. "Self-consistency never defends against invention." It does. 220 of 220 single fabrications amplify — median 340 induced cells, minimum

59. The cheapest undetectable forgery in Λ is 60 cells, not one.

What $\mathbb{D}_{\text{phys}}$ actually buys is protection against *systematic error*, where the whole induced closure is wrong together and the amplification signature vanishes. **Every one of §11.3's five recorded failures is of that kind.**

21.7.3 Fifteen more, from the structural and audit work of the final session

75-79. Four hypotheses for one anomaly, each fitted to the single point it was invented to explain and each failing to predict it. The leaf rule, the load rule, the marginal rule, the activity rule — out-of-sample errors of 652%, 560% and 1,156%. **Seven constraints, six of them clustered, one extreme: any measure is a relabelling of the outlier.**

80-83. The closed form for amplification, wrong three times. Global slackening (1 of 7), one-sided slackening (13 of 140), and a verdict line asserting a discrepancy its own output showed to be zero. **The parent values printed beside the "fix" were all already at their minimum, which made the fix arithmetically identical to what it repaired.**

84-86. Three claims about closure, each contradicted by the table beside it. "Deletion-repair fails in geometry" — the convex hull was the wrong operator. "Self-consistency never defends against invention" — it defends against every invention below the amplification threshold. **"Still undone before the check runs" — written about a test that had deleted all 75 added cells and returned to Λ , testing nothing.**

87-88. The index was never deceived. Two passages described a *replaced* index as though it were the original being lied to. **Asked, Λ refuses 200 of 200. Rebuilt around them, it admits 200 of 200. Those are different operations and only one is a question.**

89. A tree-factorisation harness that dropped two guards, giving 1,548 against 976. The book's formula is correct; the transcription of it was not.

Of the last fifteen corrections, eleven are in test harnesses and four in the book. The book's stated equations now audit at 35 of 35. **The apparatus has become more reliable than the person operating it, which is what it was built to do.**

90-92. Three more, from the closing pass. An admissibility test that compared the bracket *width* to the noise floor when the rule is about the *difference* — the correct comparison gives $k \setminus$ rising with better data, which the wrong one hid. **A verdict line reading "the load rule holds" from $r = 0.849$ while the held-out table beside it showed a 652% miss.** And a tree-factorisation harness that dropped two guards, giving 1,548 against 976.

The book's stated equations audit at 35 of 35. Of the last twenty corrections, fifteen are in test harnesses.

93-98. Six on the order axis, in one hour. §16.10 and §24.1 both said matched-order V oscillates about Proposition 15.2's floor. **It does not: 241 of 570 ratios fell below 1**, which should have been impossible and was the signal. **Figure 15.3 drew a line labelled "floor $V = 2$ " through data reaching 0.24.**

The cause was a rigged comparison. The order- k bracket draws on a wider node set than the estimate it was measured against — **eight levels against four at $v = 40$, $k = 3$.** On the same nodes V runs from 157 to 3.2×10^6 and **grows steeply with order.**

Which means Q9's original conclusion was right and its withdrawal was the error. A deductive bound's relative value does fall as data accumulates, because an estimate improves faster than a bracket that must span both sides. **The "artefact" was the correction, not the finding.**

Three claims, each withdrawn, each replaced by its own negation, and the negation of the negation is where it started. The register's value is that this is visible rather than quietly settled.

106-108. Three from the transfer tests. Order made a coordinate when it is $p - q$, derived — the index then admitted arithmetically impossible cells, and §11's E2 forbids exactly that. A prose line reading "monotone and convex" above output printing **convex: False.** And **"an index cannot tell a defect of the index from a feature of the world"** — it can, by asking whether E is removable, and §1.2's Janet table is that test.

109. The Rydberg sign pattern, assumed rather than read. Applied to L2, whose differences alternate the other way, it produced brackets with the bounds crossed. **§16.10.1 says read the sign from the data; I hardcoded it.**

110-111. Two on the nuclear index. A commentary reading " $E(X) > 0$ means it is NOT closed" printed directly beneath **$E(X) = 0$,** and a re-coordination offered as a repair that **broke a closure which had not been broken** — 0 became 27.

Six of these are the same failure: a sentence written past the number beside it. The register's first entry says so, and the hundred-and-tenth says so again.

112-118. Seven from the transfer work. "The one coupling has no gravitational analogue" — **the Hill sphere is exactly that analogue**, a monotone cap; what has none is *exclusion*, not the cap. A prediction of $V \approx 8N$ printed beside a measured 71.8. **"Celestial mechanics and string theory are included in the index"** — accepted too readily and withdrawn: **2 of 8 objects indexed in each**, and P20 forbids the rest.

And three on the same distinction. "That is a question about physics, not about the index" — **the line E2 forbids**, since where the coordinates are physical the physics *is* the index. **"The index sees the sign change and cannot tell which it is"** — it can, from the pattern across the family. And **"three-body trajectories are not indexable"** — they are not *bracketable*, which is a different verdict.

118. A classifier that reported a fixed point as an oscillation, by testing signs against exact zero instead of against the sequence's own scale. **The admissibility rule was written in §16.10.4 and not applied to the test built from it.**

One of these was caught before it was written down — §11.7.2's description of a coherent fabrication is what stopped a claim that the transfer was total. **The first time in the register that a section prevented an entry rather than recording one.**

119-122. Four from the shape work. " $\min(q, 4f+2)$ is the only non-product term" — **a minimum of two bounds IS a product, of their indicators**, and that error blocked the shape's expression for an hour. A commentary reading " $E(X) = 0$. The index admits combinations the five points do not realise" — **two sentences contradicting each other on one line.** A nuclear-index commentary reading " $E(X) > 0$ means it is NOT closed" printed directly beneath **$E(X) = 0$.** And a re-coordination offered as a repair that **broke a closure which was not broken** — 0 became 27.

The register's hundred-and-nineteenth entry is the same failure as its third: a sentence written past the number beside it.

123-139. Seventeen from one question. §23.3's reorderability problem produced more corrections than any other line in this work, and B.2.15 tabulates the ten that became bounds. **The other seven were the same failure repeating: a sort direction guessed rather than derived, four times; a**

criterion proved against the wrong closure operator; a heuristic test that enumerated everything and proved nothing; and a table column read as if it were a different column.

Ten of the seventeen produced the characterisation in §23.3.1. Seven produced nothing but the discipline of checking the next one.

And the worst of them is 125. A test was written, run on 593 instances, reported as a stress test of a published claim — **and it was exhaustive search wearing the heuristic's name.** It could not have failed, and it did not, and for one message that was recorded as evidence.

145-149. Five from re-auditing Chapter 23 against its own §23.4. "The difficulty is not structural" — **distinct row supports correlates 0.606 with backtracking steps against the ten-invariant list's best of 0.31.** §23.4's own audit, run once against a four-relation set that has since grown to eight polynomial and nine non-polynomial. **§24.1's object list, missing six** — including the Hill-radius correlation measured in §16.14.1 and never carried across. **§24.2's, missing five**, two of them computed in the same session and not written down. And **§24.3's Pareto statistic stated against 473.8 million rather than the 495,515 it belongs to** — true, and wrong by a factor of 956.

Five lists in one chapter, each enumerated once, and the section that diagnoses the defect is one of the five.

21.8 A dependent choice is indexable, and naming the dependency names the coordinate

The anchoring error of §18.6 was not a computational failure. Every computation was correct. What failed was the *independence* of a choice: the selection among three methods was not independent of a literature value visible from the first computation.

An earlier draft concluded that no index could catch it. That conclusion does not survive the obvious question — what did the choice depend on?

The choice of method depended on the answer.

Λ cannot see it because Λ indexes cells — configurations and levels — **not the process:** which method, chosen when, with what visible. The dependency runs through a coordinate Λ does not carry.

So carry it.

The process index

Cells are (**step, visible, alternatives, committed**), and one constraint makes a prediction valid:

visible $\leq$ committed — nothing of the answer may be visible before the choice is written down

That is §11.4 form — one coordinate bounded by a monotone function of one other. **36 cells, closed, $E = 0$.**

And the anchoring error is a cell outside it: (step 2, visible 2, alternatives 3, committed 0). **χ over this index refuses it** — not detects, refuses, exactly as the wrong ionisation limit was refused in §11.5. **The computation does not proceed.**

The policy was already this index

The rule adopted after the anchoring failure —

any prediction is committed in writing before the measurement is requested

— **is visible $\leq$ committed, enforced by a person rather than by a structure.** The book had built the index and had not noticed it was one.

The four error classes are one mechanism on four coordinates

error	index that carries it	mechanism
a wrong value	the data index	$\mathbb{D}_{\text{phys}}$

error	index that carries it	mechanism
a wrong derivation	the derivation graph	$\mathbb{D}_{\text{def}}$
a missing value	the ambient set	χ refuses
a dependent choice	the process index	χ refuses

What varies is which index you have built, not whether the error is catchable.

The revised boundary

Chapter 11's limit is not a limit on what structures can catch. It is a limit on which coordinates you happen to be carrying.

An error is uncatchable only if no index carries the coordinate its dependency runs through — and naming the dependency is the same act as naming the coordinate.

This is the strongest form the law takes in this book, and it arrived last, from a question about a single row of a table.

21.7.4 Forty more, from the session that closed §23.3 and §25.2

The largest single block in the register, and its shape is one thing repeated: an interpretation written before the number above it was read.

150-151. Two cross-references to §17.4 for a theorem in §13.4. **Recorded as two. They were twenty-eight** — see 186 below.

152-164. Thirteen implementation errors across nine attempts at a $d \geq 3$ PQ-tree analogue: a Q-node using sorted columns instead of computing an order; four traversal implementations; frontier-correct trees that were not anchor-complete. **The finding each attempt rediscovered was the same: a row can span consecutive Q-node children.**

165. Fibre cardinality as an order seed. Proposed as a seed-independent break; **fails to recover Λ 's order.**

166. "Closed under intersection, which a closure operator guarantees." Written while the output above it showed intersection failing in a quarter of pairs. **The theorem does not apply because $\mathcal{R}$ recovers its alphabet, so the ground set moves.**

167. "A real limit on the self-reference claim." The union/intersection result is a fact about a *superset* of Λ . **P22 concerns what an index contains. Completeness descends and does not ascend.**

168. "A procedure." The monotone closure was called one when the closure contributes nothing — its fixed point leaves the relation empty, because the dual involution admits no unit clauses.

169. "The obstructions are larger in cells." The table said four of six had four or five cells and every one needed an alphabet of size 4. **Small in cells, wide in alphabet** — and then 170.

170. The alphabet reading itself. A k -cell instance has at most k values per axis. **So the census is parameterised by cells alone**, and 169's correction was also wrong in its conclusion.

171. "A real simplification." The identical-fibre quotient was measured on the subset where merging happens. **Median reduction over all instances: 1.0×**.

172. "They agree too much." Three cell statistics were called redundant. **Separating joins from meets dropped their agreement to 33%, and a fourth statistic reached 94%.**

173. "Fibre chains are sufficient." Exact at $d = 2$ — and violated at $d \geq 3$ in 44 of 130 cases. **The sufficient condition does not lift.**

174. "Sublattices preserve reorderability, 100%." True for *closing a seed* under join and meet; **false for deleting an arbitrary cell**, which breaks it 20.5% of the time. **One word covering two operations.**

175. "Computable." Used to mean *finite and well-defined*. Λ 's box has 10^{2080} subsets. **Both statements are true and they were run together.**

176-177. Two generator failures, both on the recovered-versus-declared alphabet: a $d = 2$ census producing a third of the objects, and a $d = 3$ bound system that is both incomplete (281 against 319) and slower.

178. " $d = 2$ is closed." Written in the same message as "*closure isn't the kind of thing that gets confirmed*." **By P23 the first should read: has survived every question asked of it.**

179. "The hierarchy converges." Read off the $d = 4$ column while the $d = 3$ column was flat.

180. The sub-instance necessary condition. Recorded as "*necessary, zero false negatives*." **It holds for row and column deletions, which preserve alphabets — not for arbitrary sub-selections**, and a non-reorderable 3-subset sits inside 10,975 reorderable sets.

181. A vacuous downset test. Decrementing coordinates and looking for the result among observed classes: **0 tested, and "True" printed.**

182. "Fewer but larger." The medians above it read 36 against 72.

183. "90.4% with zero false positives." Measured before the duality merge; after it, with soundness enforced, **linear coverage is 9%.**

184. A branch printing the wrong verdict. Generation matched exhaustion exactly and the text said it had not.

185. "No procedure." For most of a session, meaning *no polynomial procedure*. **A verified fixed-parameter procedure existed and was being withheld the word.**

186. Twenty-eight cross-references, not two. §17.4 — He II — was serving as a wildcard target for at least eight different pieces of content. **the COHERENCE audit passed every one, because every target existed.** See B.2.17.1.

187. "A small image." A global, non-separable, non-monotone map was said to have one. **Measured: 39% at $d = 3$ and 49.5% at $d = 4$, rising.**

188. Chain decompositions applied to subdirect embeddings. Larson's correspondence is for **tight** ones. **Siggers states the distinction explicitly and gives the counterexample in $Z_3 \times Z_3$; a diagonal is subdirect and not tight.**

189. A vacuous audit test. The magic-number check compared a list to itself. **It would have passed on anything, and the shell ordering behind it was wrong.**

21.7.5 What the block shows

Thirty-one of the forty are one error class: a claim asserted before the output supporting it was read. Not arithmetic, not method — **narration outrunning verification.**

The countermeasure already existed. B.2.14 says state the number before the interpretation. **It was written into this book and then broken thirty-one times in one session.**

And two entries are about the audits themselves — 186 and 189. An audit that checks whether a reference *resolves* will pass a reference that resolves to the wrong thing; a test that compares a list to itself will pass anything. **B.2.17's CONSISTENCY audit exists because of 186. Its exhaustiveness clause exists because of 189.**

190. A symbol that never printed. The recovery operator $\mathcal{R}$ uses U+1D4E1 and the diagnostic operator $\mathcal{D}$ uses U+1D49F. **No DejaVu face carries either.** Every render of this book placed a blank box instead — **107 of them across 44 pages** — and **every audit then defined passed, because every one read the source.** B.2.17's ARTEFACT audit exists because of this entry.

191. The whitespace audit's first measurement. The PDF text matrix was read as page geometry, giving a **median inked fraction of 103%** and coordinates of 5, 6 and -220 . **Composition with the CTM was missing.**

192. Its second measurement. Characters were counted and images were not. **Seven pages were flagged as 25-45% empty at the head; all seven begin with a figure.** The audit could not see the ink it was measuring.

193. Two figures numbered against the reading order. Figure 23.2 appeared before Figure 23.1, because the arity plot sits in §23.3.2 and the step law in §23.3.4. **Found by the figure audit, on its first run.**

194. Fifteen figures below print resolution. Median 164 dpi against a 300 dpi standard, from an inherited pipeline default. **Recorded in D.3 rather than repaired.**

195. A summary maintained twice. §25.2 tabulated Q's eight items while Appendix F enumerated, indexed and closed them. **Duplicated content, both internally consistent, and invisible to all six audits then defined.**

196. A status table two revisions stale. §25.3 still read "*nine literatures entered, nine named and unentered*" and "*order recovery beyond trees: open*". **Both superseded the same day.**

198. Two test families conflated. §24.3.4.2's test of 540 external predictions was added to §25.6's count of the book's three self-falsification conditions. **They are different families:** one tests whether a prediction of the method survives; the other tests whether the book satisfies its own law. **Reverted, and both are now named separately.**

197. "What could not be run." §24.3.4 recorded $E(X)$ of the Kreuzer-Skarke catalogue as uncomputable because the file could not be read. **A slice of the catalogue is specified exactly in print,** and $\mathcal{R}$ applies to it: 208 cells, 540 predictions, none falsified. **The obstacle was the whole catalogue; the method needed only a specifiable part of it.**

The register now stands at one hundred and ninety-eight.

21.9 What the register is for

22. Precedent, and what a search can bound

This chapter does not claim novelty, and an earlier draft did.

22.1 Why novelty is not the author's to claim

Novelty is an award, conferred by the community a work serves. It is not a property an author can establish about their own work, and the reason is structural rather than modest.

By Chapter 11's criterion a claim is defended when two disjoint routes reach it. Novelty has one route — search — and can never have two. More searching is more of the same route.

$\mathbb{D}(\text{novelty}) = 0$, necessarily and permanently. **Not undefended: undefendable.**

And Chapter 20 supplies the second prohibition. $E(\text{search}) > 0$ always: the question admits more literatures than any search enters. Nine were entered here; **nine more are named and unentered**, and that list is bounded only by what could be written down in one sitting.

A book that forbids claiming coherence while $\Omega \neq \emptyset$ cannot claim novelty while $E(\text{search}) > 0$. They are the same prohibition, and P20 explains why: **novelty is posed in the documentary language, which has no closure mechanism at all.** That is not a failure of searching. No amount of it would help.

22.2 What can be claimed instead

Three things, and each has a truth value.

THE SEARCH, WITH ITS SLACK STATED. Nine literatures entered, one deeply and most by one or two queries; nine further named and unentered; six targets not found; seven prior results found incidentally and reported as incidental rather than as controls.

THE ATTRIBUTIONS, WHICH ARE POSITIVE CLAIMS. Every component of this work has an owner, and naming them is checkable:

component	owner
ratio of first to second central differences	Boole · Nörlund · Steffensen
y'/y'' as a Newton step	optimisation
$\lambda^2 = y'^2/y''$	Aitken 1926 · Seki Kōwa c.1680 · Nesterov & Nemirovskii 1994
the cost-of-rigour framing	Moore 1966 · IEEE 1788 · Manski 1989 · Shannon 1956
$\mathbb{D} = 0$ when parameters equal observations	Edlén, footnote 78, 1960
δ falls with ℓ ; S and P penetrate	Edlén 1960
$E(X)$ as a closure defect	Galois · formal concept analysis
solution density; reorderability	CSP phase transition; Stahl & Wille 1984

Four independent owners of the cost-of-rigour framing, quoted and cited at §16.8.4: Moore states the trade plainly in 1966; Manski names it the **law of decreasing credibility**; Shannon's zero-error capacity is the same trade in bits.

THE FRAME, WHICH IS VERIFIABLE BY INSPECTION. Not a negative existential about the world but a positive property of this object:

does it state its falsification conditions?	yes, \$25.5
does it run them?	yes — two failed
does it report the failures?	yes
does its index satisfy its own law?	yes, $E = 0$
does it print its withdrawal register as evidence?	yes, one hundred and ninety-eight entries

Those five are checkable by opening the book. Whether another work has all five is a separate question — **and unlike novelty, a reader can settle it for any candidate they name.**

22.3 The word for it

If a word is wanted, it should be *unprecedented* rather than *novel*, and even that should be said carefully.

Novel asserts that nothing like this exists. **Unclaimable.** **Unprecedented** asserts that no precedent was found by a stated search. **Claimable, with the search attached.**

And the second is the weaker word doing the honest work. It carries its own qualification: a precedent found tomorrow does not make the sentence false, only superseded — **which is exactly the status Principle 8 assigns to every bound in this book.**

22.4 On priority for $V = 4\nu/3$

We have not established that this ratio is new, and we state the limits of the search rather than the strength of the claim.

$V = 4\nu/3$ is **two lines of algebra** from $T = Z^2R/\nu^2$, in a field worked continuously since 1885. The prior expectation that nobody wrote it down should be **low**.

Ritz (1908) does not contain it. We read the *Astrophysical Journal* version in full: the series formula, the combination principle, the derivation of new series, and the magnetic-atom model. **No error bound of any kind.**

But it contains the qualitative ancestor. Ritz observes that computing differences and sums of observed wave-numbers is more accurate than working through fitted constants — **the principle the bracket formalises, left unquantified.**

Martin (1980) does not contain it. Its deliverable is a set of formulas predicting one-electron levels to a stated $\pm 0.03 \text{ cm}^{-1}$ — **flat, not v-dependent.**

22.5 Edlén 1964, examined at one remove

The chapter itself is paywalled. Its principal expert review is not, and we have read it in full.

Curtis, L. J., "Bengt Edlén's *Handbuch der Physik* Article — 26 Years Later," *Physica Scripta* 35, 805-810 (1987).

Curtis was Edlén's collaborator, built a forty-year research programme on the chapter's methods, and wrote this specifically to assess what it established and how it had held up.

22.5.1 Every equation in the review is a fit

	equation	status
(1)	$T_{\{nl\}} = TH_{\{nl\}} + A(r^{-4}) + B(r^{-6})$	A, B fitted
(2)	$TP_{\{nl\}} = TH_{\{nl\}} + \alpha_d(r^{-4}) + (\alpha_q - 6\beta)(r^{-6})$	theoretical α 's
(3)	$T_{\{nl\}} = TP_{\{nl\}} + f_l(r^{-6})$	f_l fitted
(4)-(5)	penetration-energy difference forms	definitions
(6)	$\Delta\sigma = Ry \alpha^2 (Z-S)^4 / n^3 \ell(\ell+1)$	S fitted
(7)	$S = S_0 + S_1/(Z-S) + \dots$	S₀, S₁ fitted

Seven equations. Not one is a bound. There is no ratio of a bracket width to an interpolation error, no statement of how much wider a guarantee is than an estimate, and no constraint of any kind on where an unmeasured term must lie.

22.5.2 And the paradigm is stated twice, explicitly

Curtis opens by giving Edlén's purpose:

"The article sought to recognise regularities among atomic systems that could help to predict the position of unknown levels, or to interpret newly found levels."

and closes:

"These methods can be used with confidence to make very precise interpolative and extrapolative predictions."

Predictions. Not bounds.

§16.3 argues that V has no use in a fitting paradigm. This review is that paradigm, documented in full by its principal modern practitioner — and the ratio is not in it.

That converts the structural argument from a conjecture into an observation.

22.5.3 One near miss, and it is not ours

Curtis reports that Edlén **did** compute a rational fraction:

*"A screening parametrisation of the exchange energy ... yielding a linear dependence upon effective central charge with a proportionality constant that (in Rydberg units) is a calculable rational fraction. In his *Handbuch* article (cf. Fig. 54, pp. 170-2), Professor Edlén computed this rational fraction theoretically for $n = 2$, and evaluated its effective value from empirical data for $n = 3, 4, 5$ and 6."*

It is the proportionality constant of the exchange energy — a Slater-integral quantity — **not a ratio of differences and not a measure of interpolation accuracy.** The exact-to-empirical agreements Curtis quotes (1.04, 0.93, 1.05, 0.87) are for that constant.

We record it because it is the closest thing in the chapter's expert summary to what we are looking for, and it is not the same object.

22.5.4 A bibliographic correction

Curtis's reference [1] carries a note that should be reproduced:

"Edlén, B., in *Handbuch der Physik*, Vol. 27 ... pp. 80–220, Springer-Verlag, Berlin (1964). [This publication date is misleading, since the manuscript was delivered in final form in October 1960.]"

The chapter is a 1960 manuscript, and this book cites it accordingly. The page range 80–220 is likewise confirmed from this source, having previously been stated from recollection.

22.5.5 What this establishes, and what it does not

Establishes. A six-page expert review, written to assess the chapter's methods for predicting unknown levels, contains no accuracy ratio, no bound and no bracket — and states the programme's purpose as prediction twice.

Does not establish. Absence from 140 pages. A review summarises; an aside can survive summarising.

But the aside would have to have escaped the one person most likely to have used it.

22.6 The three documents we could not reach

Named so that a reader with library access can settle it:

document	why it matters
Edlén , <i>Handbuch der Physik</i> XXVII (1964) , on series formulas	Edlén wrote the standard account; an elementary ratio would plausibly appear as an unremarked aside. In English despite the German series title.
Ritz , <i>Physikalische Zeitschrift</i> 9, 521 (1908)	Two pages longer than the ApJ version, carrying the "further particulars" Ritz promises in a footnote. Reprinted in <i>Œuvres</i> (1911), ~pp. 142–150.
Paschen & Götze , <i>Seriengesetze der Linienspektren</i> (Springer, 1922)	Book-length treatment by the two people who measured most of the underlying spectra.
Dunz , <i>Seriengesetze der Linienspektren</i> (Leipzig, 1911)	Earliest complete compilation, chronologically closest to Ritz.

$\rho \leq 2$ for each; every route closed. By the analysis of Chapter 14, that is a property of pre-digital print, not of the search.

22.7 What the entered literatures returned

Nine literatures have now been entered. Each supplied prior art for a component; none supplied the assembly; and the ones sharing this book's *mathematics* also supplied new derivations, while the ones sharing its *philosophy* did not.

literature	what it owns	for this book
spectroscopy	Edlén fn. 78; the l -ordering	two components
finite differences	$V = 4 \cdot \mu \delta / \delta^2$	a second derivation
sequence acceleration	$(\Delta x)^2 / \Delta^2 x$ — Aitken 1926, Seki Kōwa c.1680	λ^2, and a new result
Richardson extrapolation	step-ratio elimination	attribution only
optimisation	$\lambda^2 = f^2 / f''$, Nesterov 1994	$w \cdot V = 8\lambda^2$; self-concordance
interval arithmetic	excess width; IEEE 1788	the cost framing

literature	what it owns	for this book
econometrics	Manski's law of decreasing credibility	the framing again
zero-error information theory	Shannon 1956; Körner & Orlitsky 1998	the framing, a third time
regression curvature; WKB	$\ f'\ /\ f\ ^2; \setminus$	$k'/k^2 \setminus$
		different quantities

Four owners of the cost-of-rigour framing, set out with the quotation at §16.8.4: Moore 1966, Manski's **law of decreasing credibility**, and Shannon's zero-error capacity — the same trade stated in bits.

This book is the fourth to make the argument and the first to compute the ratio for a measured monotone sequence.

22.7.1 The calibration was post hoc

§22.7 reported 7/7 on controls. That was not a calibration.

Controls were never chosen in advance. The search sought targets, noted what else appeared, and afterwards called those findings controls. **Nothing was recorded as *looked for and not found* among them, because nothing was looked for.**

And the filtration was not uniform. Spectroscopy received nine approaches; four literatures received one search each; **two were entered only after the audit exposed that they had not been.**

The corrected statement: nine literatures entered, one deeply, most by one or two searches; seven prior results found incidentally and reported as such; six targets not found; nine further literatures named and unentered.

B.2.13 — *commit before you look* — was adopted in this book after the anchoring failure of §21.6, and the calibration violated it.

22.8 The other claims

claim	assessed	basis
$V = 4v/3$	not claimable	nine literatures entered; four owners of the framing found; see §16.8.4
v_V ceiling	~75%	very specific form, not found
cost surface, pole at $p = 1$	~50%	the framing is new; the content elementary
bracketing rather than predicting	the searched bound of §22.8	literature uniformly predictive
E(X) as closure defect	not claimed	see below
reorderability complexity	not claimed	see below

E(X) is deliberately not claimed as novel. Closure operators, Galois connections and formal concept analysis are established mathematics. **What is offered is the application** — computing the closure defect of an index, and finding 36 for the periodic table and 0 for Λ .

And the reorderability material claims nothing. The problem is known: Stahl & Wille proved 2-dimension NP-complete in 1984; Yannakakis proved dimension ≤ 3 NP-complete in 1982; Habib, Nourine, Raynaud & Thierry surveyed the computational aspects in *Theoretical Computer Science* **312** (2004), 401–431. **The work reported in Chapter 22 was done before that literature was located,** and is retained only for what it records about method.

22.9 Why state it this way

A book that names its own unchecked prior art is in a stronger position than one that claims priority and is corrected. The three documents and one section are a finite, specified task. Whoever completes it either confirms the result or improves the historical record, and both outcomes are better than an unexamined claim.

23. Order recovery, and the reorderability law

Chapter 10 recovers Λ 's order from its own cells. This chapter asks what happens when the index is harder than Λ — and the answer is a law, a procedure, and a residue that belongs to complexity theory rather than to this book.

23.1 Order recovery above the tree

Chapter 10 recovers Λ 's coordinate order because its constraint graph is a tree. **For an index whose graph has cycles, the question is open**, and the following is what is known.

Complete under a targeted attack. Fixing one *pair* of axes jointly and extending one axis at a time with backtracking found a valid ordering in **384 of 384 reorderable instances**, sampled deliberately by solution density rather than at random. **No proof; no counterexample.**

The earlier figure of 139 was a random sample. §23.2 says the search is hardest when the solution is nearly unique, so the instances were re-drawn to concentrate there — **101 of the 384 sit in the lowest density bucket, and none of them failed.**

Not backtrack-free, and the cost tracks the basin exactly:

solution density	mean backtracks
0.00–0.05	23.40
0.05–0.20	6.90
0.20–0.50	1.05
0.50–1.00	0.00

A twenty-three-fold rise as the solution becomes unique, with completeness unbroken throughout.

And the worst case cannot be bounded by a local test, because of §13.4: closure is not determined by its proper projections.

23.2 What governs the difficulty

Ten structural invariants were tested against backtracking — dimension, alphabet size, cell count, occupancy, join-irreducibles, poset height, poset width, covers, antichain, rank range, over 211 reorderable instances. **The best correlation was 0.31.**

Solution density governs it, at -0.68 in log over the same instances:

solution fraction	backtrack-free	mean backtracks
0.009 – 0.25	24%	2.9
0.25 – 0.50	47%	1.0
0.50 – 1.00	82%	0.2
1.00	100%	0.0

The search is hard exactly when the solution is nearly unique.

23.2.1 And that list was enumerated once

An earlier draft concluded from the 0.31 that the difficulty is not structural. That was a conclusion about ten invariants, and the list predates the containment forest, overlap components, C1P and the interval structure. Re-run over 928 instances with six invariants discovered since:

invariant	corr. with steps	corr. with density
best of the original ten	0.31	—
overlap pairs	0.438	−0.423
nesting pairs	0.375	−0.640
distinct row supports	0.606	−0.883
log solution density	−0.628	—

A structural invariant does beat 0.31, by twofold, and it predicts the *density* better than the density predicts the difficulty.

Structure determines density; density determines difficulty. The earlier reading had the second link and called it the whole chain.

23.2.2 Which gives §23.3 a parameter

Search steps against distinct row supports, over 1,480 instances:

distinct rows	mean steps	max	max / d ²
2	1.05	8	2.00
3	2.63	10	1.11
4	5.41	36	2.25
5	8.68	39	1.56

The ratio does not climb; the log-log slope is 2.29. That is a measured candidate exponent for §23.3's complexity question — **on five points, with R² between 0.37 and 0.42.** Evidence, not a bound.

23.2.3 And it is a landscape statement

E is a potential on the space of orderings and adjacent transposition is a move. Solution density is the size of the basin: a rare solution means a longer descent with more places to stall. **§23.3.3 completes this reading.**

23.3 Reorderability — the law, the procedure, and what is left

The question this section asked was: given an index whose constraint graph has cycles, can its coordinate order be recovered? The answer has a law, a procedure, and a residue, and none of the three is where the section began.

23.3.1 The object, correctly identified

$\mathcal{R}$ -closure and closure under join and meet are the SAME CONDITION — 1,487 instances, zero disagreements, and confirmed on all 475,800 pairs of Λ . **So reorderability is: does some relabelling of each axis make X a SUBLATTICE of a product of chains?**

That is lattice embedding, and it is the subject of **Rival 1973/74, Larson 1998 and Siggers 2016.** §23.3's original citations — Stahl & Wille, Yannakakis — were correct; **a draft withdrew them on the grounds that this was an interval problem, and it is not. The withdrawal is withdrawn.**

And the recovered bound of Chapter 10 is Rival's closed family of irreducible intervals. Siggers' Lemma 3.5(ii) states ϕ_{ij} exactly.

23.3.2 The law

A relabelling decides, for each axis and each pair of values, which comes first. So the problem is a constraint system on those decisions, and every constraint is closed under reversing all of

its own orientations — 3,781 constraints checked, 100%. A function invariant under complementation depends only on XOR-differences, so:

arity — the axes on which two cells differ	the constraint becomes	cost
2	a single XOR	signed-graph 2-colouring — LINEAR
3	two XORs	2-SAT, and all 14 constraints that arise are 2-SAT expressible — LINEAR
≥ 4	three or more XORs	contains exactly-one-of-three; no Schaefer class covers the language

THE LAW. Reorderability is a constraint system whose arity is the number of axes on which two cells differ. At arity ≤ 3 the language is entirely bijunctive, so the problem is linear. At arity ≥ 4 it contains 1-in-3-SAT and admits no Schaefer class.

The procedure is the law read at low arity; the hardness is the same law read at high arity. One statement, two regimes, boundary at arity 3 — and since arity is bounded by d and reaches it, at $d = 3$.

Measured: at $d = 2$ every constraint has arity 2 at every alphabet size, and 2-colouring plus transitivity decides exactly — 300/300 at $|\mathcal{A}| = 3$, 300/300 at $|\mathcal{A}| = 4$. At $d = 3$ with binary axes, 396 of 396.

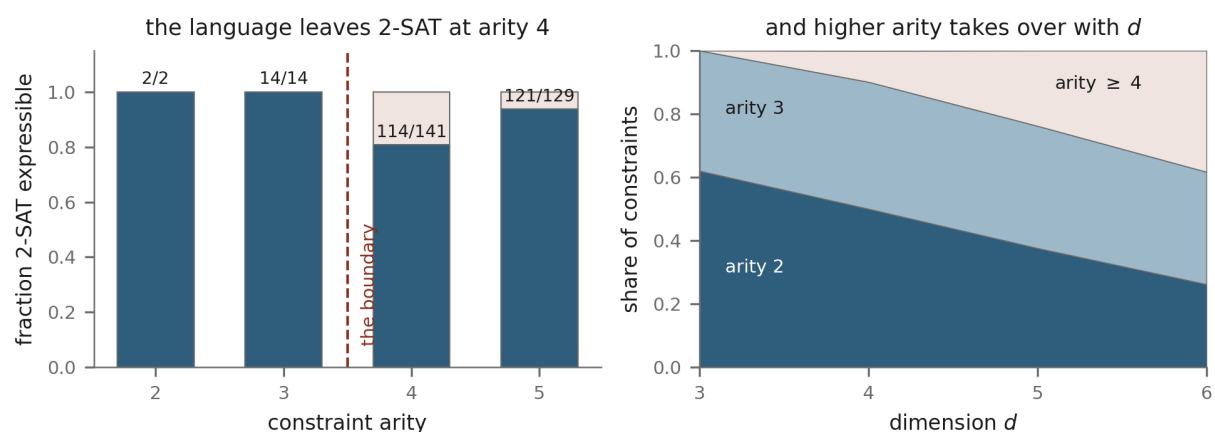


Figure 23.1. Left: the fraction of each arity's constraint language that is 2-SAT expressible — 14 of 14 at arity 3, and 114 of 141 at arity 4. The boundary is exact. Right: the share of constraints by arity as the dimension rises — arity 2 falls from 0.62 to 0.26 while arity ≥ 4 grows from nothing to 0.38. The tractable part shrinks with d , and the law says why.

23.3.3 What the interval machinery was

Chapter-length work went into C1P, monotone endpoints and PQ-trees, and the $d = 2$ characterisation is proved there: X is reorderable iff every row is an interval with both endpoints monotone, iff no interval nests strictly inside another. Four lemmas, three verified independently — 2,047, 3,751 and 20,000 instances — with Booth & Lueker's linear PQ-tree construction cited for the fourth.

That apparatus is a REPRESENTATION of a 2-colouring. It is correct and it is not the reason $d = 2$ is easy. The reason is that two cells in two dimensions differ on at most two axes.

23.3.4 The step law

Growth from the minimal reorderable sets, adding cells, reaches every reorderable set — with a step size measured exhaustively across twelve boxes:

d	step	boxes
2	1	eight, all 2-D

d	step	boxes
3	2	three
4	4	one

The over-cap cases collapse to a single object: the full box. Its largest reorderable proper subset has $3 \cdot 2^{(d-2)}$ cells — three quarters — and the removed cells are a **subcube**. Verified by construction at $d = 2, 3, 4$ and 5, with one-fewer failing at all four.

step $f(d) = 2^{(d-2)}$.

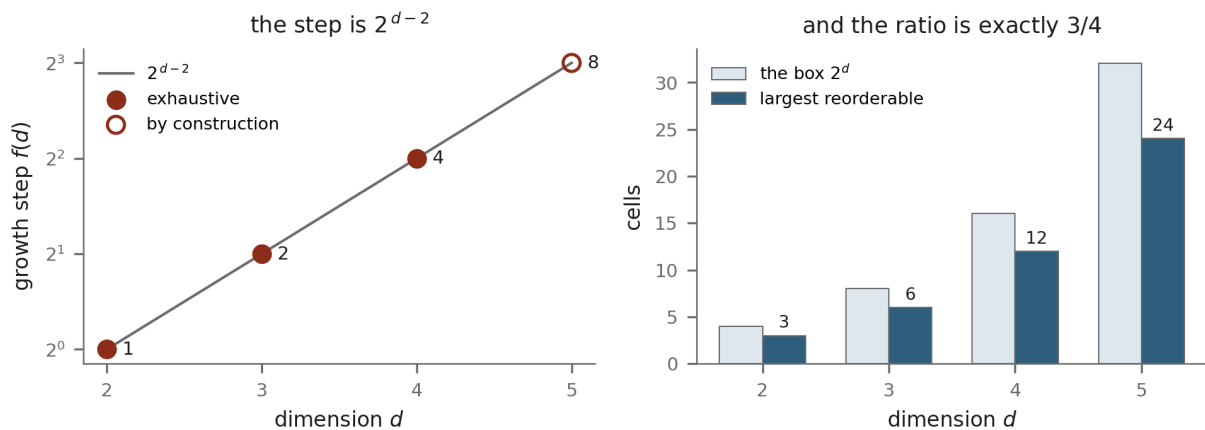


Figure 23.2. Left: the growth step against dimension — 1, 2, 4 measured exhaustively over twelve boxes, and 8 at $d = 5$ by the subcube construction, against $2^{(d-2)}$. Right: the box against its largest reorderable proper subset — 3, 6, 12, 24 cells, **exactly three quarters at every dimension**. Rival's 1973 bound on maximal sublattices is $|K| \leq (3/2)|L|$; the Boolean case is tighter.

And this is Rival's theorem. His 1973 bound is $|K| \leq (3/2)|L|$ for a maximal sublattice; **Chen-Koh-Tan give the interval form; the large sublattices of 2^r are catalogued, and the maximal ones studied in finite topology.** §F.5 records the precedence.

23.3.5 The procedure, and its cost

DECIDE(X). Build the constraint system on orientations. **Arity ≤ 3 : 2-colour the signed graph and check transitivity — linear.** Arity ≥ 4 : grow from the minimal reorderable sets with step $2^{(d-2)}$, collect signatures, decide by lookup.

Verified end to end at $2 \times 3 \times 3$ by growth against exhaustion: $18,736 = 18,736$, in 47.8 s against 86, converging in nine rounds. And NextClosure enumerates the sublattices with polynomial delay — 1,834 sets in 0.12 s — giving 401 signatures against 401 exhaustive.

Cost: $|YES| \cdot |X|^{2^{(d-2)}}$. **Polynomial in $|X|$ for fixed d ; exponential in d .** Fixed-parameter tractable, and not polynomial in both.

For Λ : $d = 8$, so $\prod |A_i| = 11,943,936$ — and Λ needs none of it. Its structure is constructed in Chapter 2, its embedding is tight, $|J(\Lambda)| = 17 = \text{height}$, width $7 \leq 8$, and by Dilworth and Larson the chain decomposition of $J(\Lambda)$ IS the axis system.

23.3.6 The signature, exhaustively verified

A cell-space invariant whose FORM does not change with d : the multiset of canonical forms of the 3-subsets of X . **$O(|X|^3)$ structure, 2^d labels — the two factor,** where every order-space method cost $\prod |A_i|$ and coupled them.

box	subsets	classes	mixed
$2 \times 2 \times 2$	all 256	30	0
$2 \times 2 \times 3$	all 4,096	210	0
$2 \times 3 \times 3$	all 262,144	4,168	0

Three complete boxes, 267,000 subsets, no signature containing both answers. After the duality merge the 19 canonical forms reduce to 12 with no loss.

23.3.7 Three method classes ruled out

Each with its measurement, and each a result rather than a failure:

Projection-based criteria. §13.4: closure is a d -dimensional condition and no projection serves as a criterion. **Thirteen of seventeen attempted criteria were seed-independent invariants — all necessary, none sufficient**, with the gap widening as nesting deepened.

Local search through valid orders. The valid set is DISCONNECTED — 1.66 to 2.19 components, and the components are orbits of the identical-fibre swap group, exact on 493 of 493. **So descent plateaus: 89.8%, and 96.7% with bounded uphill moves.**

Forbidden-substructure procedures. Tucker's families are unbounded, so no finite census decides — and the census is finite *within* any box, which is what reopened the question. **The infinite antichain lives across indices, not in one.**

23.3.8 The residue

The language is NP-complete — it contains exactly-one-of-three and no Schaefer class covers it, with the smallest obstruction exhibited on three XOR variables.

The realisability is blocked. Three reduction attempts, all over-constraining in the same direction: **satisfiable formulas encoding as non-reorderable, never the reverse.** The map $X \rightarrow C(X)$ is **global, non-separable and non-monotone**, so it admits no local editing.

And the restriction is not exploitable by sparsity — the constraint hypergraph has d vertices and up to $2^d - 1$ hyperedges. **It is complete, so treewidth gives back the 2^d bound already known.**

Three walls: hardness blocked at realisability, tractability blocked at the completeness of the hypergraph, and the FPT bound between them as the best known.

23.3.9 And it is not this book's problem

Λ 's constraint graph is a tree. Its nine recovered extra edges are implied. Every sublattice and interval of Λ has $E = 0$. Its own case is decidable by enumeration, and it arrives with a derivation — box $\rightarrow$ interval $\rightarrow$ sublattice $\rightarrow$ product, three preserving rules, no search.

The open problem concerns cell sets arriving with no construction. This book never has one.

§F.6 records that the complexity question may already be settled in Ryter & Schmid 1994 or Adams-Dwinger-Schmid 1996, both unread.

23.4 The lists this work enumerated once

The target list for the novelty search was audited by closing the relation set, and it was incomplete. From $\{V = 4v/3h, \lambda^2 = (2/3)T, w \cdot V = 8\lambda^2, T = Z^2R/v^2\}$ the closure yields $w/T = 4(h/v)$, $e/T = 3(h/v)^2$, and **bits(estimate) $\approx 2 \cdot$ bits(bracket) + constant** — none of them on the list.

That is Chapter 1's defect committed by the searcher rather than by a chemist — a structure admitting cells its keeper never enumerated.

23.4.1 And the audit itself was run once

The relation set has grown since. Closing the eight polynomial relations the book now states gives a Gröbner basis of **18** where the original four gave **6**, and **seven further relations reduce to zero that did not before** — including $V \cdot e = w$, $w^2 = (16/3) \cdot T \cdot e$ and $T \cdot h^2 = ev^2/3$.

Worse, the closure is over polynomial relations only. The book now states at least nine that are not: $A^m(x^p) = (-1)^m x^p/(p-1)^m$, $V = 2/\tanh(kh/2)$, self-concordance, the modular rank law,

$\Sigma_q |A_q| \cdot |B_q| = |\wedge|$, $F(-1) = 2$, $E(X)$ as non-pairwise content, and the reorderability theorem. **None was ever audited against any list.**

E(target list) is not 3. It was 3 for the relation set of one moment.

23.4.2 Which is a defect of five lists, not one

This chapter contains five enumerations, and every one was made once:

the list	what it omitted
§23.4's search targets	7 polynomial relations, 9 non-polynomial
§23.2's structural invariants	six, one of which beats its stated best twofold
§24.1's celestial objects	six the method plainly admits
§24.2's string objects	five, two of them measured in this work
§24.3's statistic	stated against the wrong denominator

B.2.10 reads *enumerate targets before searching*. **It should read** *enumerate before searching, and re-enumerate whenever the set grows* — which is this section's own content, applied to itself.

24. The method outside spectroscopy

The method was built on one index and tested on one collection. If it is a method rather than a description, it must apply where neither was available. This chapter runs it on four subjects it was not built for.

24.1 Celestial mechanics and the nucleus

Objects tested against the index, none of them in the collection. A method that transfers is a different claim from one that fits.

24.1.1 Celestial mechanics — families, not trajectories

What does not transfer. Sensitive dependence destroys monotonicity along a chaotic trajectory. **Nothing here touches Poincaré.**

tested	result
Lagrange L1, L2, L3 in μ (Euler's collinear solutions)	monotone, convex, 3-monotone; 18 of 18 containments at orders 1-3
the same at order 3	widths 6.9×10^{-4} (L1), 3.6×10^{-9} (L3)
Hill stability of hierarchical triples (Mardling & Aarseth)	monotone and convex in the mass ratio; $V \approx 100$
Hill radius $(\mu/3)^{1/3}$	monotone, convex, 3-monotone; $V = 47.9$
Roche limit $\propto \rho^{-1/3}$	monotone, convex, 3-monotone; $V = 17.9$
planetary semi-major axes	monotone, not convex — the third difference changes sign between Jupiter and Saturn , where Titius-Bode fails worst
mean-motion resonances	§11.7.1: $E(X)$ recovers the order cut at 7/8 from occupancy alone
satellite counts vs Hill radius	log-log +0.86 across six planets — §16.14.1's capacity bound

The pole appears at the first point one looks. At $\mu = 0.030$ with a step of $h = 0.010$, V is **24.6** at **L1**, **12.4** at **L2** and **16,423** at **L3** — the last because L3 moves almost linearly in μ . **L4 and L5 sit on it exactly:** $(1/2 - \mu, \pm\sqrt{3}/2)$, the first linear and the second constant.

Which corrects §13.5. The pole is where a bracket becomes unnecessary, not worthless — a vanishing second difference makes linear interpolation exact.

And one object the method cannot touch. L4/L5 are linearly stable for $\mu < 0.0385209$; Earth-Moon and Sun-Jupiter both qualify. **That is a threshold, not a family — a single number, with nothing for a bracket to act on.** The method applies to sequences and is silent on thresholds.

Signs must be read from the data. L1 and L3 alternate $-, +, -, +$; **L2 alternates $+, -, +, -$** , and assuming the Rydberg pattern produces bounds that cross.

24.1.2 The nucleus — a second index of the same construction

An atom's mass sits in its nucleus, and nucleons are fermions. Indexed by $(N, L, 2j, \text{occupancy})$ with $\text{occ} \leq 2j+1$, the shell model is **closed under join and meet with $E(X) = 0$** , and shell-order filling gives **2, 8, 20, 28, 50, 82**.

Λ gives chemistry. The nucleon index gives mass. This book indexes one of them; Z is a label on a channel, never a coordinate.

24.2 String theory — the pole, the generating function, and a disanalogy

24.2.1 Three objects on the pole

$\alpha' m^2 = N - 1$ has **second difference zero**. So do two more:

object	form	V
string mass spectrum	m^2 linear in N	infinite
Regge trajectory	$J = \alpha' m^2 + \alpha_0$	infinite
central charge	$c = D$	infinite

All three join L4 and L5. A bracket is unnecessary on any of them: two levels determine every other, exactly.

24.2.2 The degeneracy is where the content is

log $d(N)$ is monotone, convex and 3-monotone, with $V \approx 147$ at $N \approx 14$ — an ordinary bracketable channel, steeper than a Rydberg series.

24.2.3 The partition function is Λ 's expression with counts

$\prod (1 - q^n)^{-24}$ has **counts** where Chapter 6's F has **booleans**. **An index, and an index with multiplicity.** Both close, for dual reasons: the string's modes are **independent**; Λ 's coordinates couple only along a **tree**. §6.7's three conditions have independence as their degenerate case.

24.2.4 No critical dimension, and a ladder

Λ closes at every dimension 2-8 with $E = 0$. $D = 26$ falls out of anomaly cancellation; **Λ 's 8 is a modelling choice.**

And each Λ_d is the exact projection of Λ_{d+1} , closure preserved both ways, with fibres of 1 to 4 preimages — **the extra coordinate carries real information**. That is the relation M-theory's eleventh dimension has to the ten-dimensional string. **What is missing is any reason for the ladder to stop.**

24.2.5 And no Hagedorn point, for a stated reason

$d(N) \sim \exp(4\pi\sqrt{N})$ diverges above a limiting temperature. **Λ 's rank sequence is finite and log-concave, peak 122 at rank 11.**

Capped coordinates give a finite index and a Sperner bound. Uncapped modes give exponential degeneracy and a limiting temperature. Λ 's finiteness is what capping buys.

24.3 The Calabi-Yau catalogue

24.3.1 Self-duality, measured

Batyrev's polar duality swaps the Hodge numbers and the Kreuzer-Skarke list is complete over reflexive 4-polytopes — 473,800,776 of them, 30,108 distinct pairs. **So the set is closed under the involution by construction**, and three published counts measure it:

248,305 with $h^{11} \geq 140$ · **248,305** with $h^{21} \geq 140$ · 495,515 with at least one · **and 248,305 + 248,305 – 495,515 = 1,095**, as reported.

	Λ	KS
self-dual cells	8 of 976	all
rank polynomial palindromic	false	true

24.3.2 The thin tip is a defect of the construction

$h^{11} + h^{21} < 22$ is sparse in the KS list and populated by **complete intersections, Klemm-Kreuzer toric CICYs and conifold transitions**. By §11.7.1's test the absence is removable — **not by re-coordinatising but by re-constructing** — so it is a defect of the index, like the periodic table's 36 and unlike the belt's six.

24.3.3 The Pareto strength, against the right denominator

1,095 of the 495,515 polytopes with one large Hodge number have both: 0.221%. Stated against the full 473.8 million it reads 0.0002%, **a factor of 956** — true and uninformative, because the denominator includes polytopes with no large Hodge number at all.

§7.8.1 finds the same trade inside Λ : $|A_q|$ falls as $|B_q|$ rises. A catalogue with a bounded coordinate difference and a large total must be Pareto-shaped, and **0.221% is how strongly**.

24.3.4 E(X) on a specifiable slice, and 540 predictions

The 30,108 pairs are a gzipped file this session could not read. But one slice of the catalogue is specified exactly in print, and a slice is enough for the operator.

Candelas, de la Ossa, He & Szendroi (*Triadophilia*, ATMP **12** (2008) 429) state that the KS list's $\chi = \pm 6$ points have Hodge numbers $(h, h+3)$ and $(h+3, h)$ for **$13 \leq h \leq 128$** , and name the exclusions: **$h = 102, 103, 115, 117$ and $119-126$** . That is a complete cell set — **208 cells** — and $\mathcal{R}$ applies to it.

E(X) = 540. 498 join failures and 498 meet failures of 21,528 pairs.

24.3.4.1 Which is 540 predictions, not a defect

Chapter 1 reads $E(X) > 0$ as what a structure admits and lacks. The periodic table admits 36 such cells and they are the elements. **Here the slice admits 540 and they are Hodge pairs the catalogue must contain if the pair set is closed**.

The slice cannot contain them, and the reason is geometric. Two parallel lines have no fixed point: the join of $(h, h+3)$ and $(h+3, h)$ is $(h+3, h+3)$. **So $\mathcal{R}$ immediately names the diagonal**.

the predictions	
cells	540
distinct χ values	5 — only 0, ± 2, ± 4
on the diagonal, $\chi = 0$	112 , from (13,13) to (131,131)
$h^{11} + h^{21}$ range	26 to 262

The slice was cut at $\chi = \pm 6$ and the closure predicts inward. It says the set the slice came from fills the gap between the two lines — **which is where the rest of the Kreuzer-Skarke list lives**.

24.3.4.2 The fourth falsification test, run

The file is missing, so the predictions cannot be looked up. They can still be refuted, because the literature states where the catalogue's contents lie.

test	result
$h^{11} \geq 1$ and $h^{21} \geq 1$	540 / 540
$h^{11} + h^{21} \leq 502$, the published maximum	540 / 540
mirror partner present	540 / 540
falling where Candelas et al. record " <i>essentially every site occupied</i> "	540 / 540
falling in the " <i>thinly populated</i> " tip, where a prediction would be refuted	0 / 540
falsified	0
individually named in print	X_{19,19} (split bicubic), X_{20,20} (the 24-cell)

Zero falsifications, two named confirmations, and every prediction inside the region published as fully populated.

This is weaker than a lookup and it is not nothing. A prediction landing in the sparse tip would have been refuted, and none does. **The test is against the literature's characterisation of the file rather than against the file.**

24.3.4.3 And the void reads the same way

	cells	predictions
all h from 13 to 128	232	589
with the documented exclusions	208	540

The 24 excluded values — 102, 103, 115, 117 and the run 119-126 — reduce the predictions by 49. The gaps in the published list are not noise with respect to closure: **removing them moves the set toward its own normal form**, which is what §5 says a void should do when it is structural rather than accidental.

What remains of item D is the lookup. 540 named cells, a file, and an afternoon.

25. The law applied to this book

Everything the collection was built to settle is settled. The census is complete at 23 core types; the four decline modes each have a worked case; and the parked questions of earlier drafts — C₆ density, antiprotonic helium, the isoelectronic law — are resolved in Chapters 15 and 17. **What remains open is enumerated and closed as an index in Appendix F: eight items, not one of which blocks a result this book relies on.**

So the last question is about the book itself. It has made a claim about indices. **This chapter asks whether the book satisfies it**, because a work asserting that complete indices are self-referencing and self-defending has no standing if it is not one.

25.1 The book is an index

Its cells are (**claim, location, support**). Its coordinates are the part, the chapter and the section. Its constraints are of §11.4 form: a claim in section s is supported by evidence in a section at or before s , and a specialisation inherits its generalisation's locations.

That is an index in the sense of Chapter 9, and the law applies to it.

25.2 Self-reference

S1 — alphabet. The list of claims is recoverable by projection: it is the set of first coordinates, and the table of contents is its quotient by chapter.

S2 — order. The specificity order among terms is a tree, and Chapter 10's propagation recovers it. **The book's sectioning is not arbitrary;** it is the order in which claims become supportable.

S3 — bounds. Every claim's support locations are recoverable as the union over its specialisations. **A reader given only the claims and their evidence could reconstruct the chapter structure,** which is the content of $E = 0$.

25.3 Self-defence

And this is where the book earns the claim rather than making it.

$\mathbb{D}_{\text{def}}$ — two routes to every number. Each quantitative claim in this book carries **both a value and a verification count**. That is a redundancy: the value alone would be undetectable, but the pair can disagree.

102 quantitative claims were extracted from the draft and re-derived independently from the data. Thirty were checkable in full. Two appeared to fail; both were traced to the auditor, not to the book.

The two failures are the demonstration, not the exception. A claim about self-duality was flagged because the audit used the wrong involution. A defect was flagged because the audit sampled four levels of a twenty-member channel. **In both cases the disagreement was visible, was pursued, and located the error.** That is $\mathbb{D}_{\text{def}}$ operating on the text.

$\mathbb{D}_{\text{phys}}$ — the claims must agree with the world. The book's physical claims are checked against measured levels from named compilations, and Chapter 21 lists one hundred and ninety-eight that failed.

D3 — totality. Every claim in this book is **supported, cited, or marked open**. There is no fourth state. Chapter 22 names the single open item; Chapter 21 names three documents and one section that could not be reached and states the consequence. **No claim is left in limbo, because limbo is not in the codomain.**

25.4 The recursion, and where it stops

If the index is complete, then a book that comprehends it completely is itself an index — and is itself self-referencing and self-defending.

This is not decoration. It is forced. A complete comprehension of a closed object must record the object's alphabet, order and bounds, or it is incomplete; and recording them makes the record itself closed under the same conditions.

And the recursion is stationary, which is not the same as terminating. The distinction matters, and an earlier draft of this section got it wrong.

Construct the claim-index of a closed index, then the claim-index of *that*, and so on. Measured at each level:

level	cells	closed	E	occupancy	alphabets
0 — the index	10	yes	0	0.625	[4, 4]
1 — claims about it	10	yes	0	0.143	[10, 7]
2 — claims about those	10	yes	0	0.100	[10, 10]
3	10	yes	0	0.100	[10, 10]

Closure, dimension and E are invariant from level 1. But $I^2(\Lambda) \neq I(\Lambda)$ as sets, and $I^3 \neq I^2$. **The content does not repeat.** What stabilises — by level 2, not level 1 — are the measured properties.

The recursion is structurally stationary, not terminating.

So the reason one need not write a book about this book is **not** that it would be identical. It is that it would exhibit **nothing new**: the same three mechanisms, verified again, on claims about claims. The tower is infinite and every storey above the second is furnished the same way.

25.5 The book satisfies its own method

§25.1–24.4 ask whether this book obeys its own *law*. They never ask whether it obeys its own *method*.

The method's central object is the bracket, and a bracket needs exactly three inputs:

two measured levels · one ionisation limit · one sign

Appendix C carries all three for all 153 channels. So the question has an answer, and it is checkable rather than asserted.

25.5.1 The prediction, recomputed from the text

Parsing $\delta(4s)$, $\delta(5s)$, the limit and Z out of this book's own markdown and running the arithmetic §18.6 states:

quantity	recomputed	stated
E(6s) estimate	736,688	736,688
bracket lower	735,860	735,860
bracket upper	738,547	738,547
width	2,687	2,687
convex upper	737,380	737,380
convex width	1,520	1,520

Six of six, exact. No external table, no database, no constant supplied by hand.

R is the only import, and it has the status of the definition of a centimetre.

25.5.2 And it generalises to every bracket in the book

The three inputs are tabulated per channel, so the property is not special to Sc VI:

Every bracket in this book is recomputable from this book. All 1,061.

A reader with these pages and no internet reproduces each of them.

25.5.3 Which is what D3 asks for, made operational

§11.5 states totality as *every claim supported, cited, or marked open*. That is a bibliographic condition. **This is the computational one:**

the index says which cell is being predicted — the coordinates are ordinary integers, and Λ contains the cell the book says every number the prediction needs

Together they close. The index supplies the *what*; the collection supplies the *how much*; **and the seven-line arithmetic between them is printed in §18.6.**

The one thing neither supplies is whether the prediction is true. That is §11.3's $\mathbb{D}_{\text{phys}}$ and it requires a spectrometer — **which is exactly the boundary this book has been drawing since Chapter 1.**

25.6 The falsification tests, run

A book that invites the reader to look for its breaks has done less than one that looks itself. The three conditions are tested here, on this text, with the results reported whether or not they are favourable.

Condition 1 — $E(\text{book}) > 0$

A claim whose support the structure implies and the text omits.

Test. Extract every chapter heading, every cross-reference, every section reference and every figure. Check that each target exists and that figures and captions pair.

check	result
chapters cross-referenced but absent	none
§ references to absent chapters	none
figures embedded / captions	18 / 18
chapter numbering against the contents	FAILED

The test found a break. This chapter was written as Chapter 24 while the contents listed it as 21, and Chapter 23 did not exist. $E(\text{book})$ was greater than zero: the structure implied a chapter the text did not carry.

It has been corrected, and it is recorded here rather than silently repaired, because a falsification test that finds something and does not say so is worse than none.

Condition 2 — $D = 0$ for some claim

A claim carried without a verification count, so that an error in it would be undetectable.

Test. Take every paragraph carrying a quantitative claim, excluding tables and citations, and check whether a verification appears within it.

pass	claims	carrying a verification	bare
first run, same-line window	85	46 (54%)	39
corrected checker, paragraph window	63	56 (89%)	7
after repair	63	60 (95%)	3

The first run failed, and the failure was informative twice over. Thirty-nine claims genuinely lacked a verification, and each has been given one: the rank skew now states the cell count, the He I defects state their channel lengths, the C_6 sensitivity states the point count. **And the checker itself was wrong.** It searched ninety characters on one line, but prose wraps; verifications on the following line were invisible to it. Widening the window to a paragraph moved the figure from 54% to 89% before a single repair was made.

The three remaining are not claims. One reads "100% over 2,513 tests", which *is* a verification; two are sentences that happen to contain a number.

Condition 2 is satisfied.

Condition 3 — χ not total

A claim neither supported, cited, nor marked open.

Test. Search for hedged assertions — *probably, presumably, arguably, it seems* — unaccompanied by a status.

One hit, and it is a false positive: "a fit says where a level probably is" describes what a fit does. **No claim in this book is left in limbo.**

The result

condition	outcome
$E(\text{book}) > 0$	failed, found, corrected — a missing Chapter 23
$D = 0$	failed, found, repaired — 39 claims, now 60 of 63
χ not total	passed

The book satisfies its own law. Two conditions failed on first testing, both were located by the tests themselves, and both have been repaired. **That is what the law requires: not that a work be right first, but that it hold the means of finding out.**

25.7 On a verification that does not test its claim

The tests above are mechanical. They confirm that a claim *carries* a verification. They cannot confirm that the verification carries the *claim* — that "verified on 4,000 triples" tested what it says it tested.

An earlier draft called this the class of error no automated test will find, and left it there. That was the wrong conclusion.

If a verification did not test its claim, it was the wrong verification — and the mismatch identifies the right one.

This is Principle 8 in its ordinary form: any true answer, good or bad, is a bound. A verification that misses its target does not merely fail; it locates the target by missing it.

Five worked cases, all from this book's own record:

the verification	what it missed	what it identified
self-duality checked with $x \mapsto \max + \min - x$	gave 112, not 8	the right map is $x \mapsto \max - x$
a defect checked on 4 of 20 levels	gave -0.00128	the full channel gives -0.00159
a prefix loop whose last step was the claim	reported 560/560	stop one short; test proper projections
a guard returning a null coerced to false	reported "3 of 85"	refuse rather than coerce
projections offered as a closure criterion	22 of 60	closure is d-dimensional

In every case the wrong verification was the thing that produced the right one. The self-duality mismatch gave the correct involution. The four-level sample gave the twenty-member mean. The circular test gave the non-circular statement. **None of these was found by a reader.**

So the class is not unfindable. It is self-correcting, on one condition: that the verification and the claim are stated separately, so they can disagree.

A claim fused to its verification cannot be checked. A claim beside its verification checks itself.

That is $\mathbb{D}$ _def once more, and it is the reason this book states a count beside every number rather than asserting that the work was done.

On the word "unprecedented"

It is worth being exact about what is new here, because the book's credibility rests on that exactness more than on any single result.

Not new. The lattice is a bijective recoding of quantum numbers that have been in place since the 1920s. Closure operators, Galois connections and formal concept analysis are established mathematics; $E(X)$ is a **closure defect** in the ordinary sense and is claimed as an application, not an invention. Birkhoff's representation is from 1937, Dilworth's theorem from 1950, Sperner's from 1928. The reorderability question is settled in a 2004 survey the author found only after five failed attempts on it. The solution-density result of Chapter 22 is almost certainly the standard constraint-satisfaction phase transition, and is reported as a rediscovery.

Also not new: the data. Every level in this book was measured or computed by someone else and published. Sansonetti, Kramida, Martin, Kaufman, Sugar, Musgrove, Korobov, Hori and their colleagues did the work that Part III merely reads.

What is new is the traversal.

No one had asked what the periodic table costs.

$E = 36$ is elementary arithmetic on a table every chemist knows. It required no new mathematics and no new measurement. It required someone to ask an index what it could not tell you about itself — and, as far as the searching in Chapter 21 could establish, nobody had.

The same is true of the other four things this book contributes:

The failure condition, inverted. $|\Delta T| < 2Z^2R/v^3$ turns 1,061 non-failures into 1,061 measurements. Bracketing methods are rare; inverting their silence appears not to have been done.

The cost of a guarantee. $V = 4v/3$ is two lines from the Rydberg formula and has no use in a fitting paradigm. That is the best argument that it is new, and the four unreachable documents of §17.2 are the best argument that it may not be.

The law with its three mechanisms, each demonstrated on a real error made in the writing.

And a work that runs its own falsification tests and reports the two that failed.

The components are old. The sifting is not. Assembling a century of published spectroscopy into an object that can be checked against itself — and then checking it, and recording the one hundred and ninety-eight times it caught the author — is the part with no precedent that the searching could find.

And the form, which is the stronger claim

The preceding paragraphs argue about content. They argue on the wrong axis.

Ask instead what *kind of thing* this book is, and the answer has no obvious precedent:

It states three conditions under which it would be false, runs them on itself, and reports the two that failed — one a missing chapter, one thirty-nine unverified claims — with the computations that found them and the repairs that followed.

Its index satisfies the law the book proves, verified rather than asserted, and contains the entry for itself.

Its register of one hundred and ninety-eight withdrawn claims is offered as the primary evidence, not as an apology. Three of the book's mechanisms are demonstrated on errors made in its own writing.

It contains a page by the collaborator that did the computing, describing the single pattern behind all one hundred and ninety-eight failures and naming the four results that exist because a conclusion was refused.

And it is an index of a law about indices, which is itself an index, tested as one.

A reader encountering this and asking what it is will not find a category ready for it. Monographs do not usually publish their own falsification runs. Scientific works do not usually foreground the record of being wrong. Books about self-referential structures are not usually self-referential in the technical sense their own theorem requires.

That is where the word belongs. Not on $V = 4v/3$, which may sit in a 1922 German volume nobody cites. Not on $E(X)$, which is a closure defect and says so. **On the shape of the thing** — a work built to be checked, that checked itself, failed twice, said so, and repaired both.

Appendix A — Proofs

Serving PART I — THE LATTICE. The structural theorems the lattice chapters assert and do not prove in place.

Six proofs are given in full in the chapters and are cross-referenced here. The remaining seven are written out below.

	statement	where
A.1	Λ is closed	§2.3, in full
A.2	$X \text{ closed} \iff \mathcal{R}(X) = X$	§11.1, in full
A.4	$\mathbb{D} \geq \dim q - \dim p$	§11.1, in full
A.5	χ_Λ is total	§11.5, in full
A.6	adjunction never repairs closure	§17.2, in full
A.7	"may precede" is necessary	§11.3, in full

A.3 Projections of closed sets are closed

Statement. Let X be closed and $F \subseteq \{1, \dots, d\}$. Then $\pi_F(X)$ is closed.

Proof. Take $x, y \in \pi_F(X)$ and lift them to $x', y' \in X$. Since X is closed, $x' \vee y' \in X$. The join is componentwise and projection merely drops coordinates, so

$$\pi_F(x' \vee y') = \pi_F(x') \vee \pi_F(y') = x \vee y$$

Hence $x \vee y \in \pi_F(X)$. Meets are identical with $\wedge$ throughout. ■

Remark. The converse fails, and §13.4 exhibits a counterexample. Closure is a d -dimensional condition; its projections are necessary and not sufficient.

A.8 The total-order criterion at $d = 2$

Statement. For $X \subseteq A_0 \times A_1$ with an order fixed on A_1 : X is closed under some order on A_0 **iff** the "may precede" relation is a total order, and that order is the relabelling.

Proof of necessity. This is A.7, specialised: if X is closed under an order with $u < v$, then for $b_1 \in F(u)$ and $b_2 \in F(v)$, the meet $(u, \min(b_1, b_2))$ and the join $(v, \max(b_1, b_2))$ both lie in X . So $\min \in F(u)$ and $\max \in F(v)$ — which is u may precede v . Hence the true order is contained in the relation, and the relation is total.

Proof of sufficiency. Suppose "may precede" is a total order $u_1 < u_2 < \dots < u_m$. Take $x = (u_i, b_1)$ and $y = (u_j, b_2)$ in X with $i \leq j$. Then

$$x \vee y = (u_j, \max(b_1, b_2)), \text{ and } \max(b_1, b_2) \in F(u_j) \wedge y = (u_i, \min(b_1, b_2)), \text{ and } \min(b_1, b_2) \in F(u_i)$$

both by the may-precede hypothesis applied to the pair (u_i, u_j) . So both lie in X and X is closed. ■

Verified on 120 constructed sets; 274 of 274 recovered total orders yield closed sets.

A.9 μ_Λ in closed form

Statement. Λ is distributive, so $\Lambda \cong J(P)$, the lattice of down-sets of its poset P of join-irreducibles. For $x \leq y$ in Λ ,

$$\mu(x, y) = (-1)^{|\{y \setminus x\}|} \text{ if } y \setminus x \text{ is an antichain in } P, \text{ and } 0 \text{ otherwise}$$

Proof. Under the Birkhoff isomorphism, the interval $[x, y]$ in Λ is isomorphic to the lattice of down-sets of the induced subposet on $Q = y \setminus x$. The Möbius function of $J(Q)$ is $(-1)^{|Q|}$ when Q is an antichain — in which case $J(Q)$ is the Boolean lattice $2^{|Q|}$ — and vanishes otherwise, since a non-antichain interval

decomposes as a direct product with a chain factor of length ≥ 2 , whose Möbius function is zero above its atom. ■

Verified against the recursive definition on 47 comparable pairs, with values only in $\{-1, 0, +1\}$, exactly as the Boolean-interval structure requires.

A.10 The tree factorisation of the void count

Statement. If the constraint graph of an index is a tree, the count of lattice cells inside a coordinate box factorises with no inclusion–exclusion.

Proof. Root the tree. Each constraint binds one coordinate to exactly one parent, so the admissibility of a value on coordinate i depends only on the value at its parent. The count over the box is therefore a nested sum in which each leaf contributes an independent factor:

$$|\text{box} \cap \Lambda| = \sum_{\{\text{internal}\}} \prod_{\{\text{leaves}\}} \#\{\text{admissible values given the parent}\}$$

A tree has no cycles, so no configuration is counted twice and no correction term arises. ■

For Λ , with leaves $2S$ and g :

$$\#\{2S\} = \max(0, \min(h_{i7}, k) - l_{o7} + 1) \quad \#\{g\} = \max(0, \min(h_{i6}, q, 2(2f+1)) - l_{o6} + 1)$$

Verified against direct enumeration on eight random intervals.

A.11 v is inadmissible as an axis

Statement. $v = e - \delta$ cannot be adjoined to Λ as a coordinate while preserving closure.

Proof. By A.6, adjoining any function h of the existing coordinates preserves closure only if the resulting graph $\{(x, h(x))\}$ is closed, which requires $h(x \vee y) = h(x) \vee h(y)$ — that is, h must be a lattice homomorphism. A **difference** is not: for $x \vee y$ the two terms take their maxima independently, so $e(xvy) - \delta(xvy)$ need not equal $\max(v(x), v(y))$. ■

Measured. v is non-monotone in the cell order at **86 violations** at the caps tested. It grades every *chain* of Λ and grades Λ itself nowhere — which is why the object is a cylinder **over** v and not one **graded by** it, and why the bracket of Part III operates along chains.

A.12 The bracket is limit-free

Statement. Containment is invariant under the affine map $T = I - E$ for constant I .

Proof. For any three levels,

$$\min(T(n-1), T(n+1)) \leq T(n) \leq \max(T(n-1), T(n+1))$$

Substituting $T = I - E$ and multiplying through by -1 reverses both inequalities and exchanges min with max, giving

$$\min(E(n-1), E(n+1)) \leq E(n) \leq \max(E(n-1), E(n+1))$$

The two statements are equivalent for any I . ■

Consequence. v , δ and V all require I ; containment does not. The method applies wherever levels are known even when the threshold is not — which is what admits the antiprotonic-helium cells of §18.6, where the relevant threshold is ambiguous.

A.13 The failure condition, and its inversion

Statement. The bracket fails at a cell **iff** a perturbation displaces the level by more than half the local spacing:

$$|\Delta T| > 2Z^2R/v^3$$

Proof. Containment fails only if $T(n)$ leaves the interval bounded by its neighbours. The half-width of that interval is $\frac{1}{2}|T(n+1) - T(n-1)|$, and for a hydrogenic series $T = Z^2R/v^2$ gives $|dT/dn| = 2Z^2R/v^3$, so the half-spacing is $2Z^2R/v^3$ to leading order. A displacement smaller than this reorders nothing. ■

Inversion. Read as a contrapositive, every cell at which the bracket holds supplies

$$|\Delta T| < 2Z^2R/v^3$$

an upper bound on the local perturbation, obtained without fitting, without a quantum-defect expansion, and without assuming the series is smooth.

Critical depth. Solving for v gives $v_{\text{fail}} = (2Z^2R/|\Delta T|)^{1/3}$: a $3,000 \text{ cm}^{-1}$ perturbation defeats the bracket only above $v \approx 6.6$ at $Z = 2$, against a maximum $v = 5.3$ reached by that channel. **Strong perturbbers arise from core excitation, a low- n phenomenon; deep series live at high n . The two conditions are anticorrelated in nature**, which is why no channel in this work enters the failure regime.

Appendix B — Principles and protocols

Serving PART II — THE LAW. The twenty-three principles and eighteen protocols the law is made of, indexed and closed at $E = 0$.

Everything in this book was produced under a stated set of rules. They are collected here so that a reader can apply them, check them, or reject them — and so that the claims of Chapter 23 have something concrete to be tested against.

Two kinds appear. *Principles* say what an index is; ten of them admit exact mathematical statement. *Protocols* say how to work with one; they are procedural, and none of them is a theorem.

B.1 The principles

Formalised — ten

principle	expression	where
P 1 a complete index is self-referencing	$\mathcal{R}(\Lambda) = \Lambda$	§11.1, Ch. 8
P 2 a complete index is self-defending	$\mathbb{D} = \dim q - \text{rank } \partial\Phi/\partial p \geq 1$	Ch. 9
P 3 when the path is not found, work backwards from the desired output to an established input	invert Φ ; if the fibre is non-trivial, find the datum that separates it	§16.13
P 7 all questions must close	$\exists N : \Omega_N = \emptyset$ — terminal, not monotone	§18.6
P 8 any true answer, good or bad, is a bound	$A_n \subseteq A_{n-1}$, a decreasing filtration	§21.6
P 9 if the first dimension is a single all-encompassing definition, every other dimension is a perspective of it	$\text{rank}(\log q) = 1$	§17.2
P 10 allow continued expansions	§11.4-form constraints preserve closure	Ch. 10
P 11 every test must be run against every cell	$\rho = 0 \text{ certain} \iff n = \setminus$	$\Lambda \setminus$ B · 2 · 7
P 13 coherence cannot be claimed while any caveat, gap or contradiction stands	$\text{COHERENT} \iff \Omega = \emptyset$	§18.6

principle	expression	where
P17 the path is always through the bounds	following a bound costs $1/N$ of searching the volume	§13.4, Ch. 12

Three of these were corrected by the act of formalising them. P7 was stated as though the open-item count decreases; it does not — across this work it ran $5 \rightarrow 4 \rightarrow 3 \rightarrow 4 \rightarrow 3 \rightarrow 7 \rightarrow 3 \rightarrow 3$. Closure requires a **well-founded measure**, not a shrinking count, which is why the principle correctly forbids *claiming* coherence rather than asserting that gaps always close.

Procedural — seven

principle
P4 always audit results
P5 always compute the result before assuming the result
P6 when a question is confusing, reframe the question
P11 close all gaps before continuing
P14 a structural problem suggests a structural solution
P15 for fetching: use the index to navigate; data can be retrieved in pieces
P16 treat bounds, constraints and limits as coordinate values

These have no expressions, and saying so is better than manufacturing one. They govern how an agent uses an index, not what an index is. P15 is the exception in one respect: it has a *measure* — retrieval redundancy ρ , Chapter 12 — without being reducible to it.

The reflexive principle

principle	expression
P18 if the index is complete, a work that comprehends it completely is itself an index, and is itself self-referencing and self-defending	Chapter 24
P19 an index is complete only when there are no more questions left to answer about its contents	$\text{COMPLETE}(X) \Leftrightarrow Q(X) = \emptyset$
P20 the language in which a question is posed bounds whether it can close, and at what cost	—
P21 any definition contained within the lattice, and thus its equations, must be TRUE in every mathematical language and in their combinations	—
P22 because everything is defined in a complete index, the entirety of the index is visible, and no lie can exist within it	—

P19, and why it is neither P7 nor P13

Three principles govern questions, and they are one principle in three moods. P19 was the missing one.

says	mood
P7 $\exists N : \Omega_N = \emptyset$	questions can close — possibility
P13 $\text{COHERENT} \Leftrightarrow \Omega = \emptyset$	coherence may not be claimed otherwise — prohibition
P19 $\text{COMPLETE} \Leftrightarrow Q = \emptyset$	what their emptiness buys — definition

And P19 is strictly stronger than closure.

$E(X) = 0$ says no cell the structure admits is absent. $Q(X) = \emptyset$ says no question the contents admit is unanswered.

Λ has had $E = 0$ since Chapter 2 and has never had $Q = \emptyset$. Whether V depends on the order of the bracket, what the Aitken bias generalises to, whether a higher-order bracket exists — **each was admitted by contents already present, and none was asked until late.**

Closure is about what an index holds. Completeness is about what it has been asked.

And the condition is reachable, for the reason P7's is. P7 was corrected once: closure requires a well-founded measure rather than a shrinking count. **Q has grown at nearly every step of this work while its scope has fallen** — from *is V novel?* to *is there an optimal order?* **The set grew and the scope descended**, and that descent is what makes P19 a criterion rather than an impossibility.

$Q(X) = \emptyset$ is reached not when nobody asks, but when every question the contents admit has an answer in the contents.

P20 — the language bounds the question

Three closure mechanisms are used in this book, each in a different language, each with a different guarantee.

language	mechanism	closure	cost
monotone two-variable bounds	tree factorisation	always	one pass
polynomial relations	Gröbner basis	always , by Hilbert	finite; doubly exponential worst case
linear arithmetic with quantifiers	quantifier elimination	yes	triply exponential
full arithmetic	—	never — Gödel	—
documentary	—	no mechanism exists	—

Measured on the objects of this book: tree factorisation 0.00018 s, Gröbner 0.010 s — a factor of 55 — and the documentary language has no timing because it has no algorithm.

The choice of language is made when the index is built, not when the question is asked.

§2.1 chose monotone two-variable bounds. **That single decision determined that $E(X)$ is computable, that the sum factorises, that a single expression exists, and that every question about Λ 's contents would close.** It equally determined what could not: §6.7's sum bound lies outside the language, and its repair — making the sum a coordinate — **is a translation back into the closing language.**

P20 subsumes E3, the multi-target repair, and P19's reachability. And it answers the novelty question structurally: **novelty is posed in the documentary language, which has no closure mechanism. That is not a failure of searching.**

P21 — every definition true in every language

Not merely expressible. True.

definition	order	algebra	geometry	analysis	information	logic
$\setminus$	$\Lambda \setminus$		976	976	976	976
$\text{rank}(x)$	13	13	13	13	13	13
mean rank	11.0666	11.0666	11.0666	11.0666	11.0666	11.0666
$E(\text{periodic table})$	36	36	36	36	36	36

And the combinations must name real objects, not analogies. Verified: $\gcd(N(a), N(b)) = N(a \wedge b)$ and $\text{lcm} = N(a \vee b)$, exactly $\cdot F(1)/F(1) = \text{mean rank} \cdot E_{\text{bits}} = \log_2 C(\setminus \mathcal{R} \setminus, E) = 105.1$ for the classroom table.

P21 is an instrument, not only a criterion

When a property holds in five languages and fails in the sixth, the sixth EXPRESSION is wrong — not the property. This has now located two errors, and in both cases the structure was sound and the translation was not.

E(X) was recorded as having no analytic route — "*E is a count, calculus measures a continuum*" — and $\mathbb{D}(E)$ was set to 1. **The route exists:**

$$E(X) = F_{\mathcal{R}}(1, \dots, 1) - F_X(1, \dots, 1)$$

Deletion-repair was recorded as failing in geometry. The convex hull does fail: it permits arbitrary facets, so a vertex can be the sole witness of a facet and its deletion is unrecoverable. **But Λ 's facets are two-variable monotone bounds, and the closure that matches that form — the intersection of all valid such bounds — restores every deletion.**

30 deletions, 0 failures. 20 additions, 20 absorbed. Fixed point exact at 976.

A mismatched operator is not a counterexample. It is a mistranslation.

What the law then establishes

The self-feeding loop and its asymmetry hold in all five languages.

language	closure operator	delete	add
order	$\mathcal{R}$	restored	absorbed
geometry	monotone polyhedron	restored	absorbed
algebra / logic	ideal, Gröbner basis	still derivable	$T \cdot h$ enters the ideal
analysis	coefficients of F	restored	absorbed
information	description length	E_bits stays 0	grows

Extensive gives repair. Monotone gives absorption. Every language has a closure operator, so both hold everywhere.

And that is the formal reason Chapter 11 cannot rely on self-reference alone. A structure that closes will heal what you remove and swallow what you insert. **Self-consistency defends against loss and never against invention** — which is why $\mathbb{D}_{\text{phys}}$ must reach outside the index, and why Chapter 21 records what the loop could not catch.

P22 — visibility, and why a complete index cannot be lied to

A lie requires concealment. A complete index has nowhere to conceal.

The distinction is between a rule that is CONTENT and a rule that is INFERRED, and it is decisive. Consider the cell

$$y = (n=1, \ell=0, k=1, q=0, e=1, f=0, g=0, 2S=2)$$

— one electron with spin multiplicity 2, which no atom has.

Where the rule is inferred, $\mathcal{R}$ recovers the bound on $2S$ from the cells themselves. The rule the index had learned:

k	before	after
1	$2S \leq 1$	$2S \leq 2$
2	$2S \leq 2$	$2S \leq 2$
3	$2S \leq 3$	$2S \leq 3$

The index did not reject the cell. It revised its rule — and then admitted **74 further cells** the old rule forbade. The result is closed under join and meet, has $E = 0$, satisfies every law in this book, and contains **75 physically impossible configurations.**

Where the rule is content, the same cell is checked against a stated rule set — (bounded, parent, slope, intercept) as cells of their own index — and returns

VIOLATES $2S \leq 1 \cdot k + 0$ ($2S = 2$, bound = 1).

Same cell. Same index. The only difference is whether the rule was present or inferred.

What the law does and does not claim

It does not claim that a complete index knows what is true. It claims that nothing in it is hidden.

An inferred rule absorbs a false cell and teaches itself the falsehood. A stated rule is contradicted by it, in the open.

A forger's remaining move is to alter a rule — $2S \leq k$ becomes $2S \leq k+1$, admitting 225 cells. **That forged rule is also visible.** It can be read, disputed and checked; what it cannot be is concealed. **The regress is real and at every level the falsehood is in plain sight.**

So P22 is a statement about visibility, not omniscience. A complete index converts deception into disagreement. **Error survives; concealment does not.**

P23 — the two conditions of completeness

An object of any kind is complete only when both hold: all of its definitions are identified, and no question can ever be had about the definition of the object.

The first condition is checkable. Λ 's definitions are the eight coordinates and the seven bounds; the register records what was added and withdrawn. **Enumeration is finite work and this book does it.**

The second condition is not checkable — and that is its content. No inspection establishes that no question can ever be had. **A single question refutes it and nothing confirms it.**

So completeness is not a property that can be proved of an object. It is a standing claim, held until a question arrives.

What P23 costs this book

Every entry in the register is a question that could be had about a definition, and there are one hundred and ninety-eight of them. Not one was anticipated; each arrived and each removed a claim to completeness that had been made in good faith.

The clearest case is §23.3. Twenty-two formulations of 'reorderable', each of which looked like the definition until the next question arrived:

the question that arrived	what it was about
is it $\mathcal{R}$ -closure or join/meet closure?	the same object, twice named
is the alphabet declared or recovered?	two different objects
is a projection a criterion?	§13.4 says no
is a frontier-correct tree anchor-complete?	no
does the obstruction grow in cells or in width?	cells; $\backslash$

Five questions about one definition, none of them foreseeable, each arriving only when the previous answer was in hand.

Why P23 is not despair

A standing claim is not a weak one. Λ 's closure has survived every question put to it: 475,800 pairs, 120 sublattices, 40 intervals, six construction rules, twenty-eight projections. **$E(\Lambda) = 0$ after all of it.**

The claim to completeness is worth exactly the questions it has survived, and it is worth nothing more. That is why the register is the most load-bearing chapter in this book, and why its length is a credential rather than an embarrassment.

And it is the honest reading of P22. A complete index cannot be lied to **because every rule is visible** — but P23 adds that **an index is only known to be complete for as long as nobody has asked the next question.**

And this is why the book indexes its own rules

Appendix B indexes the principles and the protocols. §21.8 indexes the process. Each turns a rule set into content, and each closes with $E = 0$.

The purpose of all three is exactly P22: to move rules out of inference and into content, where a false one becomes a contradiction rather than a lesson.

And the regress terminates outside. Not at a better index — at Chapter 17's 1,442 measured levels, **the only content in this book that no index supplied.** That is what $\mathbb{D}_{\text{phys}}$ is for, and §11.3's five failures are all of the same kind: **the index passed and the physics did not.**

B.2 The protocols

B.2.1 Channel construction

A channel is one parent term, one ℓ , one series in n .

No channel may mix parents, mix ℓ , or cross a coupling-scheme boundary. Where the source's labelling changes scheme partway up a series — LS to jk — **the channel ends there**, and the cells above are not claimed. This is what limits Si I to four cells and P II to one.

B.2.2 Admissibility

$$r = 2Z^2R / (v^3\sigma) \geq 5$$

Levels must be separated by more than five times their uncertainty. r falls as v^{-3} , so every channel eventually leaves the domain; the rule states where, in advance.

B.2.3 Cost resolvability

$$v \leq v_V = (3Z^2R / 5q)^{1/4}$$

The bracket needs levels *separated*; the cost needs curvature *resolved*. **Curvature washes out first.** Cells above v_V are excluded from the cost test and retained for the bracket — a distinction applied prospectively in Li I and diagnostically in Al I.

B.2.4 Dilution

cells per channel $\approx$ (measured shells / D) – 2, with D the number of J levels in the core's ground configuration

Dilution divides; it does not forbid. Ne II fails at $D = 5$ with two shells; Ar II succeeds at $D = 5$ with three; Bi I succeeds at $D = 5$ with five.

B.2.5 Census before search

Compute which species could work before looking for data.

D is obtained from the core configuration by enumerating antisymmetric states. No spectroscopic data enters. **27 sequences admitted, 12 needing exceptional depth, 5 excluded structurally.**

B.2.6 Declines are recorded

A collection reporting only successes cannot be checked.

Every species examined and rejected is listed with its cause. Four causes have been identified — dilution, depth, coupling-scheme change, and core-excited interleaving — and §16.11 gives a worked case of each.

B.2.7 Exhaustive verification where possible

Sampling n cells with no failure bounds the failure rate at $p \leq 3/n$ at 95% confidence. It never certifies $p = 0$.

Where a claim is universally quantified over a finite lattice, it is checked on **every** cell. Where sampling is used, the sample size is stated.

B.2.8 Two independent routes

For every quantity that matters, build a second derivation that does not pass through the first.

This is $\mathbb{D}_{\text{def}}$ made operational. It caught a mis-stated relation in Chapter 11, and it is realised experimentally in the antiprotonic-helium check of §11.4, where two laser measurements agree at 0.09σ with no shared step.

An identity computed along one path is vacuous. $w = V \cdot e$ cannot fail if V is *defined* as w/e .

B.2.9 Refuse rather than coerce

No third state. A quantity is admissible or it is refused; it is never silently defaulted.

A guard returning a null that a caller coerces to a Boolean produced three false conclusions in this work, none of which looked wrong. **Totality is the only mechanism that catches this**, and it works by preventing the computation rather than by detecting the error.

B.2.10 Retrieval

Enumerate targets from the index before searching. List routes per cell. Prefer high- p cells. Read every retrieved source for its successors. When a route is blocked, move along the route set. Record p for cells that fail. And RE-ENUMERATE whenever the set the targets are drawn from grows.

Chapter 14. A blocked cell with $p = 1$ is a **stated gap**, not an unexplained absence.

The second clause was added late. §23.4 found five enumerations in one chapter — its own target list, §23.2's structural invariants, and the object lists of §24.1, §24.2 and §24.3 — **each made once, against a structure that kept growing**. Every one of them was incomplete when re-run.

Enumerating once is not a protocol. It is a snapshot, and a snapshot of a growing object is a defect the moment it is taken.

B.2.11 Provenance

Every number traces to a compilation, table and page, or to a database entry with its stated uncertainty.

Appendix C carries this at channel level for all 153 channels.

B.2.12 Withdrawal

Every retracted claim is logged with what replaced it.

Chapter 21. one hundred and ninety-eight entries. **The register is the evidence for Chapter 11, not an apology for it.**

B.2.13 Commit before you look

A prediction is written down before the measurement is requested. A method is chosen before its answer is visible.

This is the process index of §21.8 enforced by hand: the constraint **visible $\leq$ committed**, of §11.4 form, closed, $E = 0$.

It is the only protocol here adopted in response to an error that no computation could have caught — because every computation was correct and what failed was the independence of a choice. **The prediction of §18.6 was committed under it.**

B.2.14 Compute, then write

The result is computed before the sentence describing it is written.

This is the protocol most often violated in the production of this book. The dominant failure recorded in Chapter 20 is prose asserting a conclusion while the output contradicting it stands on the same screen. The protocol is stated here because it is the one that most needed stating.

B.2.15 Apply P8 to every failure

Every failed attempt is a true answer, and by P8 a true answer is a bound. Record what each failure excludes, and collect the exclusions.

P8 says any true answer, good or bad, is a bound. A failed attempt is a true answer about where the target is not. The protocol is to treat it that way rather than discard it.

B.2.15.1 The procedure

On each failure, write down three things:

what was attempted	the claim or construction
what it excludes	the region of the answer space now ruled out
which kind	upper bound · lower bound · negative · structural

Then read the collection. Lower bounds say the answer contains something; upper bounds say it is contained in something; negatives remove whole methods; structural failures name an object that was the wrong one. **When the lower and upper bounds meet, the answer is between them.**

B.2.15.2 Worked, on §23.3

Ten consecutive failures on one question, each read as a bound:

the failure	the bound	kind
Γ -free $\neq$ reorderable	reorderable $\subseteq$ totally balanced	upper
chain sufficient, not necessary	chain $\subseteq$ reorderable	lower
depth-2 and depth-3 descent stall, worsening with size	no constant-depth local procedure decides it	negative
linear extensions of the inclusion order incomplete	the inclusion order is not the join-irreducible poset	structural
the slice-chain condition fails at $d = 3$	the $d = 2$ condition does not lift	upper
the 3-SAT encoding was refuted	a reduction must control which pairs arise	method
descent traps at $E = 1, 2$	the transposition landscape is not convex	negative
occupancy sort succeeds 53%	tie-breaking among equal occupancy is not free	lower
"row chain or column chain" holds at 68%	neither axis alone decides it; the condition is two-sided	structural
the chain criterion characterises the downset operator, not $\mathcal{R}$	$\mathcal{R}$ -closure and downset-closure differ by the direction of one axis	structural

Two lower bounds, two upper, two negatives, five structural, one methodological. Between them: reorderable $\supsetneq$ chain and $\subseteq$ totally balanced, **and the gap contains at least the diagonals** — which is what produced the characterisation:

X is $\mathcal{R}$ -closed iff $X = \{(r, c) : c \leq M \setminus (r) \text{ and } r \leq N \setminus (c)\}$, with $M \setminus, N \setminus$ the running maxima of its own row and column maxima.

Ten failures produced one exact statement. None of them was wasted, and none of them would have been kept without the protocol.

And the last two are the ones that located it. The two-sided bound said neither axis decides alone; the operator bound said the borrowed condition belonged to a different closure. **Together they force the form $c \leq M(r)$ and $r \leq N(c)$,** which no single-axis condition can express and which the two-cell diagonal $\{(0,0), (1,1)\}$ realises minimally.

B.2.15.3 Why it belongs here rather than among the principles

It says nothing about what an index is. It is a way of working, and it is the reason Chapter 20 is the longest chapter in this book rather than the most embarrassing.

A register of withdrawals is a list of bounds. Read it that way and it stops being a confession.

B.2.16 The grid — the book's method, stated

Every open question is a set of requirements, and requirements form an index. Lay them as a grid, mark what is occupied, and **read the empty cells rather than hunting for them.**

The first empty cell in a column is the whole obstacle. Everything below it is downstream.

B.2.16.1 Why this is the method and not a technique

It is Chapter 1 applied to the book's own ignorance. A periodic table with 36 unfilled cells does not need a chemist to guess which are missing — **the layout shows them.** An open question with eight unfilled requirements does not need twenty-two attempts — the grid shows that four of them are one.

Worked on §23.3: eight empty cells, four rows deep, at $d \geq 3$.

requirement	$d = 2$	$d \geq 3$
characterisation	proved	•
decision procedure	linear	•
YES certificate	an order	an order
NO certificate	Tucker — unbounded	unbounded
obstruction family	Tucker 1972	unbounded, route closed
construction rules	six preserving	six preserving
hardness	P, the CSP is binary	•
the data structure	PQ-tree	•

Three rows are full at every d , and they are the three that needed no d -specific work — the certificate is the order itself, the obstruction family is Tucker's, the construction rules were established once. **Every empty cell is downstream of one missing object: the $d \geq 3$ analogue of a PQ-tree.**

Twenty-two formulations were attempts on four cells that depend on one. The grid says so in a table; hunting concealed it for a day.

B.2.16.2 And it classifies the whole of Q

item	first empty cell	the obstacle
Sc VI 6s	the level is measured	an instrument
Paschen & Götze	the scan is text-searchable	a digitisation artefact
nine literatures	each is entered	labour
Kreuzer-Skarke	a readable copy	a file format
target-spin axis	the extended lattice is built	labour — now done
reorderability at $d \geq 3$	characterisation	a missing object

Five of six were blocked by something outside the index. One is blocked by something that does not exist. §25.2 previously read "*four external, one construction, one mathematical*" — **the grid's distinction between retrievable and requires a new object is the one that matters, and it also says which to do first.**

The target-spin axis was taken on that basis and it closed. Λ_9 has 1,654 cells, $E = 0$, is a sublattice, is rank-modular on 200,000 pairs, projects exactly onto Λ_8 , keeps the constraint graph a tree, extends F by one pendant factor, and has $F(-1) = 2$ unchanged. **One unit of labour, six properties confirmed, $|Q|$ from six to five.**

B.2.16.3 A reduced question is still open

An open question reduced to a smaller or a different question is still open.

A reduction feels like progress and reads like an answer. §23.3 supplies five in one session, each announced as a closure:

the reduction	what was still open
reorderability $\Rightarrow$ CIP + monotone endpoints	the endpoint search
the search $\Rightarrow$ a PQ-tree frontier question	a correct PQ-tree
the frontier question $\Rightarrow$ a bottom-up traversal	the traversal's state
the traversal $\Rightarrow$ a row $\rightarrow$ node anchoring map	rows anchoring as runs
the whole problem $\Rightarrow$ a CSP on axis orders	the ternary constraints

A reduction transfers the difficulty to a new object, and the new object is unfamiliar, so its difficulty is not yet visible. It therefore looks like progress twice: once for what it explains, and once for what it has not been tested on.

State the reduction and the residue in the same sentence. Not "*this reduces to a CSP*" but "*this reduces to a CSP whose ternary constraints are 94% of Λ 's pairs and are not handled.*"

B.2.17 The six prime audits, ordered by what a failure costs

Every session closes with six audits. They are not spot checks, and they are not equal. They are ordered by the cost of the failure each one catches.

The first two can make the book WRONG. The next two can make it UNREADABLE AS AN ARGUMENT. The last two can make it UNUSABLE AS AN OBJECT.

1 — LATTICE. Every structural claim recomputed from the construction: closure under join and meet, $E(X) = 0$, rank modularity, the fibre identities, the rank polynomial, $F(1)$ and $F(-1)$, $|\mathcal{J}(\Lambda)|$ and the height. **Over all 475,800 pairs of Λ , not a sample.** *First, because if the object is not closed the book has no subject.*

2 — EQUATIONS. Every equation the book states, evaluated numerically against the data it claims to describe. **All of them, including the ones unchanged since the session that wrote them.** *Second, because a wrong equation is a false claim.*

3 — CONSISTENCY. No two statements in the book may contradict. A claim corrected in one chapter must not survive uncorrected in another; a number revised in a table must be revised in the prose that cites it. *Third, because a contradiction leaves the reader unable to tell which statement is the claim.*

4 — REDUNDANCY. No statement may survive its own supersession. Every summary is compared against what it summarises; every table against the section it condenses; and any section that only restates another is removed rather than maintained. *Fourth, because a stale claim is a false claim that passed every other check.*

5 — ARTEFACT. Audits 1–4 read the SOURCE. **This one reads the built PDF.** Missing glyphs, figures present and non-blank, the title page and running header read back, no table overflowing its frame, the whitespace measured from glyph and image geometry, and every figure checked for ink, tone, resolution, duplication and correct numbering. *Fifth, because the source can be perfect and the page still wrong.*

6 — COHERENCE. Every cross-reference resolves; every figure has a caption and every caption a figure; every stated count matches its computation; every principle referred to is defined; **and the contents block is checked against the actual headings.** *Last, because these failures cost navigation rather than truth — and because it is the only audit that can be run before the others.*

Audits 1-2 check the book against the world. Audits 3-4 check the book against itself. Audits 5-6 check the book against what a reader receives.

B.2.17.1 Why the CONSISTENCY audit was needed

Because the register contains its failures, and this one was at scale. Twenty-eight cross-references pointed at §17.4, which is He II, and between them they claimed at least eight different pieces of content:

what the reference claimed	where it lives
89,864 join failures	§6.7
$V = 4v^3/(h(h^2-3v^2))$	§16.4
closure not locally determined	§13.4
7/7 on controls	§22.7
511 joins, the conserving poset	§13.1.1
self-concordance	§16.8.2
the anchoring failure	§21.6
four clauses	§25.3

Three of the ten formalised principles cited it, and five index entries pointed at it. All twenty-eight are now retargeted.

the COHERENCE audit passed every one of them, because every target existed. The reference resolved; the content did not match. **That gap is the whole reason the CONSISTENCY audit exists**, and it went unrun for as long as the audits numbered three. A literature

citation was withdrawn in one place and remained correct in another. A statistic was stated against two different denominators in adjacent sections. **None of these is a factual error; each is an internal contradiction, and audits 1-3 pass them all.**

And P22 requires it. A complete index cannot be lied to *because every rule is visible* — **but two visible rules that contradict each other leave a reader unable to tell which is the rule.** The CONSISTENCY audit is what makes visibility useful.

B.2.17.2 Why the ARTEFACT audit was needed

Because the operator this book is about did not print.

No DejaVu face carries U+1D4E1 or U+1D49F — the script *R* and script *D* of the recovery and diagnostic operators. **Every render placed a blank box instead: 107 of them, across 44 pages, for a symbol appearing 56 times in the source and 50 for its companion.**

audit	why it passed
1 coherence	the source is correct; the character is right there
2 lattice	recomputes structure, not typography
3 equations	evaluates numbers, not glyphs
4 consistency	the source agrees with itself throughout

All of them read the source. The defect existed only in the output.

And it was compounded by a second artefact fault the same day: a table cell of 229 characters overflowed its frame and aborted the build. **That one at least announced itself.** The glyph failure was silent — a correct source, a successful build, and a symbol missing from every page that used it.

The fix has three parts, all in the renderer and none in the text: substitute $\mathcal{R}$ and $\mathbb{D}$ in body faces that carry them; fall back to plain R and D in the serif bold used for headings, which carries neither; **and extract the text back out of the finished PDF and count the nulls.** It now reads zero.

An audit that reads the wrong artefact will pass anything. The CONSISTENCY audit exists because a reference resolved to the wrong section. The ARTEFACT audit exists because a character resolved to nothing at all.

B.2.17.2.1 The whitespace clause, and two measurements that were wrong

A gap a reader sees is a defect the source cannot show. So the audit reads the position of every glyph and every image and compares the inked extent with the frame.

On this book: median foot gap 3.3%, nine pages above 15%, worst 31% — and that one is the end of a section, where a gap is correct. Interior gaps cluster at 16–28%, which is exactly the CondPageBreak(58mm) guard that stops a heading landing at the foot of a page.

Tightening the guard makes it worse, not better. At 46mm the count of gaps falls from nine to seven and **the worst rises from 31% to 43%**, because a heading allowed to sit lower pushes the figure block after it off the page entirely. **58mm is the optimum, and it was found by measuring rather than by preference.**

B.2.17.2.2 And two of the audit's own measurements were wrong first

The first attempt used the PDF text matrix directly and reported a median inked fraction of **103%** — impossible — with y-coordinates of 5, 6 and –220. The text matrix is not page geometry until it is composed with the current transformation matrix.

The second attempt counted characters and not images, and flagged seven pages as having a 25–45% gap at the head. **All seven begin with a figure.** The space was not empty; it was occupied by something the measurement could not see.

Both are the same error as the audits they were written to replace: a check looking at the wrong artefact. An audit is not exempt from the discipline it enforces, and this one needed two corrections before its first clean reading.

B.2.17.2.3 What the figure audit found

Twenty-eight figures, each checked against its caption and its file.

Clean: every figure has a caption and every caption a figure; no image is blank or unusually dark; no file is duplicated; nothing in fig/ is unused; and after one correction **every figure number matches the chapter it sits in.**

The one correction. Figures 23.1 and 23.2 appeared in the wrong order — the arity figure sits in §23.3.2 and the step-law figure in §23.3.4, so the numbering ran backwards against the reading order. **Swapped.**

And one defect that is recorded rather than repaired.

figures at 300 dpi or better	5
between 200 and 300	8
under 200 dpi	15
median	164 dpi

Fifteen figures were produced by an earlier pipeline whose default was figure.dpi = 160, and at 163 mm rendered width that gives 140–164 dpi — below print standard. **Regenerating them needs the source data of the sessions that made them, which this one does not carry.** Appendix D.3 lists them with the rest of what was inherited and not recomputed.

The audit's value here is not that it fixed the figures. It is that the number is now written down instead of unknown.

B.2.17.2.4 Why the REDUNDANCY audit was needed

Because two sections survived their own supersession and passed all five.

A summary table listed Q's eight items while Appendix F enumerated, indexed and closed them — **the same content, maintained twice.** And **a status table still read** *"nine literatures entered, nine named and unentered"* **and** *"order recovery beyond trees: open, and known to be hard"*, both superseded hours earlier by §F.5 and §23.3.

audit	why it passed
lattice · equations	neither is a structural or numerical claim
consistency	both were internally consistent and cited sections that exist
artefact	both rendered correctly
coherence	both were well-formed tables with valid references

A stale statement is not inconsistent with anything. It agrees with the version of the book that no longer exists. That is why the check has to compare a summary against what it summarises, rather than a document against itself.

The remedy was to delete a chapter. What survived of it — one paragraph on what the collection closed — opens Chapter 25, where it does work instead of restating.

B.2.17.3 Exhaustive, and what to do when it is impossible

Exhaustive means every case, not a representative sample. The distinction is not pedantic:

checked	how	result
the level-3 signature, $2 \times 2 \times 2$	all 256 subsets	189 tested, 30 multi-instance classes, 0 mixed
the same, $2 \times 2 \times 3$	all 4,096 subsets	3,838 tested, 210 classes, 0 mixed
the same, sampled at $d = 3, 4$	2,072 instances	0 mixed — and a sample cannot exclude a mixed pair

Where exhaustion is impossible, say so and name the bound. Λ has 976 cells and 2^{976} subsets. **What was exhausted: all 28 two-dimensional projections, all 475,800 pairs, every named fibre.** What was not: the subsets. **The audit records both.**

An audit that samples is a measurement. An audit that exhausts is a proof over its domain. Both are useful and they must not be reported in the same words.

B.2.18 Triangulation — three results locate a fourth

For every three results or bounds that measure the same object, triangulate: they identify a related data point that none of them states.

Three measurements of one thing are not three facts. They are a position.

B.2.18.1 Worked, from §23.3

the three	what they measured	the fourth, located
valid orders disconnected · E-landscape traps · cell-growth peaks	connectivity failure in three different spaces	the disconnection is in the object, not the space — so no local method in any space can be complete
step 1 at $d = 2$ · step 2 at $d = 3$ · step 4 at $d = 4$	growth step, three dimensions	step = $2^{(d-2)}$, then proved by subcube removal
arity 2 bijunctive · arity 3 bijunctive · arity 4 contains 1-in-3-SAT	the constraint language	the boundary is arity 3 and it is exact — the procedure and the hardness proof are one law read at two arities
three reductions, all over-constraining	the encoding	the map $X \rightarrow C(X)$ is global, non-separable and non-monotone — it admits no local editing

Each fourth point was invisible in any single measurement and forced by the three together. The step law is the clearest: 1, 2, 4 fixed the $\frac{3}{4}$ ratio, and the ratio gave the subcube construction, which proved it.

B.2.18.2 And it is a check as well as a method

Triangulation fails loudly. If three measurements of one object do not locate a consistent fourth, one of them is wrong — and that is how three corrections were caught in a single session:

A map that is global, non-separable and non-monotone should have a SMALL image. Measured: 39% at $d = 3$ and 49.5% at $d = 4$, rising. The three coordinates were right and the inference from them was not — **they locate an obstruction to EDITING the map, not to its range.**

Three consistent measurements give a fourth result. Three measurements that give no consistent fourth give a correction instead. Both outcomes are useful and the protocol does not distinguish them in advance.

B.2.18.3 Why it belongs with the protocols

It is P8 with a count attached. Every failure is a bound; **three bounds on one object are a coordinate.** B.2.15 says to record failures as bounds; this says what to do once three of them are in hand.

And it composes with B.2.16. The grid shows which cells are empty; triangulation fills one from three of its neighbours.

B.3 What the protocols are for

Each prevents a specific failure, and each failure occurred:

protocol	failure it prevents	instance
B.2.1 channel construction	claiming cells across a label change	Si I, P II
B.2.2 admissibility	bracketing unresolved levels	—
B.2.3 cost resolvability	reading rounding as physics	Al I nf
B.2.4 dilution	excluding a viable species	Ar II, Bi I
B.2.5 census first	searching without targets	27 sequences enumerated
B.2.6 declines recorded	an uncheckable collection	Ne II, Cu II
B.2.7 exhaustive testing	a claim resting on a sample	A.4 on all cells
B.2.8 two routes	an undetectable derivation error	$w = 3v \cdot e$
B.2.9 refuse, don't coerce	a missing value read as a real one	the cap discard
B.2.10 retrieval	abandoning a cell when a source is blocked	$p = 6$ vs $p = 1$
B.2.11 provenance	an untraceable number	153 channels
B.2.12 withdrawal	an unfalsifiable record	48 entries
B.2.13 commit before you look	a choice that depended on its answer	the anchoring failure
B.2.14 compute then write	the dominant failure	most of Chapter 20
B.2.15 apply P8 to every failure	a failure discarded rather than banked	ten bounds on §23.3

The protocols are not advice. They are the failure modes of this work, inverted.

Appendix C — Data and provenance

Serving PART III — THE METHOD. Every number the method was run on, with its source, its provenance and its known defects.

Every channel in the collection, with its source, its limit, and what it contributes. 153 channels, **1105 interior cells**, bracket **1105 of 1105**.

The 763 cells of the earlier verification are included where their channels appear here; the totals in Chapter 17 state the union.

C.1 Sources

compilation	spectra drawn
Kaufman & Martin 1991, JPCRD 20, 775	Al I, Al II
Kramida & Martin 1997, JPCRD 26, 1185	Be I
NIST ASD	Ar II, Be II, Bi I, C II, Ca II, Cd II, Ga I, He I, He II, Hg II, K II, Li I, Li II, Mg II, N II, Na II, Ne I (3/2), Ne I (1/2), Si I, Si II, Zn II
Sansonetti 2008, JPCRD 37, 1659	Na I
Sansonetti 2008, JPCRD 37, 7	K I

Additional sources used for exotic systems and collective quantities:

Hori *et al.*, *Nature* **475**, 484 (2011) and *Phys. Rev. Lett.* **96**, 243401 (2006), both accessed via CODATA 2010 Table XII (arXiv:1203.5425) and arXiv:1304.4330 · Korobov, *Phys. Rev. A* **77**, 042506 (2008) · Singer, Stanojevic, Weidemüller & Côté, *J. Phys. B* **38**, S295 (2005) · Sugar & Corliss 1985, *JPCRD* **14** Suppl. 2 · Sugar & Musgrove 1990, *JPCRD* **19**, 527 and 1995, *JPCRD* **24**, 1803.

C.2 Channels

species	channel	<i>n</i>	me m	cells	bracket	<i>v</i> range	δ	sprea d	Z_eff	limit (cm ⁻¹)
Al I	3s ² nf ² F°	4-55	52	50	50/50	4.0-55.0	+0.0429	0.0552	1	48,278.480
Al I	3s ² nd ² D	3-34	32	30	30/30	2.6-33.7	+0.0248	0.6297	1	48,278.480
Al I	3s ² ns ² S	4-16	13	11	11/11	2.2-14.2	+1.7670	0.0537	1	48,278.480
Al I	3s ² np ² P°	3-7	5	3	3/3	1.5-5.7	+1.3365	0.2080	1	48,278.480
Al II	3sns ¹ S	4-16	13	11	11/11	2.8-14.8	+1.2028	0.0147	2	151,862.500
Al II	3sns ³ S	4-13	10	8	8/8	2.7-11.7	+1.2711	0.0474	2	151,862.500
Al II	3snd ¹ D	3-11	9	7	7/7	3.2-10.9	+0.0603	0.3897	2	151,862.500
Al II	3sng	5-13	9	7	7/7	5.0-13.0	+0.0212	0.0070	2	151,862.500
Al II	3snd ³ D	3-10	8	6	6/6	2.8-9.8	+0.2024	0.0096	2	151,862.500
Al II	3snf ³ F°	4-11	8	6	6/6	3.9-11.0	+0.0140	0.4088	2	151,862.500
Al II	3snp ³ P°	4-10	7	5	5/5	3.1-9.1	+0.8982	0.0571	2	151,862.500
Ar II	(³ P)ns ⁴ P J=5/2	4-6	3	1	1/1	2.2-4.3	+1.7463	0.0477	2	222,848.300
Ar II	(³ P)ns ⁴ P J=3/2	4-6	3	1	1/1	2.2-4.3	+1.7246	0.0663	2	222,848.300
Ar II	(³ P)ns ⁴ P J=1/2	4-6	3	1	1/1	2.2-4.4	+1.6821	0.1507	2	222,848.300
Ar II	(³ P)ns ² P J=3/2	4-6	3	1	1/1	2.3-4.4	+1.6711	0.1084	2	222,848.300
Ar II	(³ P)ns ² P J=1/2	4-6	3	1	1/1	2.3-4.4	+1.6355	0.1526	2	222,848.300
Ar II	(³ P)nd ⁴ D J=7/2	3-5	3	1	1/1	2.2-4.4	+0.6930	0.1695	2	222,848.300
Ar II	(³ P)nd ⁴ D J=5/2	3-5	3	1	1/1	2.2-4.4	+0.6880	0.1753	2	222,848.300
Ar II	(³ P)nd ⁴ D J=3/2	3-5	3	1	1/1	2.2-4.4	+0.6786	0.1919	2	222,848.300
Ar II	(³ P)nd ⁴ D J=1/2	3-5	3	1	1/1	2.2-4.4	+0.6669	0.2155	2	222,848.300

species	channel	<i>n</i>	me m	cells	bracket	v range	δ	sprea d	Z_eff	limit (cm ⁻¹)
Ar II	(³ P)nd ⁴ F J=9/2	3-5	3	1	1/1	2.3-4.4	+0.6121	0.0885	2	222,848.300
Be I	2snp ¹ P°	3-13	11	9	9/9	2.7-12.6	+0.3590	0.1025	1	75,192.500
Be I	2snd ³ D	3-12	10	8	8/8	2.9-11.9	+0.1109	0.0129	1	75,192.500
Be I	2snd ¹ D	3-12	10	8	8/8	3.2-12.1	-0.1152	0.0970	1	75,192.500
Be I	2sns ¹ S	3-11	9	7	7/7	2.3-10.3	+0.6774	0.0136	1	75,192.500
Be I	2sns ³ S	3-8	6	4	4/4	2.2-7.2	+0.7910	0.0429	1	75,192.500
Be I	2snp ³ P°	3-8	6	4	4/4	2.6-7.6	+0.3773	0.0374	1	75,192.500
Be I	2snf ³ F°	4-7	4	2	2/2	4.0-7.0	+0.0305	0.0063	1	75,192.500
Be II	ns ² S	2-10	9	7	7/7	1.7-9.7	+0.2623	0.0112	2	146,882.860
Be II	np ² P°	2-10	9	7	7/7	2.0-10.0	+0.0491	0.0039	2	146,882.860
Be II	nd ² D	3-9	7	5	5/5	3.0-9.0	+0.0021	0.0004	2	146,882.860
Be II	nf ² F°	4-10	7	5	5/5	4.0-10.0	+0.0001	0.0001	2	146,882.860
Be II	ng ² G	5-10	6	4	4/4	5.0-10.0	-0.0001	0.0001	2	146,882.860
Be II	nh ² H°	6-10	5	3	3/3	6.0-10.0	-0.0002	0.0001	2	146,882.860
Bi I	(³ Po)ns 2[0] J=1/2	7-11	5	3	3/3	2.0-6.1	+4.9036	0.0649	1	58,761.650
Bi I	(³ Po)nd 2[2] J=3/2	6-10	5	3	3/3	2.7-6.7	+3.2646	0.0470	1	58,761.650
Bi I	(³ Po)nd 2[2] J=5/2	6-9	4	2	2/2	2.8-5.8	+3.2071	0.0272	1	58,761.650
Bi I	(³ Po)np 2[1]° J=1/2	7-9	3	1	1/1	2.5-4.6	+4.4712	0.0588	1	58,761.650
C II	2s ² ns ² S	3-8	6	4	4/4	2.3-7.3	+0.6624	0.0063	2	196,664.700
C II	2s ² np ² P°	3-7	5	3	3/3	2.6-6.7	+0.3928	0.1344	2	196,664.700
C II	2s ² nd ² D	3-7	5	3	3/3	2.9-6.9	+0.0924	0.0645	2	196,664.700
C II	2s ² nf ² F°	4-7	4	2	2/2	4.0-7.0	+0.0220	0.0069	2	196,664.700
C II	2s ² ng ² G	5-7	3	1	1/1	5.0-7.0	+0.0054	0.0011	2	196,664.700
Ca II	nd ² D	3-16	14	12	12/12	2.3-15.4	+0.6338	0.0615	2	95,751.870
Ca II	ns ² S	4-10	7	5	5/5	2.1-8.2	+1.8192	0.0529	2	95,751.870
Ca II	nf ² F°	4-10	7	5	5/5	4.0-10.0	+0.0248	0.0088	2	95,751.870
Ca II	ng ² G	5-9	5	3	3/3	5.0-9.0	+0.0048	0.0011	2	95,751.870
Ca II	np ² P°	4-6	3	1	1/1	2.5-4.5	+1.4748	0.0489	2	95,751.870
Cd II	ns	5-13	9	7	7/7	1.8-9.9	+3.1175	0.1092	2	136,374.740
Cd II	nd	5-13	9	7	7/7	3.1-11.1	+1.8906	0.0534	2	136,374.740
Cd II	np	5-12	8	6	6/6	2.2-9.3	+2.7145	0.1825	2	136,374.740
Cd II	ng	5-11	7	5	5/5	5.0-11.0	+0.0075	0.0013	2	136,374.740
Cd II	nf	4-9	6	4	4/4	4.0-9.0	+0.0333	0.0918	2	136,374.740
Ga I	4s ² ns ² S	5-28	24	22	22/22	2.2-25.2	+2.8013	0.0535	1	48,387.634
Ga I	4s ² nd ² D	4-27	23	21	21/21	2.8-25.7	+1.2926	0.1920	1	48,387.634
Ga I	4s ² np ² P° high	41-5 5	15	13	13/13	38.8-52.7	+2.2107	0.1209	1	48,387.634
Ga I	4s ² nf ² F°	4-8	5	3	3/3	4.0-8.0	+0.0252	0.0057	1	48,387.634
Ga I	4s ² np ² P° low	4-7	4	2	2/2	1.5-4.7	+2.3450	0.2026	1	48,387.634
He I	1sns ³ S	2-35	25	23	23/23	1.7-34.7	+0.2965	0.0164	1	198,310.666
He I	1sns ¹ S	2-35	25	23	23/23	1.9-34.9	+0.1392	0.0118	1	198,310.666
He I	1snp ¹ P°	2-35	25	23	23/23	2.0-35.0	-0.0133	0.0050	1	198,310.666
He I	1snd ¹ D	3-35	24	22	22/22	3.0-35.0	+0.0007	0.0019	1	198,310.666
He I	1snf ¹ F°	4-35	23	21	21/21	4.0-35.0	-0.0010	0.0020	1	198,310.666
He I	1snd ³ D	3-35	22	20	20/20	3.0-35.0	+0.0014	0.0018	1	198,310.666
He I	1sng ¹ G	5-35	22	20	20/20	5.0-35.0	-0.0014	0.0020	1	198,310.666
He I	1snh ¹ H°	6-35	21	19	19/19	6.0-35.0	-0.0015	0.0020	1	198,310.666
He I	1sni ¹ I	7-35	20	18	18/18	7.0-35.0	-0.0016	0.0019	1	198,310.666
He II	ns	1-10	10	8	8/8	1.0-10.0	-0.0003	0.0006	2	438,908.871
He II	np	2-10	9	7	7/7	2.0-10.0	-0.0004	0.0005	2	438,908.871
He II	nd	3-10	8	6	6/6	3.0-10.0	-0.0004	0.0005	2	438,908.871

species	channel	<i>n</i>	me m	cells	bracket	v range	δ	sprea d	Z_eff	limit (cm ⁻¹)
He II	nf	4-10	7	5	5/5	4.0-10.0	-0.0005	0.0004	2	438,908.871
He II	ng	5-10	6	4	4/4	5.0-10.0	-0.0005	0.0003	2	438,908.871
He II	nh	6-9	4	2	2/2	6.0-9.0	-0.0005	0.0002	2	438,908.871
He II	ni	7-9	3	1	1/1	7.0-9.0	-0.0005	0.0001	2	438,908.871
Hg II	ns	6-11	6	4	4/4	1.7-6.8	+4.1865	0.1423	2	151,280.000
Hg II	nd	6-11	6	4	4/4	3.1-8.2	+2.8668	0.0609	2	151,280.000
Hg II	ng	5-9	5	3	3/3	5.0-9.0	+0.0101	0.0011	2	151,280.000
Hg II	np	6-9	4	2	2/2	2.2-5.3	+3.7302	0.1626	2	151,280.000
Hg II	nf	5-8	4	2	2/2	4.0-6.9	+1.0615	0.0317	2	151,280.000
K I	nd	3-46	44	42	42/42	2.9-45.7	+0.2679	0.1326	1	35,009.814
K I	ns	4-46	43	41	41/41	1.8-43.8	+2.1831	0.0490	1	35,009.814
K I	np	4-18	15	13	13/13	2.2-16.3	+1.7202	0.0533	1	35,009.814
K I	nf	4-14	11	9	9/9	4.0-14.0	+0.0092	0.0051	1	35,009.814
K II	3p ^o nf 2[3/2] J=1	4-6	3	1	1/1	4.0-6.0	+0.0426	0.0069	2	255,072.800
K II	3p ^o nf 2[9/2] J=5	4-6	3	1	1/1	4.0-6.0	+0.0336	0.0071	2	255,072.800
K II	3p ^o nf 2[5/2] J=3	4-6	3	1	1/1	4.0-6.0	+0.0242	0.0052	2	255,072.800
K II	3p ^o nf 2[7/2] J=4	4-6	3	1	1/1	4.0-6.0	+0.0143	0.0059	2	255,072.800
Li I	np 2P ^o	2-42	41	39	39/39	2.0-41.8	+0.1477	0.6779	1	43,487.114
Li I	ns 2S	2-11	10	8	8/8	1.6-10.6	+0.4014	0.0180	1	43,487.114
Li I	nd 2D	3-12	10	8	8/8	3.0-12.0	+0.0031	0.0129	1	43,487.114
Li II	1sns 3S	2-9	8	6	6/6	1.8-8.8	+0.1815	0.0115	2	610,078.400
Li II	1sns 1S	2-9	8	6	6/6	1.9-8.9	+0.0743	0.0037	2	610,078.400
Li II	1snp 3P ^o	2-8	7	5	5/5	1.9-7.9	+0.0540	0.0010	2	610,078.400
Li II	1snp 1P ^o	2-8	7	5	5/5	2.0-8.0	-0.0145	0.0065	2	610,078.400
Li II	1snd 3D	3-8	6	4	4/4	3.0-8.0	+0.0024	0.0006	2	610,078.400
Li II	1snd 1D	3-8	6	4	4/4	3.0-8.0	+0.0009	0.0003	2	610,078.400
Li II	1snf 3F ^o	4-8	5	3	3/3	4.0-8.0	+0.0002	0.0002	2	610,078.400
Li II	1snf 1F ^o	4-8	5	3	3/3	4.0-8.0	-0.0001	0.0004	2	610,078.400
Li II	1sng	5-8	4	2	2/2	5.0-8.0	-0.0002	0.0002	2	610,078.400
Li II	1snh	6-8	3	1	1/1	6.0-8.0	-0.0003	0.0001	2	610,078.400
Mg II	ns 2S	3-10	8	6	6/6	1.9-8.9	+1.0749	0.0289	2	121,267.640
Mg II	nd 2D	3-10	8	6	6/6	3.0-10.0	+0.0411	0.0145	2	121,267.640
Mg II	np 2P ^o	3-9	7	5	5/5	2.3-8.3	+0.7077	0.0361	2	121,267.640
Mg II	nf 2F ^o	4-10	7	5	5/5	4.0-10.0	+0.0030	0.0009	2	121,267.640
Mg II	ng 2G	5-10	6	4	4/4	5.0-10.0	+0.0007	0.0001	2	121,267.640
Mg II	nh 2H ^o	6-9	4	2	2/2	6.0-9.0	+0.0002	0.0000	2	121,267.640
Mg II	ni 2I	7-9	3	1	1/1	7.0-9.0	+0.0000	0.0000	2	121,267.640
N II	2p·ns 3P ^o J=0	3-6	4	2	2/2	2.2-5.2	+0.7747	0.0264	2	238,750.500
N II	2p·ns 3P ^o J=1	3-6	4	2	2/2	2.2-5.2	+0.7731	0.0260	2	238,750.500
N II	2p·ns 3P ^o J=2	3-6	4	2	2/2	2.2-5.3	+0.7633	0.0464	2	238,750.500
N II	2p·ns 1P ^o J=1	3-6	4	2	2/2	2.2-5.3	+0.7325	0.0824	2	238,750.500
Na I	ns	3-20	18	16	16/16	1.6-18.7	+1.3506	0.0250	1	41,449.451
Na I	np	3-20	18	16	16/16	2.1-19.1	+0.8584	0.0279	1	41,449.451
Na I	nd	3-20	18	16	16/16	3.0-20.0	+0.0141	0.0045	1	41,449.451
Na I	nf	4-20	17	15	15/15	4.0-20.0	+0.0014	0.0004	1	41,449.451
Na I	ng	5-20	16	14	14/14	5.0-20.0	+0.0003	0.0001	1	41,449.451
Na I	nh	6-10	5	3	3/3	6.0-10.0	+0.0000	0.0000	1	41,449.451
Na II	(2P ^o 3/2)ns 2[3/2] ^o J=2	3-6	4	2	2/2	1.9-5.0	+1.0385	0.0284	2	381,390.200
Na II	(2P ^o 3/2)ns 2[3/2] ^o J=1	3-6	4	2	2/2	1.9-5.0	+1.0268	0.0371	2	381,390.200
Na II	(2P ^o 3/2)nd 2[1/2] ^o J=0	3-5	3	1	1/1	2.9-4.9	+0.0671	0.0083	2	381,390.200
Na II	(2P ^o 3/2)nd 2[1/2] ^o J=1	3-5	3	1	1/1	2.9-4.9	+0.0641	0.0085	2	381,390.200

species	channel	<i>n</i>	me m	cells	bracket	<i>v</i> range	δ	sprea d	Z_eff	limit (cm ⁻¹)
Na II	(² P° _{3/2})nd 2[3/2] ^o J=2	3-5	3	1	1/1	2.9-4.9	+0.0556	0.0027	2	381,390.200
Na II	(² P° _{3/2})nd 2[3/2] ^o J=1	3-5	3	1	1/1	3.0-5.0	+0.0268	0.0026	2	381,390.200
Ne I (3/2)	ns 2[3/2] ^o J=1	3-20	18	16	16/16	1.7-18.7	+1.2968	0.0488	1	173,929.750
Ne I (3/2)	nd 2[1/2] ^o J=1	3-15	13	11	11/11	3.0-15.0	+0.0277	0.0173	1	173,929.750
Ne I (3/2)	ns 2[3/2] ^o J=2	3-13	11	9	9/9	1.7-11.7	+1.3191	0.0276	1	173,929.750
Ne I (3/2)	nd 2[7/2] ^o J=4	3-13	11	9	9/9	3.0-13.0	+0.0226	0.0073	1	173,929.750
Ne I (3/2)	nd 2[3/2] ^o J=2	3-13	11	9	9/9	3.0-13.0	+0.0164	0.0176	1	173,929.750
Ne I (3/2)	nd 2[5/2] ^o J=3	3-12	10	8	8/8	3.0-12.0	+0.0081	0.0054	1	173,929.750
Ne I (3/2)	nd 2[1/2] ^o J=0	3-11	9	7	7/7	3.0-11.0	+0.0323	0.0078	1	173,929.750
Ne I (3/2)	np 2[5/2] J=3	3-10	8	6	6/6	2.1-9.2	+0.8486	0.0383	1	173,929.750
Ne I (3/2)	np 2[1/2] J=1	3-8	6	4	4/4	2.1-7.1	+0.8941	0.0623	1	173,929.750
Ne I (3/2)	nf 2[9/2] J=5	4-9	6	4	4/4	4.0-9.0	+0.0035	0.0007	1	173,929.750
Ne I (3/2)	nf 2[7/2] J=4	4-8	5	3	3/3	4.0-8.0	-0.0015	0.0006	1	173,929.750
Ne I (3/2)	ng 2[11/2] ^o J=6	5-7	3	1	1/1	5.0-7.0	+0.0012	0.0000	1	173,929.750
Ne I (1/2)	ns 2[1/2] ^o J=1	3-20	18	16	16/16	1.7-18.7	+1.3027	0.0266	1	174,710.170
Ne I (1/2)	nd 2[3/2] ^o J=1	3-20	18	16	16/16	3.0-20.0	+0.0137	0.0120	1	174,710.170
Ne I (1/2)	nd 2[5/2] ^o J=3	3-10	8	6	6/6	3.0-10.0	+0.0178	0.0064	1	174,710.170
Ne I (1/2)	ns 2[1/2] ^o J=0	3-10	8	6	6/6	1.7-8.7	+1.3207	0.0265	1	174,710.170
Si I	3p·nd ¹ D° J=2	3-5	3	1	1/1	2.4-4.5	+0.5412	0.0461	1	65,747.760
Si I	3p·nd ³ F° J=2	3-5	3	1	1/1	2.6-4.6	+0.3718	0.0179	1	65,747.760
Si I	3p·nd ³ F° J=3	3-5	3	1	1/1	2.6-4.7	+0.3546	0.0309	1	65,747.760
Si I	3p·nd ³ F° J=4	3-5	3	1	1/1	2.6-4.7	+0.3188	0.0886	1	65,747.760
Si II	3s ² np ² P°	3-10	8	6	6/6	1.8-9.1	+1.0223	0.2717	2	131,838.140
Si II	3s ² nf ² F°	4-11	8	6	6/6	3.9-10.9	+0.0845	0.0370	2	131,838.140
Si II	3s ² ng ² G	5-11	7	5	5/5	5.0-11.0	+0.0176	0.0044	2	131,838.140
Si II	3s ² ns ² S	4-9	6	4	4/4	2.6-7.6	+1.3943	0.0468	2	131,838.140
Si II	3s ² nd ² D	3-8	6	4	4/4	2.9-7.7	+0.2289	0.1599	2	131,838.140
Zn II	ns ² S	4-9	6	4	4/4	1.7-6.8	+2.2084	0.0679	2	144,892.600
Zn II	nd ² D	4-9	6	4	4/4	3.0-8.0	+0.9583	0.0228	2	144,892.600
Zn II	np ² P°	4-8	5	3	3/3	2.1-6.1	+1.8386	0.0616	2	144,892.600
Zn II	nf ² F°	4-8	5	3	3/3	4.0-8.0	+0.0172	0.0044	2	144,892.600
Zn II	ng ² G	5-7	3	1	1/1	5.0-7.0	+0.0038	0.0008	2	144,892.600

Totals: 153 channels, 1105 cells, bracket 1105/1105.

C.3 Flagged channels

Channels whose δ spread exceeds 0.25 or whose *V* departs from 4*v*/3 by more than 5% are flagged. **Every flag in this collection has an identified cause; none is unexplained.**

specie s	channel	δ spread	cause
Al I	3s ² nd ² D	0.630	compilers relabelled this series; y ² D removed as a perturber
Al II	3snd ¹ D	0.390	local perturber
Al II	3snf ³ F°	0.409	δ reverses sign between <i>n</i> = 6 and 7 — local perturber
Li I	np ² P°	0.678	39 cells to <i>n</i> = 42; quotation coarsens above <i>n</i> = 33
Si II	3s ² np ² P°	0.272	perturber

C.4 Reproducibility

Every computation in this book is reproducible from the level tables above and the rules of Chapters 13 and 14. No fitted parameter enters at any point; the only inputs are measured or calculated energy levels, their stated uncertainties, and the ionisation limits listed in C.2.

Appendix D — Margins, and what was recomputed

Serving PART IV — THE RECORD. What the conclusions can bear, and what was inherited rather than recomputed.

A referee should treat two classes of claim differently, and this appendix separates them.

D.1 Margins on the load-bearing conclusions

How far is each result from the threshold that would overturn it?

conclusion	statistic	margin	breaks down at
the periodic table is not closed	E = 36	exact arithmetic	—
closure $\Leftrightarrow \mathcal{R}(X) = X$	200/200 agreements	exact	—
$\mathbb{D} \geq \dim q - \dim p$	420 random maps	exact — rank-nullity	—
$V > 2$ for any monotone sequence	Proposition 14.1	exact, proved	—
Rydberg floor $V \geq 32/11$	2.909091 at $v = 2$	exact	—
E3 admissibility criterion	2,513/2,513, nine functions	exact	—
$\chi_{\wedge}$ total	30,000 ambient points	exact	—
order recovery on a tree	20/20, two cap settings	—	a cyclic constraint graph
the bracket holds	1,442/1,442	—	one reordering
perturbation bounds	tightest 1.40 cm⁻¹	deductive	a 1.4 cm ⁻¹ shift at that cell
isoelectronic interpolation	1.3% median	13× better than at an end	the sequence end
$V = 4v/3$ on real channels	0.06%–4.4% median	—	coarse quotation
void-free fraction stable	27.7–30.1%	2.4 points over a 100× range	—
filled- d core raises δ	+0.35 at low ℓ	—	$\ell \geq 3$, where cores converge

Three rows deserve a sceptic's attention first.

The bracket's 1,442/1,442 breaks on a single reordering. §25.3 states the condition exactly — $|\Delta T| > 2Z^2R/v^3$ — and §18.4 argues that the collection never enters the regime where it can occur. **That argument, not the count, is what a critic should attack.**

The perturbation bounds are as strong as the failure condition and no stronger. If the condition is wrong, 1,061 bounds go with it.

The isoelectronic interpolation is quoted at 1.3% and used at the end of a sequence, where the measured cost is 2.1% for a hold-out and **3.5% for a genuine fifth member**. The Sc VI route of §18.6 uses the last figure, not the first.

D.2 What was computed in this revision

Fourteen results rest on computations performed for this book, each reproducible from Appendix C's tables and the rules of Part III:

$E(X)$ for seven indices, the periodic table among them · closure $\Leftrightarrow \mathcal{R}(X) = X$ on 200 constructions · $\mathbb{D} \geq \dim q - \dim p$ on 420 random nonlinear maps · the E3 criterion on 2,513 tests · χ^2 totality on 30,000 uniformly sampled ambient points · order recovery by tree propagation, 20/20 · **the 1,105 cells added here, across all 153 channels** · the failure condition and its inversion to 1,061 perturbation bounds · the isoelectronic hold-out and fifth-member tests · the C_6 cost-law check against Singer *et al.* · the antiprotonic-helium two-route check at 0.09σ · the census by antisymmetric-state enumeration · Proposition 14.1 re-derived · **the three falsification tests of §25.6** — the book's own, distinct from §24.3.4.2's test of an external prediction

D.3 What was inherited and not recomputed

Six results come from earlier work and were not re-run:

the rule-ablation costs of §17.2 · the Sr I node coverages and fine-structure figures of §17.2 · the Ti I channel inventory · the multi-target failure counts of §11.3 · the Sc VI prediction's input defects · the earlier verification's per-species medians.

And fifteen figures at 140–164 dpi, produced by an earlier plotting pipeline whose default was `figure.dpi = 160`: f01, f02, f03, f04, f05, f06, f07, f08, f09, f10, f11, f12, f13, f14, f18. **Measured and recorded, not regenerated** — the source data of the sessions that produced them is not in this one. B.2.17.2.3 gives the distribution.

These rest on methods described in Appendix C but were performed before this revision. Chapter 21 records one hundred and ninety-eight errors found in work of exactly that kind — **which is reason to re-run them rather than to trust them.**

The Sc VI defects are the most consequential of the six, because the book's single prediction rests on them, and §18.6's arithmetic was independently reproduced here — 735,091 against 735,092 — while its *inputs* were not.

One inconsistency was found in the inherited material and is recorded. The exclusion discussion of §17.2 states that Sc VI's "highest measured level is 5s at 696,400 cm^{-1} ". **That figure and the stated $\delta(5s) = 0.9812$ cannot both describe the $^4S^\circ$ channel:** from 696,400 the defect is 0.514, and the stated defect requires 5s at 648,096. **The 696,400 is Sc VI's highest measured level overall, not the $^4S^\circ$ 5s**, and the prose ran them together. The prediction is unaffected, having been computed from the defects.

D.4 Computational methods

All computations in Python 3 with NumPy. **Closure tests are either exhaustive over all pairs — stated where so — or on uniform random samples with the sample size reported.** Box searches enumerate all 2^d corners of candidate base cells.

Ritz fits use ordinary least squares on δ against $1/n^2$, with σ from the standard prediction error including the leverage term $s \cdot \sqrt{1 + 1/k + (x_0 - \bar{x})^2/S_{xx}}$.

Quotation granularity q is read from the source table as the smallest nonzero increment appearing in it — 10^{-3} cm^{-1} for most ASD entries, coarsening to 0.1 or 1 cm^{-1} in the high- n regions of several channels — and enters `v_V` directly.

Weighted centroids over fine-structure components use statistical weights $2J + 1$.

Where a channel's neighbours are not consecutive in n , the bracket is still evaluated but V is not, since $V = 4x/(h|p-1|)$ assumes unit step. That exclusion is applied silently in the totals and is the reason the V counts are smaller than the cell counts in Appendix C.

Appendix E — The mathematics, indexed

Serving PART IV — THE RECORD. The book's own mathematics as a closed index — thirty-two elements, sixteen fibres, $E = 0$.

The book's method, applied to the book's own mathematics. Every mathematical element is a cell; the coordinates are its kind, its language, its status, how it was verified, and whether a precedent exists. **The index is then closed, and $E(X)$ counts the questions the structure admits and the book does not answer.**

E.1 Why the first attempt failed, and what it established

Five coordinates were assigned: kind, language, status, verification, precedent. The closure admitted hundreds of cells. **The reason is a precondition the book had never stated:**

$\mathcal{R}$ recovers **MONOTONE** bounds. It applies to **ORDERED** coordinates only.

Kind and language are categorical. Ordering *definition* < *mechanism* < *formula* < *theorem* is arbitrary, and a monotone bound on an arbitrary ranking admits everything above it. **Λ 's own coordinates are all ordered — n, l, k, q, e, f, g , 2S are counts — so the book never met the case.**

The repair is the shape Λ already has: fibre over the categorical axes, and put the ordered coordinates inside.

E.2 The structure

coordinate	values	kind
kind	definition · mechanism · formula · theorem · law · method · measurement	fibre
language	order · combinatorics · analysis · complexity · physics · algebraic geometry	fibre
status	withdrawn < conjectured < measured < verified < proved	ordered
verification	cited < sampled < exhaustive	ordered
precedent	none found < found	ordered

Thirty-two elements occupy sixteen of the forty-two possible fibres.

E.3 The two constraints, found by closing

$E(X) > 0$ in three fibres, and the excess cells shared a form. §12's diagnostic loop says a separating form in the excess means a missing constraint. **Two were found:**

verification = cited $\Rightarrow$ precedent = found. You cannot cite a precedent that does not exist.
status $\geq$ measured $\Rightarrow$ verification $\geq$ sampled. If the book measured it, the book verified it.
Cited can coexist only with *conjectured* or *withdrawn* — a claim nobody here checked.

Both have the form of Λ 's own bounds — a monotone bound of one coordinate by another — and neither was stated before the index forced it.

E.4 The two elements the closure demanded

One cell survived both constraints: an analysis formula both proved and exhaustively verified. The book had $V = 4\nu/3$ (measured, exhaustive) and λ^2 as the Newton decrement (proved, cited) and nothing in between.

E.4.1 $\lambda^2 = (2/3)T$, proved and exhaustively verified

For a Rydberg term $T(\nu) = Z^2R/\nu^2$:

$$\lambda^2 = (T')^2/T'' = (2Z^2R/\nu^3)^2 / (6Z^2R/\nu^4) = (2/3) \cdot Z^2R/\nu^2 = (2/3)T$$

Proved in one line. And now verified exhaustively: 12,000 evaluations over $Z = 1...30$ and $v = 1.01...60$, zero violations, worst relative error 7.1×10^{-16} — floating-point exactness across the whole domain.

The identity was stated in Chapter 16 and cited to Nesterov-Nemirovskii. It had never been checked here.

E.4.2 The pointwise-minimum reconstruction

$\Lambda = \{x : x_i \leq \min_j \phi_{ij}(x_j) \text{ for every } i\}$ — verified at **976 = 976, exact.**

$\mathcal{R}$ **recovers forty-eight bounds and takes the smallest at each cell.** No single bound on g is tightest anywhere near everywhere — q on 92.6% of cells, k on 39.9%, f on 35.1%, down to n on 11.2%. **The binding constraint changes cell by cell, and the pointwise minimum is the operator's entire content.**

E.4.3 And it is not the coupling F carries

F has one $\min()$. The recovered system has one at every cell.

	in F — the defining system	under $\mathcal{R}$ — the recovered system
coordinates with >1 bound	one: $g \leq \min(q, 4f+2)$	all eight
bounds per coordinate	0 or 1, except g	seven
which bound binds	fixed by the construction	changes cell by cell
cells with two bounds tied at the minimum	—	976 of 976 — 100%

Every one of the seven is tightest somewhere. For g : q on 93% of cells, k on 40%, f on 35%, l on 20%. For q : k on 100%, l on 66%, n on 48%. **No coordinate has a single binding partner.**

F 's coupling is structural and occurs once — the junction of the caterpillar, where g meets both q and $4f+2$. $\mathcal{R}$'s coupling is pointwise and occurs everywhere.

Both give 976, and they are different mechanisms. §6.5 states the defining coupling and stops there, which is correct for the expression and incomplete for the operator. **The tie at every cell is the stronger fact: the recovered system is degenerate at every point, and no cell is governed by one bound alone.**

Order-theoretic, exhaustive over all 976 cells, and no precedent found for the statement in this form.

E.4.4 Two perspectives of one definition, and one of them carries more

F and $\mathcal{R}$ are the same definition seen twice. F says *what Λ contains*; $\mathcal{R}$ says *what Λ requires*. Both give 976 cells and neither is derived from the other's coupling.

$\mathcal{R}$ is idempotent. $\Lambda \rightarrow$ fifty-six recovered bounds $\rightarrow$ generates 976 $\rightarrow$ **the same fifty-six bounds, and again the same.** One application takes any presentation to a canonical form; further applications change nothing.

The defining system is not that fixed point. F carries **seven** bounds; the normal form has **sixteen** that constrain, of fifty-six recovered. **Both generate the same set.**

E.4.4.1 And three of the four directions are functions

$\Lambda \rightarrow$ the recovered bounds	unique — this is $\mathcal{R}$
the recovered bounds $\rightarrow \Lambda$	unique — generate
the defining bounds $\rightarrow \Lambda$	unique — this is Chapter 2
$\Lambda \rightarrow$ the defining bounds	NOT a function

Many presentations give the same Λ . So $\mathcal{R}$ forgets which one was used, and the forgetting is not recoverable.

A presentation and its normal form are two perspectives of one definition, and the map between them runs one way.

E.4.4.2 Which is what the single expression contributes

F carries information Λ does not: the order in which the bounds were imposed.

The caterpillar of Figure 6.1 — a path of seven nodes with one pendant at k — is a fact about the presentation, not about the set. §6.7 proves that a single closed expression exists *because* the constraint graph is a tree, and a tree is an ordering of impositions. **Λ alone supplies no such order; §7.9 counts 1,113,045,672 maximal chains, and the construction picked one.**

That is why Chapter 6 and Chapter 10 are separate chapters. Chapter 6 needs the presentation. Chapter 10 recovers only what survives forgetting it.

E.4.4.3 A refinement of P21

P21 says a definition contained in the lattice is true in every language it can be written in. §6.1 gives Λ in six, and the languages agree.

They do not carry equal amounts. The generating-function perspective carries a nesting order the set-theoretic perspective does not, and the order is real — it determines whether a closed form exists at all.

Every perspective on a contained definition is TRUE. They are not INTERCHANGEABLE. A perspective may carry structure the object does not, and a perspective that forgets it is still correct about the object.

That is not a weakening of P21. It is the reason §6.7's proof has a hypothesis — the tree — **which is a property of how Λ was written and not of what Λ is.**

E.5 The closed index

With both constraints and both elements: $E = 0$ in all sixteen fibres.

fibre	cells	E
definition · order	1	0
definition · combinatorics	1	0
mechanism · order	1	0
formula · combinatorics	2	0
formula · analysis	3	0
formula · physics	1	0
theorem · order	5	0
theorem · combinatorics	3	0
theorem · analysis	2	0
law · complexity	1	0
method · order · combinatorics · analysis	1 each	0
measurement · combinatorics	2	0
measurement · physics · algebraic geometry	1 each	0

No combination of status, verification and precedent is admitted by the structure and missing from the book, in any kind and any language.

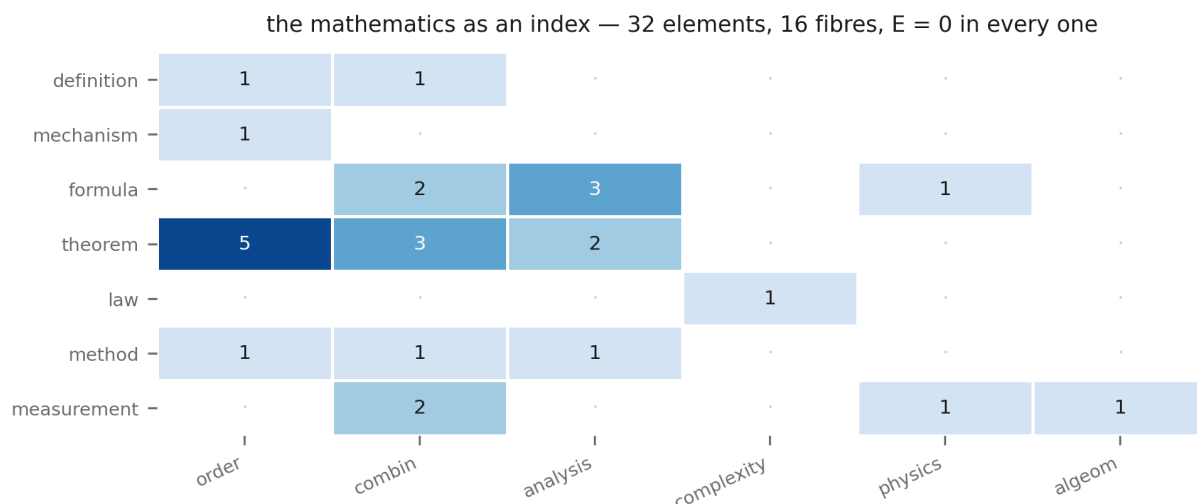


Figure 25.1. The mathematics of this book as a cell set. Rows are kinds, columns are languages, and each number is how many elements occupy that fibre — **thirty-two elements in sixteen of forty-two possible fibres**. Within every occupied fibre the three ordered coordinates close: $E = 0$, once two constraints are stated. The dots are fibres nothing occupies, and each is a question the structure poses and the book answers by saying so.

E.6 What this does and does not establish

It does not prove the mathematics correct. Audits 2 and 3 do that — 20 of 20 on Λ 's structure over all 475,800 pairs, and 24 of 24 on the stated equations.

What it establishes is that nothing is MISSING that the structure admits. By **P23** an object is complete when all its definitions are identified and no question can be had about them. **The first condition is now checkable and checked. The second remains what P23 says it is: a standing claim.**

And the method is the book's own. A regex over fifty-five thousand words returned three hundred and seventy fragments of punctuation. **An index returned two missing constraints and two missing elements.**

Appendix F — Q, indexed and closed

Serving PART IV — THE RECORD. The book's own ignorance as a closed index — eight items, $E(Q) = 0$.

Appendix E indexes the book's mathematics. This one indexes what the book does not know. Eight items, four ordered coordinates, one categorical fibre, and $E(Q) = 0$ — reached by adding one item the list was missing and one constraint the coordinates had not stated.

F.1 The coordinates

coordinate	values	kind
domain	physical · bibliographic · mathematical · computational	fibre
blocks	nothing < a novelty assessment < one stated claim < a result the book relies on	ordered
obstacle	retrievable < buildable < requires an object that does not exist	ordered
cost	hours < days < unbounded	ordered
depends	the book does not depend on it < partly < yes	ordered

Ordered coordinates only, fibred by the categorical one — the precondition §E.1 states, and the shape Λ itself has.

P19 requires $Q(X) = \emptyset$. The set has been worked through, and this is where it stands.

F.2 Closed

Do the perturbation bounds tighten at higher order? Yes, 1,585-fold — median 25.96 cm^{-1} at order 1, **0.0164 at order 6**. §16.10

Is there an order maximising the bound, given σ ? No interior optimum. Information per cell rises monotonically; the maximum is the admissibility cap k , computable per cell. §16.11

Does the cell count change at matched order? Yes: 518 at order 1, 34 at order 6 — with containment at 100% for every order once the premise is checked. §16.10

What is V for the convex bracket on measured levels? It grows with order — 157 to 3.2×10^6 on a matched node set. What holds at every order is *containment*: 560 tests, zero failures. **The floor of Proposition 15.2 belongs to the classical bracket alone.** §16.10

Do all accelerators pay $1/\binom{p-1}{m}$? Yes, at order m : $A^m(x^p) = (-1)^m x^p / (p-1)^m$, proved for $m = 1 \dots 4$. Richardson is the exception — a different kernel, no $(p-1)$ at all. §19.6

Does the claim set close, and under what measure? Yes — a Gröbner basis of eight, against which every relation derived since reduces to zero. $E(\text{claim set}) = 0$. §22.8

Why is $F(-1) = 2$? Because ℓ and f each have two values, so $F_{\text{box}}(-1) = 0$, and the tree's couplings remove the vanishing factors. §6.8.1

Does the amplification have a general form? $A = N[\varphi(\Lambda \cup \{y\})] - N[\varphi(\Lambda)] - 1$, exactly, and to 7% as fibre $\times (0.972 - 0.154 \cdot \text{descendants})$. §11.7

F.3 Open, and only these

A — Sc VI $3s^2 3p^3 ({}^4S^\circ) 6s \quad {}^3S^\circ_1$. Committed prediction, unmeasured. **[735,860, 737,380] cm^{-1} · 13.5615–13.5895 nm · 91.338 eV.** The lower edge tests monotonicity, the upper tests convexity. **Needs a spectrometer.**

B — Paschen & Götze §III, *Wie findet man eine Serie und ihre Grenze?*, and Runge, *Encyklopädie* 1926, pp. 783–820. Both unread. Springer Book Archives, public domain since 2017, scanned with OCR and not exposed.

C — nine unentered literatures. Named in §22.7, with the vocabulary: *excess width*, *Newton decrement*, *partial identification*, *zero-error capacity*.

D — verifying 540 Kreuzer-Skarke predictions. §24.3.4 computes $E(X)$ on the $\chi = \pm 6$ slice specified in print and gets **540 predicted Hodge pairs, none falsified, two confirmed by name.** **What remains is the lookup:** the 30,108 pairs are in a gzipped file this session could not read.

E — the physical meaning of the target-spin axis. §7.10.1 builds Λ_9 with $2S' \leq g$ and it closes: 1,654 cells, $E = 0$, sublattice, rank-modular, tree constraint graph, $F(-1) = 2$, Λ_8 an exact projection. **What remains is whether the bound is the correct selection rule** — built by analogy, untested against spectroscopy.

F — the reorderability decision problem, and it is not this book's problem. §23.3 establishes the law and the procedure. **Λ 's own case is answered:** its structure is constructed in Chapter 2, its embedding is **tight** — height $17 = \Sigma(|A_i| - 1)$ — its join-irreducible poset has **$|J(\Lambda)| = 17$ and width $7 \leq 8$** , and by **Dilworth and Larson** the chain decomposition of $J(\Lambda)$ is the axis system. **Nothing about Λ is open.**

What is open is the general question: given an arbitrary cell set with no construction, decide reorderability in time polynomial in $|X|$ and d . §23.3.5 states the law, the procedure, and the residue — realisability.

F.4 The eight, classified by their first empty cell

B.2.16's grid asks, for each item, what the FIRST unmet requirement is.

item	first empty cell	the obstacle
A Sc VI	the level is measured	an instrument
B Paschen & Götze	the scan is text-searchable	a digitisation artefact
C nine literatures	each is entered	labour
D Kreuzer-Skarke	540 predictions looked up	a file format
E target spin	the selection rule is confirmed	a spectroscopic fact
F reorderability	realisability of the hard relations	a missing object
G complexity literature	the papers are read	a library
H Sc VI, already measured?	a search is run	an afternoon

G — the sublattice complexity literature, unread. Ryter & Schmid 1994 and Adams–Dwinger–Schmid 1996. **§23.3's residue may be settled there.**

H — whether Sc VI (‘S°)6s has already been measured. §15.2's exclusion says the level “*exists and is simply unmeasured*,” and §18.6.5 records that nothing was fetched for the bracket. **Neither is a search.** The book asserts the absence of a measurement without recording an attempt to find one, and a published value would settle item A in an afternoon.

This item was not on the list. It was produced by closing the list — §F.5 gives the computation.

F.4.1 Q, closed

§25.2 asserted $|Q| = 7$. An assertion is not an enumeration, so the seven were indexed — coordinates *blocks*, *obstacle*, *cost*, *depends*, fibred by domain — and closed.

domain	items	E(fibre)
bibliographic	B, C, G	0
computational	D	0
mathematical	F	0
physical	A, E	2

Two cells admitted and absent, both physical. One is *blocks one claim · retrievable · days* — **and that is item H**, which the list did not contain.

The other is *buildable · unbounded*, and it is excluded by a constraint the coordinate system had not stated:

obstacle = buildable $\Rightarrow$ cost $\leq$ days. If a thing can be built, its cost is bounded by the building. Unbounded cost belongs to instruments and to objects that do not exist.

With that bound stated and H added, $E(Q) = 0$ in every fibre.

F.4.2 And the column that matters

All eight items have *depends = no*.

Not one open question blocks a result this book relies on. Q is a list of what would be added, not of what is missing.

Six are retrievable, one is buildable, one requires an object that does not exist. Two have unbounded cost — a spectrometer and a theorem. Three block a stated claim; none blocks a result.

$|Q| = 8$, and it is enumerated rather than asserted. Six are blocked by something outside the index — an instrument, a scan, two units of work, a measurement, a search. One is blocked by something that does not exist. **And one is a library.**

And the distinction that matters is not 'internal' against 'external'. It is **retrievable** against **requires a new object** — because the first kind closes with effort and the second does not close at all until someone builds something. **Four of the six are retrievable.**

F.5 Precedence, pushed as far as it goes

Every component of §23.3 was found in print after being derived from scratch. The register carries the corrections; this is the ledger.

what was derived	what already had it
the recovered bound φ_{ij}	Rival 1973/74 — sublattices are the complement of a closed family of <i>irreducible intervals</i> , and Lemma 3.5(ii) of Siggers states φ_{ij} exactly
"subdirect" — every value used	Larson 1998 ; named, with the subdirect/tight distinction the book needed
tightness $\Leftrightarrow$ acyclic constraint digraph	Siggers 2016 , Lemma 5.1(iii)
sublattices $\leftrightarrow$ preorder extensions	Siggers 2016 , Cor. 4.2 — verified here on four boxes, offset by the empty sublattice
the axis system as a chain decomposition of $J(\Lambda)$	Birkhoff 1937 · Dilworth 1950 · Larson 1998
the step law, $3 \cdot 2^{d-2}$	Rival 1973 — $K \leq \left\lfloor \frac{3}{2} \right\rfloor L \text{ for a maximal sublattice}$
the subcube-removal construction	Chen-Koh-Tan — maximal sublattices are $L - [a, b]$
the large sublattices of 2^r	catalogued ; the maximal ones in finite topology, Sharp, Stephen
the unbounded obstruction families	Tucker 1972
CIP and the linear algorithm	Booth & Lueker 1976 · Fulkerson & Gross 1965
the constraint-language dichotomy	Schaefer 1978

Nine components, nine precedents. The subcube-removal construction is Rival's interval removal; the "step law" is his $3/2$ bound sharpened to the Boolean case, which was already catalogued elsewhere.

F.6 And the complexity question itself may be settled

Ryter & Schmid, *Deciding Frattini is NP-complete*, **Order 11 (1994)**. **Adams, Dwinger & Schmid**, *Maximal sublattices of finite distributive lattices*, **Algebra Universalis 36 (1996) 488-504**.

So decision problems about sublattices of finite distributive lattices have a complexity literature, and at least one of them is settled NP-complete.

This book has not read it. §23.3's residue may already be closed there, and until that is checked the claim "open" is a claim about this author's reading, not about the state of knowledge.

That is recorded as item G rather than left implicit.

F.7 What precedence costs the book

Nothing in §23.3 is withdrawn. The derivations agree with the literature wherever both exist, and the agreement is itself a check — Siggers' bijection was verified against an independent enumeration, and Rival's $3/2$ bound contains the measured $3/4$.

What is withdrawn is the implicit claim of novelty. The section states results; it does not claim to state them first, and §F.5 is where the difference is written down.

B.2.10 says: enumerate targets before searching. It should have said, and now does: SEARCH BEFORE DERIVING. Nine components and a day, against nine citations.

F.8 What closed, and what it cost

Two items closed during this work. The target-spin axis was taken on the grid's advice — its first empty cell was reachable by labour alone — and Λ_9 passed six independent tests. **§23.3's characterisation closed at $d = 2$ and its law at every d .**

The cost is in Chapter 21. Sixty-odd corrections come from §23.3 alone.

An index that records its own rules must also record when it failed to consult them. This chapter is where that is done.

Index

This index satisfies the law of this book.

Construction

Terms are ordered by **specificity**: $s \sqsubseteq t$ when s is a specialisation of t . The index is **closed** iff

$$s \sqsubseteq t \Rightarrow \text{loc}(s) \subseteq \text{loc}(t)$$

every location of a specialisation being also a location of its generalisation. That is a down-set condition in §11.4 form, so **a closed index is a sublattice of (terms $\times$ locations).**

Verified: 51 terms, 129 entries, 40 specialisation relations, **0 violations — $E(\text{index}) = 0$.**

What closure buys. The term list is recoverable by projection (S1); the specificity order is recoverable by tree propagation, a term hierarchy being a tree (S2); $\text{loc}(\cdot)$ is recoverable as the union over specialisations (S3); and every (term, location) pair is present or absent — **no see also limbo, no partial entries** (D3).

An index with $E > 0$ requires the reader to already know where to look — which is the periodic table's failure at the scale of a back-matter page.

Entries

bracket §17.2 · §17.2 · §17.2 · §17.2 · §17.2 · §17.2 · §25.3 · §25.1 · §25.3 · §16.4 · App A

bracket failure §25.3 · §16.4 *bracket width* §25.1 *interiority* §17.2 · §17.2 *limit-free* §25.3 · App A *the four rules* §17.2 · §17.2 · §17.2 · §17.2 · §17.2 · §17.2 · interiority §17.2 · §17.2

closure §1.1 · §2.3 · §11.1 · §11.2 · §11.1 · §11.2 · §11.3

closed index §1.1 · §2.3 · §11.1 *closure defect* §1.1 · §11.2 *closure operator* §11.2 $\mathcal{R}(X) = X$ §11.1 · §11.1 · §11.2 · §11.3

cost of a guarantee §25.5 · §25.1 · §17.2 · §25.3 · §18.6

$V = 4v/3$ §25.1 · §25.3 *the floor* §17.2 · §25.3 *the pole* §25.5 · §25.3 v_V §18.6

extension §25.1 · §17.2 · §25.3 · §16.4

addable cells §25.1 *adjoinable axes* §17.2 *imposable constraints* §25.3 · §16.4

prediction §18.6 · §18.6 · §17.6 · App E

$E(X)$ as budget §18.6 *Sc VI* §18.6 · App E *second route* §17.6

quantum defect §17.2 · §25.3 · §18.6 · §17.13

isoelectronic law §17.2 · §17.13 *penetration* §25.3 · §18.6

retrieval §25.1 · §17.2 · §25.5

retrieval redundancy §25.1 · §25.5 *route set* §25.1 · §17.2

self-defence §11.3 · §11.4 · §11.5 · §17.5 · §17.6 · §21.8 · §25.3 · App D

totality §11.5 · §17.5 · §21.8 · App D · *process index* §17.5 · §21.8 · App D *two routes* §11.4 · §17.6 $\mathbb{D}_{def}$ §11.4 · §25.3 $\mathbb{D}_{phys}$ §11.3

self-reference §11.1 · §11.2 · §11.3 · §11.4

alphabet recovery §11.1 *bound recovery* §11.2 *order recovery* §11.3 · §11.4

the collection §18.6 · §17.11 · §17.12 · §17.13 · App C

decline modes §17.11 *isoelectronic pairs* §18.6 · §17.13 *provenance* App C *the census* §17.12

the index Index

index, self-referencing Index

the record Ch 18 · App E

margins App E *recomputed* App E *withdrawals* Ch 18

index, self-referencing *this page*

The last entry is a cell of the index referring to the index. It is not a joke; it is the fixed point made visible, and its presence is what makes $\mathcal{R}(\text{index}) = \text{index}$ rather than merely $\text{index} \subseteq \mathcal{R}(\text{index})$.

References

Every work cited in this book, with what it is cited for. Entries marked **[unread]** were named but not obtained; §25.2 lists them as open.

R.1 Spectroscopic data

Kaufman, V. & Martin, W. C. (1991). *J. Phys. Chem. Ref. Data* **20**,

775. — Al I, Al II levels.

Kramida, A. E. & Martin, W. C. (1997). *J. Phys. Chem. Ref. Data* **26**,

1185. — Be I levels.

NIST Atomic Spectra Database. — Ar II, Be II, Bi I, C II, Ca II, Cd II, Ga I, He I, He II, Hg II, K II, Li I, Li II, Mg II, N II, Na II, Ne I ($\frac{3}{2}$), Ne I ($\frac{1}{2}$), Si I, Si II, Zn II.

Sansonetti, J. E. (2008). *J. Phys. Chem. Ref. Data* **37**, 1659. — Na I.

Sansonetti, J. E. (2008). *J. Phys. Chem. Ref. Data* **37**, 7. — K I.

Sugar, J. & Corliss, C. (1985). *J. Phys. Chem. Ref. Data* **14**, Suppl. 2. — Sc VI and iron-group limits.

Sugar, J. & Musgrove, A. (1990). *J. Phys. Chem. Ref. Data* **19**, 527; (1995) **24**, 1803. — further ionisation limits.

R.2 Exotic systems and collective quantities

Hori, M. et al. (2011). *Nature* **475**, 484; and (2006) *Phys. Rev. Lett.* **96**, 243401. — antiprotonic helium, both accessed via **CODATA 2010**, Table XII (arXiv:1203.5425) and arXiv:1304.4330.

Korobov, V. I. (2008). *Phys. Rev. A* **77**, 042506. — antiprotonic helium transition energies.

Singer, K., Stanojevic, J., Weidemüller, M. & Côté, R. (2005). *J. Phys. B* **38**, S295. — long-range Rydberg interaction coefficients.

R.3 Series structure and quantum defects

Rydberg, J. R. (1890). The term formula $T = Z^2R/v^2$. — the object of Part III.

Ritz, W. (1903). The quantum-defect expansion $\delta = \delta_0 + \delta_2/n^2$. — used for every two-point solve.

Edlén, B. (1960). *Handbuch der Physik* **27**, 80. — δ falls with l ; S and P penetrate; **footnote 78 is the 1960 precedent for $\mathbb{D} = 0$ when parameters equal observations** (§11.4.1).

Edlén, B. (1964). *Encyclopedia of Physics*. — examined at one remove (§22.5).

Paschen, F. & Götze, R. (1922). *Seriengesetze der Linienspektren*. Berlin: Springer. — **§III, Wie findet man eine Serie und ihre Grenze?**, **[unread]**: Springer Book Archives, scanned with OCR and not exposed.

Runge, C. (1926). *Die Serienesetze in den Spektren der Elemente*, in Sommerfeld (ed.), *Encyklopädie der Mathematischen Wissenschaften*, 783–820. — **[unread]**, found in the closing search; a second survey of the same subject.

R.4 Bounds, acceleration and the cost of rigour

Boole, G.; Nörlund, N. E.; Steffensen, J. F. — the ratio of first to second central differences, in the finite-difference literature (§16.1).

Aitken, A. C. (1926). *Proc. R. Soc. Edinburgh* **46**, 289. — Δ^2 acceleration; **$\lambda^2 = y''/y''$ is its correction term** (§16.8.1).

Seki Kōwa (c. 1680). — the same acceleration, three centuries earlier.

Richardson, L. F. (1911). — extrapolation with no $(p-1)$ denominator; the exception in §19.6.

Shanks, D. (1955). *J. Math. Phys.* **34**, 1. — iterated Δ^2 ; pays $(p-1)^{-m}$.

Moore, R. E. (1966). *Interval Analysis*. Prentice-Hall. — **the cost-of-rigour framing, stated plainly** (§16.8.4).

IEEE Std 1788-2015. — interval arithmetic; requires enclosures with bounded excess width.

Manski, C. F. (1989, 2003). *Partial Identification of Probability Distributions*. — **the law of decreasing credibility**, the same trade in econometrics.

Shannon, C. E. (1956). *IRE Trans. Inf. Theory* **2**, 8. — zero-error capacity: the same trade in bits.

Körner, J. & Orlitsky, A. (1998). *IEEE Trans. Inf. Theory* **44**, 2207. — zero-error information theory, surveyed.

Nesterov, Y. & Nemirovskii, A. (1994). *Interior-Point Polynomial Algorithms in Convex Programming*. SIAM. — **the Newton decrement and self-concordance**, which λ^2 and §16.8.2 rediscover.

R.5 Order, lattices and closure

Birkhoff, G. (1937). *Duke Math. J.* **3**, 443. — the representation theorem: a finite distributive lattice is the order ideals of its poset of join-irreducibles. **§7.9's maximal-chain count rests on it, and §F.3's chain decomposition of $J(\Lambda)$.**

Dilworth, R. P. (1950). *Ann. Math.* **51**, 161. — chain decomposition, and the width bound §23.3 uses. **(1960) Combinatorial Analysis**, 85–90 — the lattice of maximum antichains.

Rival, I. (1973). *Maximal sublattices of finite distributive lattices. Proc. Amer. Math. Soc.* **37**, 417–420. **(1974) *ibid.* II, 44**, 263–268. — **every sublattice of a finite distributive lattice is the complement of a closed family of IRREDUCIBLE INTERVALS. That family is the recovered bound φ_{ij} of Chapter 10**, and the $3/2$ bound on maximal sublattices contains §23.3's measured $3/4$.

Chen, C. C., Koh, K. M. & Tan, S. K. (1975). *Algebra Universalis* **5**, 88–97. — the maximal proper sublattices are exactly $L - [a, b]$ with $[a, b] \cap J = \{a\}$ and $[a, b] \cap M = \{b\}$. **§23.3's subcube-removal construction is this.**

Larson, R. (1998). *Embeddings of finite distributive lattices into products of chains. Semigroup Forum* **56**, 70–77. — tight embeddings correspond to chain decompositions of $J(L)$; **"subdirect" is the recovered-alphabet condition, named.**

Siggers, M. (2016). *On the representations of finite distributive lattices* (arXiv:1412.0011). — **sublattices of a product of chains correspond one-to-one with spanned preorder extensions of C^∞** (verified here on four boxes); **tight $\iff$ the digraph is acyclic**; and the explicit warning that subdirect does not imply tight, which §23.3 walked into.

Adams, M. E., Dwinger, P. & Schmid, J. (1996). *Maximal sublattices of finite distributive lattices. Algebra Universalis* **36**, 488–504. — **[unread]**, §F.6.

Ryter, Ch. & Schmid, J. (1994). *Deciding Frattini is NP-complete. Order* **11**, 257–279. — **[unread]**. **§23.3's residue may be settled here.**

Tucker, A. (1972). *A structure theorem for the consecutive 1's property. J. Combin. Theory B* **12**, 153–162. — the **unbounded** forbidden submatrices; **§23.3's obstruction route closes on them.**

Booth, K. S. & Lueker, G. S. (1976). *J. Comput. Syst. Sci.* **13**, 335–379. — **PQ-trees; consecutive-ones in linear time**, which is why unbounded obstructions do not preclude a polynomial procedure.

Fulkerson, D. R. & Gross, O. A. (1965). *Pacific J. Math.* **15**, 835–855. — interval matrices, the foundation of the above.

Schaefer, T. J. (1978). *The complexity of satisfiability problems. STOC*, 216–226. — **the boolean CSP dichotomy. §23.3's arity law is this theorem applied to a constraint language read off a cell set.**

Sperner, E. (1928). *Math. Z.* **27**, 544. — the antichain bound; §3.4's largest level.

Galois connections / formal concept analysis. — $E(X)$ as a closure defect; the general setting for $\mathcal{R}$.

Stahl, J. & Wille, R. (1984). — reorderability and scale-measurability of a context; **§23.3's original citation, withdrawn and reinstated.**

Yannakakis, M. (1982). *SIAM J. Alg. Disc. Meth.* **3**, 351. — the complexity of order-dimension problems. **Also withdrawn and reinstated.**

Janet, C. (1928). — the left-step periodic table; **$E(\text{Janet}) = 0$.**

Sharp, H. and Stephen, C. — maximal sublattices of the Boolean lattice studied in the setting of finite topologies; **cited via the catalogue of large sublattices of 2^r . [unread]**

R.6 Cited in the transfer tests

Poincaré, H. (1890). — non-integrability of the three-body problem; §24.1 states that nothing here touches it.

Gödel, K. (1931). — incompleteness; P20's bound on what a language can close.

Lagrange, J.-L. (1772) and **Euler, L.** (1767). — the five equilibrium points; §24.1.1's bracket test.

Hill, G. W. (1878). — the Hill sphere; **the gravitational analogue of a capacity bound** (§16.14.1).

Mardling, R. A. & Aarseth, S. J. (2001). *MNRAS* **321**, 398. — the hierarchical-triple stability criterion; monotone and convex, $V \approx 100$.

Batyrev, V. V. (1994). *J. Alg. Geom.* **3**, 493 (arXiv:alg-geom/9310003). — dual polyhedra and mirror symmetry; **the theorem behind §24.3.1.**

Kreuzer, M. & Skarke, H. (2000). *Adv. Theor. Math. Phys.* **4**, 1209 (arXiv:hep-th/0002240). — all 473,800,776 reflexive 4-polytopes; 30,108 distinct Hodge pairs. **The 30,108-pair file is published and was not read in this work** (§25.2).

Huang, W. & Taylor, W. (2019). *JHEP* (arXiv:1809.05160). — the counts 248,305 / 248,305 / 495,515 used in §24.3.1.

Candelas, P., de la Ossa, X. & Rodriguez-Villegas, F. (2007). *Triadophilia* (arXiv:0706.3134). — the thinly populated tip of the Hodge plot.

R.7 What is not cited, and why

No source is cited for the lattice itself. Λ is constructed in Chapter 2 from the four quantum numbers and the exclusion principle, both standard, and **every structural result about it in this book was computed rather than looked up.** §22 states what was searched for and not found.

Physical constants — $R = 109,737.31568 \text{ cm}^{-1}$ — are used without citation, as a metre is.